Roadmap for Lectures 1-5

- Models of Supply and Demand
- Elasticities
- Optimal consumer choice: indifference curves, utility functions, budget constraints, etc.
- Applications of Consumer choice
- Choice under uncertainty
- Production functions (Thursday - screencast)
- Opportunity costs and sunk costs
- Derivation of textbook cost functions
- Other cost concepts (e.g., EoS, learning by doing)
- Profit maximization
- Deriving short-run and long-run supply for firms and industry
- Source of economic profit (i.e., producer surplus) in competitive

Day 1:
Supply, demand, elasticities, revenue

What is Microeconomics?

- Basic economic problem:
  - about how scarce resources are allocated
- Microeconomics:
  - about individual decision making and resulting aggregate effects focused on single markets
- Macroeconomics:
  - concerns aggregate (multi-market) phenomena, often built upon microeconomic foundations

Scope of Microeconomics is huge!

- Studying individual choice to determine the allocation of scarce resources in “markets” is broad
- The Chicago School has been at the frontier of applying economic principles to non-traditional markets (e.g., Friedman, Becker, Levitt):
  - Marriage
  - Crime and gangs
  - Addiction
  - Environment
- Bottom line: microeconomics is about studying how incentives (e.g., prices, laws) influence economic decisions and markets

Mathematics, models, and making

- We use simple economic models to focus attention on specific economic effects
- Mathematics provide precise statements
- Economists (like natural scientists) make simplifying assumptions
- Some assumptions are patently untrue and at best an approximation. But the test is the value of the model in understanding and predicting economic phenomena (Friedman)

Recurring Themes

- Trade-offs are everywhere
- Marginal analysis is useful
- Incentives matter (e.g., prices, laws, etc.)
- Trade benefits everyone
- Prices reflect marginal values/costs
- Markets are powerful (often beneficial) forces
  - Stigler: ‘The more one studies markets, the more one respects them.’

The Supply & Demand model

- Most famous of all economic models
- Applicable in **perfectly-competitive markets**
  - well-defined market with many buyers and sellers
  - buyers and sellers are price takers
  - market price equates supply and demand
- Examples: commodity markets
- Inappropriate model for when either firms or buyers strategically influence the market price
  - S&D model appropriate for DRAM but less so for CPU market

Demand functions

- A demand function is a mathematical relationship (usually empirically determined) that transforms underlying economic variables into units demanded:

$$Q^d = D(p, p_1, \ldots, p_m, I, \ldots)$$

- Q^d is measured in units of consumption (over a fixed period of time for a defined market)
- Demand depends upon the good's market price
- Demand also depends upon other or "cross" prices: p $p_1, \ldots, p_m$, which include prices of substitutes and complements
- Demand also depends upon income, I .

Demand functions

- Implicit in the definition of any demand function is
  - the **measured units** of output (e.g., *bushels* of wheat, *kilograms* of copper, etc.)
  - the **duration** of the market period characterized by the demand function (e.g., bushels of wheat *per month*, kilograms of copper *per year*, etc.)
  - the **boundaries** of the market (e.g., Chicago, US, World).
- Thus, to be precise, we might say something like "the monthly demand for bushels of wheat in Chicago," etc.

Demand functions

- Also implicit in the demand function is the **time of adjustment** for a given change in variables.
  - For example, a *short-run* demand function will tell us how demand changes this period in response to a change in price that occurs in this period.
  - A *long-run* demand function would tell us how demand changes this period in response to a change in price that may have occurred many periods in the past.
  - Because more adjustment time will generally lead to larger responses to a given economic change, long-run demand functions will generally exhibit greater price sensitivities than short-run demand curves.

Units of measurement

- It is important and useful to keep in mind the units of measurement for each economic variable, just as in the physical sciences
- Prices are always "unit of money/unit of output" (e.g., €/kg, $/bushel, etc.)
- The measurement of output should correspond to the denominator of the price measurement
  - e.g., if you are plotting bushels of corn as the output (or even millions of bushels of corn), then the price should be in money/bushel (as opposed to money per kilogram).

Units of measurement

- Whenever you combine economic variables, the resulting combination should be consistent with the original dimensions
  - e.g., Price x Quantity = Revenue

    If price is measured by "$/bushel" and quantity by "bushels", then revenue is measured by

    ($/bushel) x (bushels) = $.
- This dimensional analysis provides a useful consistency check on your calculations

“Law” of demand

- We expect a negative relationship between quantity demanded of a good and its price
- This is more an empirical regularity than a law
- Violations?
  - Giffen goods
  - information revealed in price (Akerlof, et al.)
  - network or social effects in consumption (e.g., Becker, et al.)

Other relationships in demand

- We say that two goods are **demand substitutes** if an increase in the price of one raises the demand for the other
  - e.g., French and Australian wines are substitutes
- Two goods are **demand complements** if the relationship is reversed: an increase in price for one good reduces demand for the other
  - e.g., French wine and French cheese
- NB: two goods may have neither relationship or their relationship may change with prices

Other relationships in demand

- We say that a good is **normal** if an increase in a consumer's income raises demand
  - e.g., restaurant meals are normal goods
- A good is **inferior** if the relationship is reversed: an increase in income leads to a reduction in demand
  - e.g., subsistence-level food stuffs (rice, potatoes)
- A good may change from normal to inferior as incomes change
  - e.g., cheap table wine (normal if poor)

Market versus consumer demand functions

- We can easily imagine deriving a demand function for a single consumer or household:

$$q_i^d = D_i(p, p_1, \ldots, p_m, I_i)$$

where we have used i as a subscript to distinguish a particular consumer's demand, q_i^d, and her income, I_i

- Market demand would then satisfy

$$Q^d = \sum_{i=1}^{N} q_i^d = \sum_{i=1}^{N} D_i(p, p_1, \ldots, p_m, I_i)$$

A few remarks

- Note that the market demand function should depend upon the profile of all household incomes rather than the simply the sum of incomes; this poses problems in estimation
- Statistical estimation of market demand curves typically proceeds using regression analysis (which you'll learn in statistics) taking advantage of variation in supply factors.

Corn example

- Demand function for corn (in US per year)

$$Q^d_{\text{corn}} = 5 - 2p_{\text{corn}} + 4p_{\text{potatoes}} - 0.25p_{\text{butter}} + 0.0003I$$

- Potatoes are substitutes; butter is a complement; corn is a normal good.
- Note this is only illustrative; it is not a real demand function.

Demand curves

- A demand curve is a plot of a demand function with quantity demanded on the horizontal axis and own price on the vertical axis, holding fixed all cross prices and income.
- Corn example:

$$Q^d_{\text{corn}} = 5 - 2p_{\text{corn}} + 4p_{\text{potatoes}} - 0.25p_{\text{butter}} + 0.0003I$$

  - Suppose potatoes are \$0.50/lb, butter is \$4/lb, and annual average income is \$30,000. Then

Or alternatively, $$Q^d_{\text{corn}} = 15 - 2p_{\text{corn}}.$$

$$p_{\text{corn}} = 7.50 - 0.50Q^d_{\text{corn}}$$

Remarks:

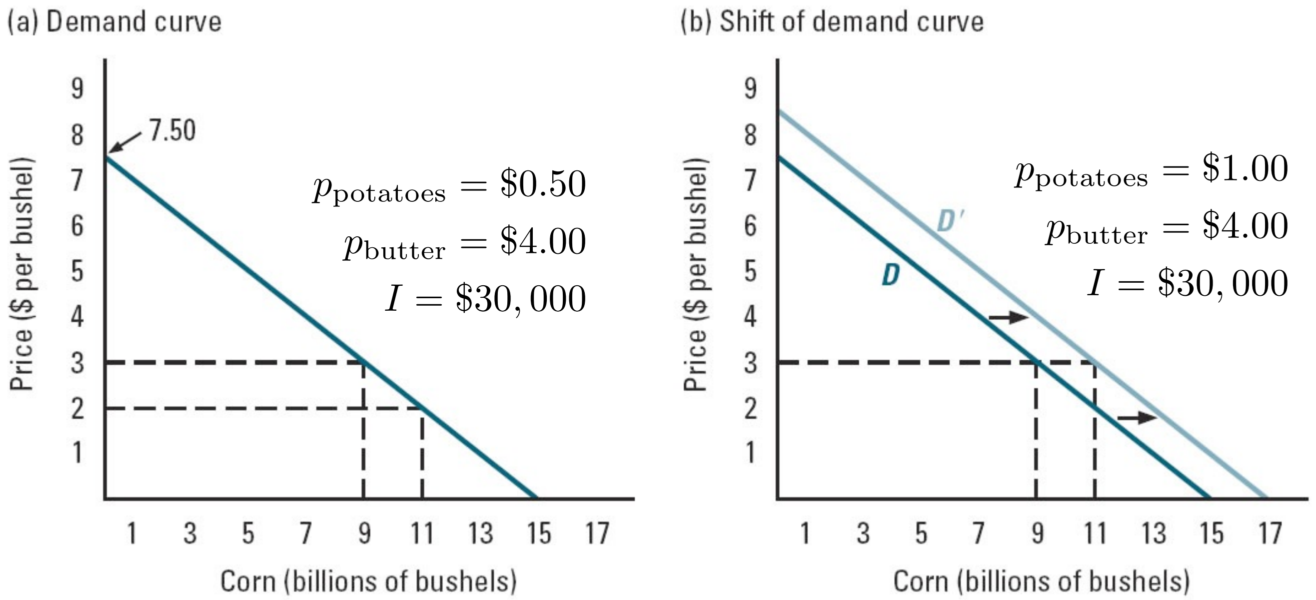
- Demand curves are always graphed with quantity on the horizontal axis and price on the vertical axis.
- For this reason, the mathematical expression for the demand curve, e.g.

$$p_{\text{corn}} = 7.50 - 0.50Q^d_{\text{corn}}$$

  is called an **inverse demand function**. Sometimes we will refer to it generically as a function $p = P(Q^d)$.
- Note: The slope of a demand curve as we graph it is $\dfrac{\Delta p}{\Delta Q^d}$,

  while the "slope" of a demand function is $\dfrac{\Delta Q^d}{\Delta p}$.
  Be careful!

Demand Curve for U.S. Corn Market



$$p_{\text{potatoes}} = \$0.50 \quad p_{\text{butter}} = \$4.00 \quad I = \$30,000$$

$$p_{\text{potatoes}} = \$1.00 \quad p_{\text{butter}} = \$4.00 \quad I = \$30,000$$

$$p_{\text{corn}} = 7.50 - 0.50Q^d_{\text{corn}}$$

$$p_{\text{corn}} = 8.50 - 0.50Q^d_{\text{corn}}$$

(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

Demand curve shifts/movements

- A demand curve represents the relationship between demand and own price.
- Consequently, if some other variable changes (cross price, income, government regulation), the demand curve must shift appropriately
  - e.g., price of potatoes and the demand for corn
- The effect of a change in own price on demand, however, is directly determined by moving along the demand curve

Supply functions

- A supply function is a mathematical relationship that maps economic variables into units supplied:

$$Q^s = S(p, w, r, \ldots)$$

- Supply is measured in units of output (over a fixed period of time to a defined market)
- Supply depends upon the good's market price
- Supply also depends upon input prices, $w, \ r$ which we use for the prices of labor and capital.
- Other factors such as technology, etc., also relevant

Relationships in supply

- Product's supply curve shows:
  - How much sellers of the product want to sell at each possible price
  - Holding fixed all other factors that affect supply
- On a graph: vertical axis shows $ per unit of the good, horizontal axis shows quantity supplied per unit of time
- Upward sloping (selling the product is less attractive when the price is low than when the price is high)

Relationships in supply

- The higher the market price, the larger supply, holding other variables constant.
- Do higher input prices imply lower supply?
  - Negative relationship is an empirical regularity
  - But not always true. The key we will learn is how input prices affect marginal costs.

Inputs and time frames

- In general, we can think of production having many time frames and many inputs.
- In the shortest run, only a subset of inputs can be varied; within the next period, a larger set of inputs can be varied, ..., and in the longest time frame, all inputs can be varied.
- This complex setting will be much clearer to understand once we develop the model with two inputs and two time frames.

Inputs and adjustment times

- Two productive inputs will illustrate the basic ideas; more than two in reality
- We also assume there are two time frames for simplicity
  - Short run (e.g., this year), we assume only “labor” is variable; “capital” is fixed
  - Long-run (e.g., next year), “capital” is also variable; capital requires one period of advanced planning before any change can be made
  - Actual lengths of periods depend upon product
- In reality, many time frames and many inputs.

Market versus firm supply functions

- We can also denote the supply function for a single firm:
  $$q_i^s = S_i(p, w, r, \dots)$$
  where we have used i as a subscript to distinguish a particular firm’s supply
- Market supply (by all firms) then satisfies

$$Q^s = \sum_i q_i^s = \sum_i S_i(p, w, r, \dots) = S(p, w, r, \dots)$$

Corn example, continued

- Supply function for corn (in US per year)

$$Q^s_{\text{corn}} = 9 + 5\, p_{\text{corn}} - 2\, p_{\text{fuel}} - 1.25\, p_{\text{soybeans}}$$

- Higher prices for corn induce greater supply
- Fuel is an input. The price of soybeans represents an opportunity of the land
- If it takes one year to change crops from corn to soybeans, then this supply function captures the long-run response of farmers
- What would short-run supply depend upon?

Supply curves

- A supply curve is a plot of a supply function with quantity supplied on the horizontal axis and own price on the vertical axis, holding fixed all other prices, technology, etc.
- Corn example:
  - suppose that fuel costs \$2.50/gallon, and soybeans are \$8/bushel:

$$Q^s_{\text{corn}} = 5\, p_{\text{corn}} - 6,$$

  or alternatively

$$p_{\text{corn}} = 0.20\, Q^s_{\text{corn}} + 1.20$$

Remarks:

- Supply curves always are graphed with quantity on the horizontal axis and price on the vertical axis.
- For this reason, the mathematical expression for the supply curve, e.g.

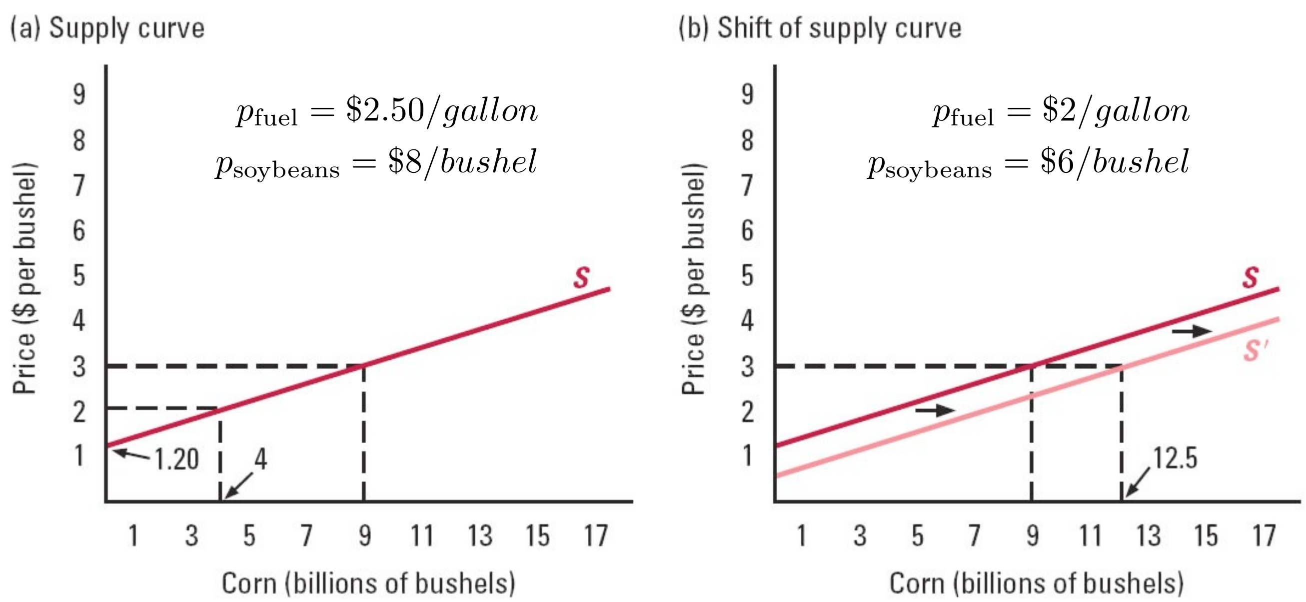
$$p_{\text{corn}} = 0.20\, Q^s_{\text{corn}} + 1.20$$

is called an inverse supply function.

- The slope of a supply curve is $\dfrac{\Delta p}{\Delta Q^s}$,

while the "slope" of a supply function is $\dfrac{\Delta Q^s}{\Delta p}$.

Supply Curve for U.S. Corn Market



$p_{\text{corn}} = 0.20\, Q^s_{\text{corn}} + 1.20$ $\quad$ $p_{\text{corn}} = 0.20\, Q^s_{\text{corn}} + 0.50$

(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

Supply curve shifts/movements

- A supply curve represents the relationship between supply and own price.
- Consequently, if some other variable changes (input prices, technology, regulation), the supply curve must shift appropriately
  - e.g., price of fuel and the supply of corn
- The effect of a change in own price on supply, however, is directly determined by moving along the supply curve

Market Equilibrium

- A market is in equilibrium if the price is such that the supply offered by the firms equals the demand of the buyers. Such a price is an equilibrium price.
- Mathematically, if the equilibrium is at (Q^*, p^*)

  $$Q^* = D(p^*, p_1, \ldots, p_m, I) = S(p^*, w, r)$$

  given cross prices, income, and input prices.
- Graphically, the equilibrium point is at the intersection of the supply and demand curves.

Corn example

- Suppose that the exogenous data are

$$\begin{aligned} p_{\text{potatoes}} &= 0.50/bushel \\ p_{\text{butter}} &= \$4/pound \\ I &= \$30,000 \\ p_{\text{fuel}} &= \$2.50/gallon \\ p_{\text{soybeans}} &= \$8/bushel \end{aligned}$$

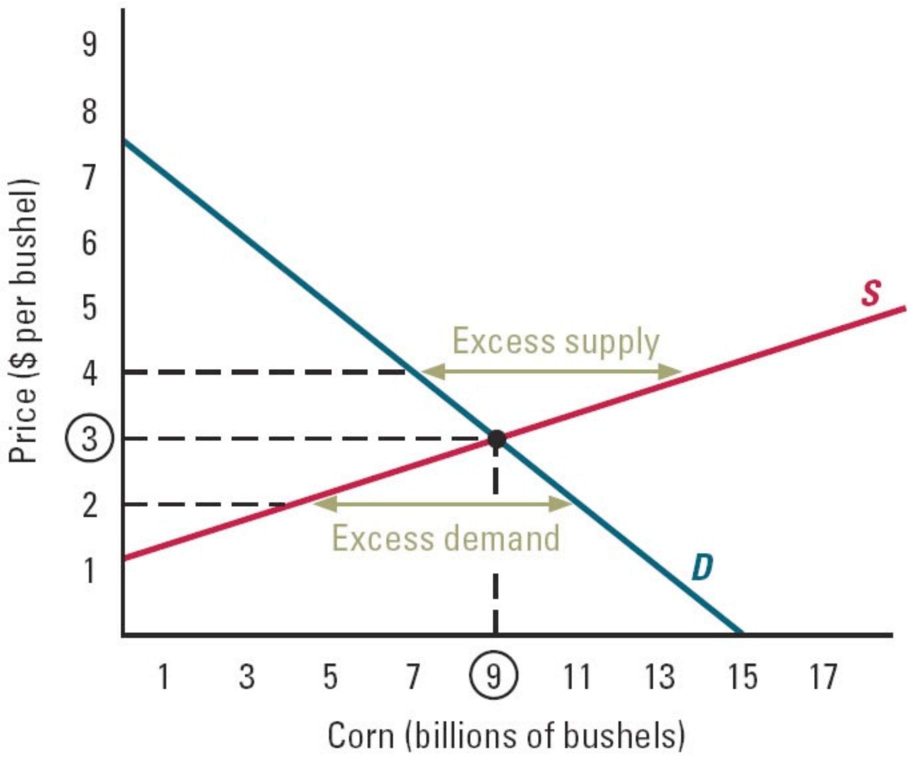
- Then the demand and supply curves are

$$p_{\text{corn}} = 7.50 - 0.50 Q^d_{\text{corn}} \qquad p_{\text{corn}} = 0.20\ Q^s_{\text{corn}} + 1.20$$

- Solving these, we have

$$p^*_{\text{corn}} = \$3/bushel,\ Q^*_{\text{corn}} = 9 \text{ million bushels}$$

Equilibrium in the Corn Market



(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

What are the equilibrium forces?

- If price is above equilibrium price then ...
  - more output is supplied than demanded (excess supply)
  - and as products remain unsold, sellers lower prices to reduce inventories, etc.
- If price is below the equilibrium price then ...
  - demand is greater than available supply (excess demand)
  - and as buyers queue to purchase limited quantities, sellers have an incentive to raise prices
- Only point at which there is no tension on the market price is at the point where demand equals supply.

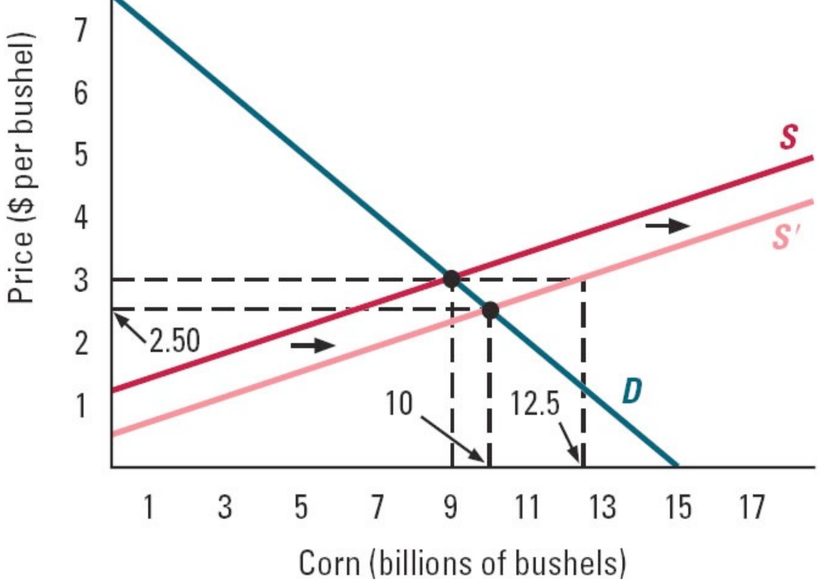
Changes in Market Equilibrium

- Suppose that some exogenous variables (e.g., cross-prices, income, input prices, etc.) in our demand and supply model change.
- Such a change will impact either the supply or demand function (or both).
- This in turn will impact the equilibrium price.
- Example: in the case of Corn, a decrease in the price of fuel will shift the supply curve outward, lowering the equilibrium price of corn and increasing output.

Change in Market Equilibrium

$$\text{original supply: } Q^s = 5p - 6 \iff p = 1.20 + 0.20Q^s$$

$$\text{new supply: } Q^s = 5p - 2.5 \iff p = 0.50 + 0.20Q^s$$



(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

A simple example of Demand & Supply

- Suppose there are 300 buyers and 300 sellers in a market. Each seller has exactly one bushel of apples available to sell; each buyer has demand for at most one bushel of apples to buy.
- Suppose 200 sellers have a cost of $10 per bushel and 100 sellers have a cost of $30 per bushel. These costs represent what the sellers give up by selling a bushel of apples (i.e., this is their *opportunity cost* of the apples).
- Suppose 200 of the buyers have a value for apples of $20 per bushel and 100 buyers have a value of $40 per bushel. These values represent the most that the buyers are willing to pay.
- What is the equilibrium price in this market and how many bushels of apples are traded?

Further questions from our example

(We'll discuss these questions in class)

- How much "profit" do the sellers collectively make?
- How much "consumer surplus" do the buyers obtain?
- Suppose we tax the sellers \$5 per bushel sold. What is the new equilibrium (pre-tax) price?
- Which side of the market actually ends up paying the tax? Would it matter if we placed the tax on the buyers rather than the sellers?

Simple trading example

- The market in equilibrium:

Surpluses in simple apple example

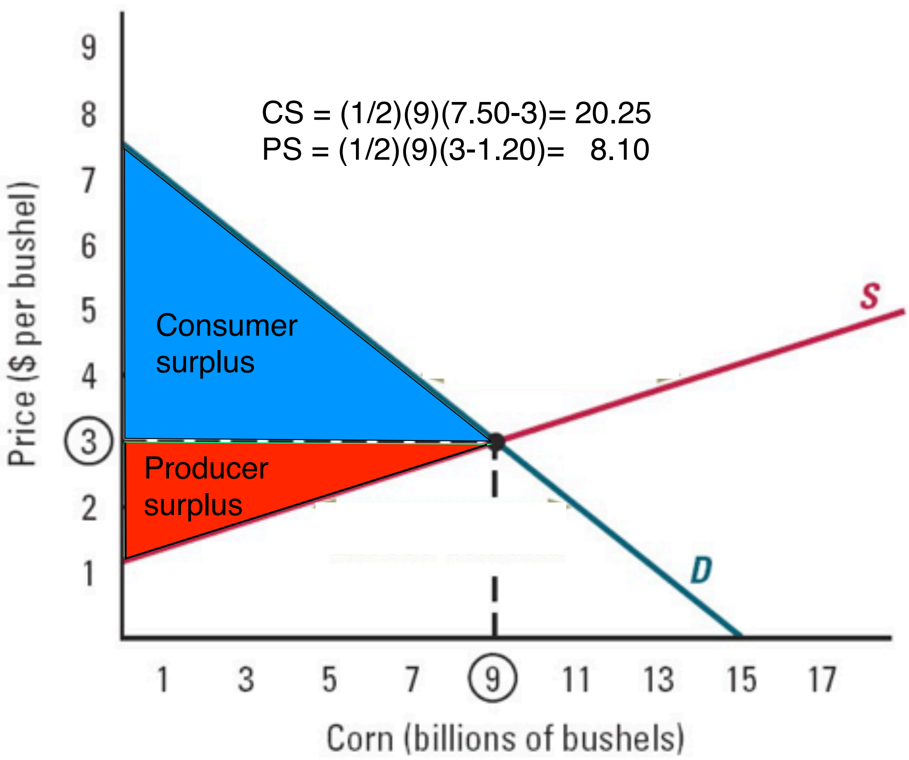
Consumer and producer surplus

- In the simple apple example, ***producer surplus*** (another term for ***economic profit***) is measured by the sellers' received price minus the unit cost for each unit sold.
- **Consumer surplus** is measured by the (monetary) value of the good (i.e., the consumer's willingness to pay) minus price for each unit purchased.
- **Total surplus** or **market welfare** is the sum of these two.

Consumer and producer surplus

- Each point on the demand curve is associated with some buyer's (marginal) value for the good at that point
- Also, each point on the supply curve is associated with some producer's (marginal) cost of producing the good at that point
- Hence: ***Demand and supply curves summarize the marginal benefits and costs of the market.***
- If there is a single market-clearing price, then these surpluses have a simple geometry.
  - **Producer surplus** is the area between the equilibrium price and the market supply curve
  - **Consumer surplus** is the area between the market demand curve and the equilibrium price
- Note: If curves are linear, then each surplus is **area of a triangle**

Remarks



General versus partial equilibrium

- In this course, we will mainly concern ourselves with partial equilibrium analysis.
- That is, when studying a market, we will take some variables as exogenously given.
- In a full General Equilibrium (GE) model, every economic variable is determined together
  - For the Corn example, a GE model would consider the price of soybeans, butter, potatoes, fuel and incomes as jointly determined with the price of corn.   Can get very complex and GE prone to huge mis-specification errors.
  - in this is more complex and small mistakes in one area can make huge mistakes in predictions …
  - … but it is more realistic

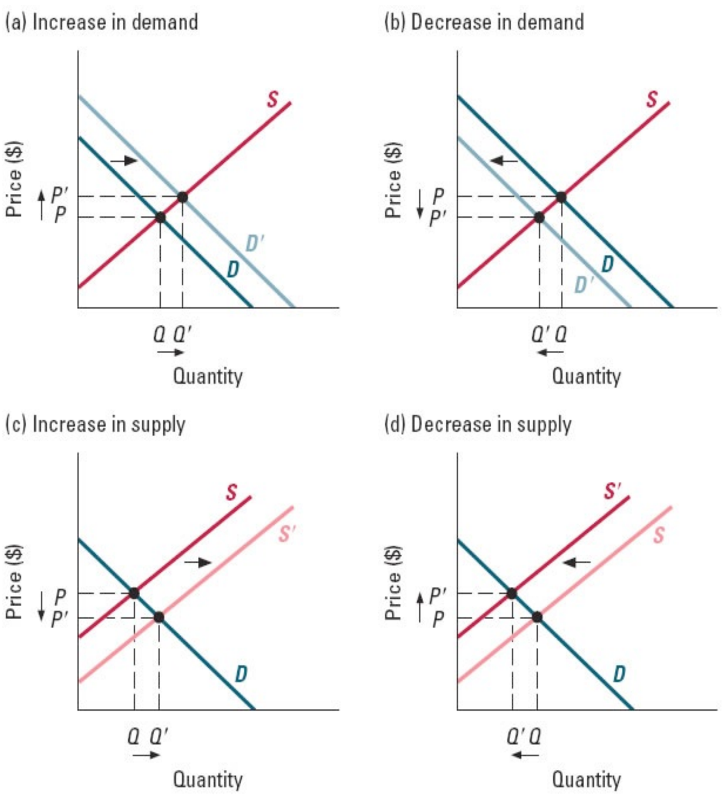
General versus partial equilibrium

- It seems reasonable to take a step or two in the direction of GE models in some cases
  - E.g., in the case of corn, we may want to jointly model the markets for soybeans and corn since each are demand substitutes and each are related to the opportunity costs of production. Maybe add market for farm equipment, etc.
- This would still be a partial equilibrium analysis, but closer in the spirit to GE models.
- The more realistic and general the model, the more complex and error-prone is the analysis because we cannot rely as much on our economic intuition.

Changes in Market Equilibrium

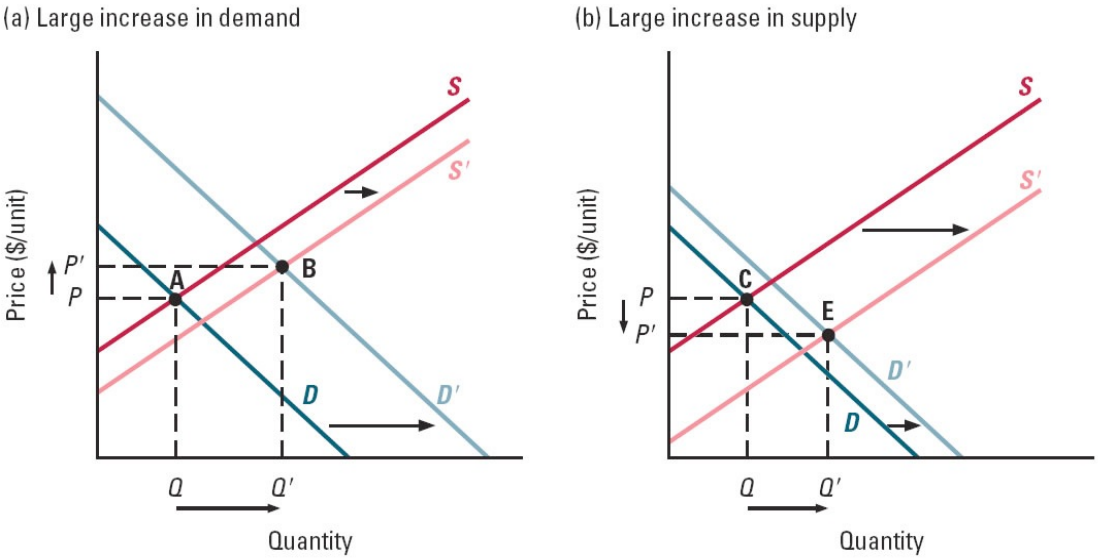
- There are four distinct changes to consider
  - an inward or outward shift in supply or demand
- It is possible that both shift simultaneously
  - E.g., suppose that the price of soybeans appears in the demand for corn (as well as the supply)
- When both curves shift simultaneously, we may be unable to determine the net output or price effect using only theory. We may need an empirical study to sort out the net effect.

Changes in Market Equilibrium



(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

Increase in Both Demand and Supply



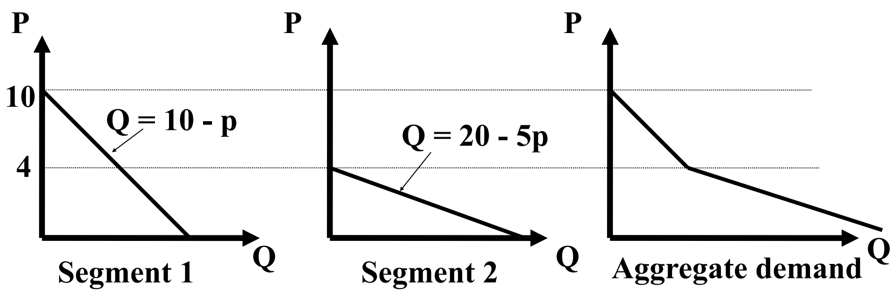
(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

Aggregating curves

- Sometimes it is useful to decompose demand or supply curves into smaller segments
  - E.g., world oil demand is sum of US and non-US demand
  - E.g., different buyer segments make up the demand for a monopolist
- Algebraically: add output segments (e.g., demand)

$$Q^d = \sum_{i=1}^{N} q_i^d = \sum_{i=1}^{N} D_i(p, p_1, \ldots, p_m, I_i)$$

- Graphically: add curves horizontally (e.g., demand)



(note - same idea is used for supply functions and curves)

Aggregating curves, continued

Application: World cotton supply

(see handout)

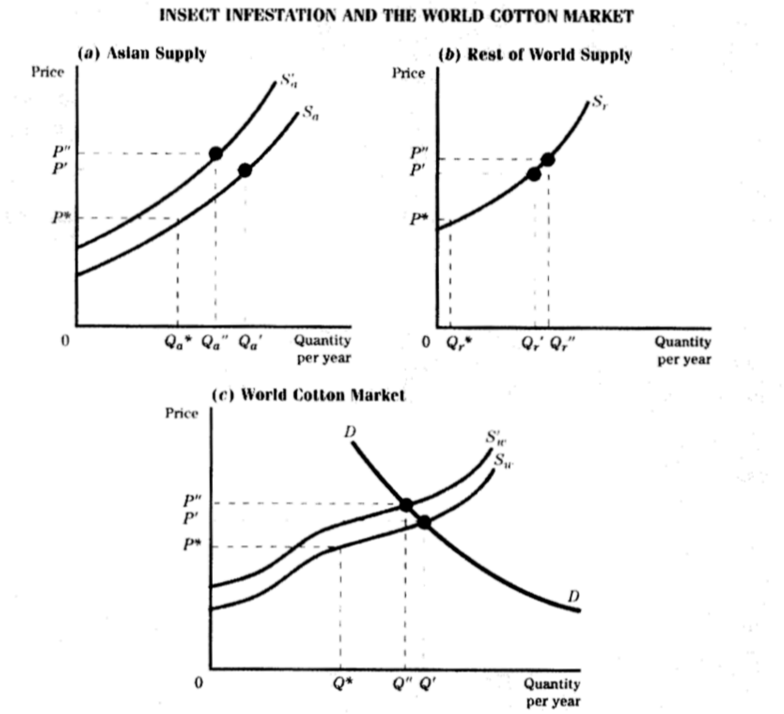


Figure 1-10 The insect infestation shifts the Asian supply function inward (panel a) and the world supply function shifts inward and becomes S_w' (panel c). The equilibrium price rises from P' to P''. Because they receive a higher price, suppliers in the rest of the world expand output from Q_r' to Q_r'' (panel b).

Source: Peter Pashigian, Applied Price Theory, 2nd ed.

Aggregating curves, continued

- Application: What would happen to world C02 levels if US mandated all cars use only pure Ethanol (which produces 50% less carbon/mile) and subsidized Ethanol so that fuel price per mile remains unchanged to US drivers?

- Suppose a billionaire has a love for chickens (he had one as a pet) and wants to spend his money to save chickens from being slaughtered. He decides to spend \$1 billion to buy chickens this year to release onto his private reserve where they will live a life free from slaughter. Suppose that the original price of chickens is \$1/chicken. If 5 billion chickens would have been slaughtered without the billionaire's plan, how many will now be slaughtered if the billionaire implements his plan? What do you need to know?

Short-run versus Long-run Adjustment Times

- Because consumers can choose among more alternatives when given more time to respond to a price change, long-run demand curves will tend to be flatter (more responsive) than short-run curves.
- Because firms can choose to vary more inputs when given a longer time to respond to a price change, a firm's long-run supply curve will tend to be flatter (more responsive) than its short-run supply curve.
- **Note:** There are subtle issues involved with durable goods that requires we modify the above statements. As a practical matter, however, one can generally expect long-run curves to be flatter than short-run curves on either side of the market.

Graphing short-run and long-run equilibria

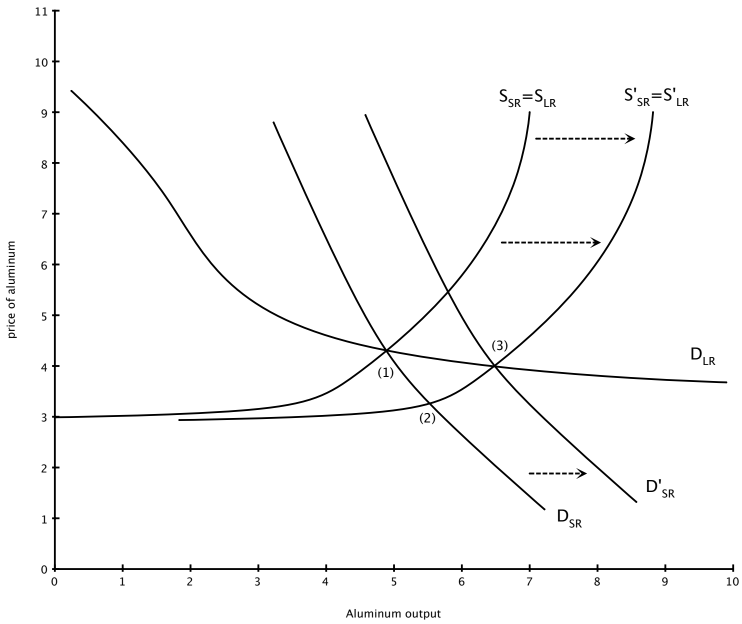
- Suppose that the short-run and long-run demand and supply functions are different, as is likely.
- The short-run equilibrium price is the price which equates the short-run demand and supply curves without any adjustment time.
- The long-run equilibrium price is the price which equates the long-run demand and supply curves, allowing firms and consumers to fully adjust to the change in prices.
  - when the long-run equilibrium emerges, it will also be a new short-run equilibrium
- If these prices differ, we predict the initial short-run equilibrium price for the present period, and the long-run equilibrium price for following next period.

Example of short-run and long-run equilibria

- Suppose that large reserves of bauxite (the input for aluminum) are suddenly discovered and the aluminum supply curve shifts outward
  - the short-run equilibrium price of aluminum decreases from (1) to (2), where short-run demand equals short-run supply
  - over time buyers find more uses for aluminum at its lower price, so the short-run demand curve shifts outward along the long-run demand curve
  - eventually, long-run (as well as short-run) equilibrium arises at (3)

Aluminum supply increase

- (1) is a short-run and long-run equilibrium before the change in supply
- (2) is a short-run equilibrium, but not long-run equilibrium
- At (3) we are again in a short-run and long-run equilibrium after demand has fully adjusted.



Elasticities of Demand and Supply

- Slopes of demand and supply curves play an important role in economic analysis
- But "slopes" are unit specific: slopes depend upon the unit of currency and the measure of output
- Elasticity is a measure of responsiveness that is in percentage terms and is exactly what we want to measure price sensitivity in economic analysis
- As a bonus, the connection between revenue and price along a demand curve is entirely dictated by the elasticity

General idea of elasticity

- An elasticity is a slope in percent terms.
- The general formula for an elasticity of *y* with respect to *x*, is

$$\varepsilon_{y,x} = \varepsilon_x^y = \frac{\%\Delta y}{\%\Delta x}$$

- E.g., elasticity of quantity demanded with respect to price is

$$\varepsilon_{Q^d,p} = \varepsilon_p^d = \frac{\%\Delta Q^d}{\%\Delta p}$$

- This is inversely related to the slope of the demand curve,

$$\frac{\Delta p}{\Delta Q^d}$$

Elasticity of demand

- There are many elasticities related to the demand function, but the (own-price) elasticity of demand is most used:

$$\varepsilon_{Q^d,p} = \varepsilon_p^d = \frac{\%\Delta Q^d}{\%\Delta p}$$

- What does it mean? How is it used?
  - Suppose that a tax will raise prices by 10% and the elasticity of demand is -0.5. What will happen to quantity demanded?

$$\%\Delta Q^d = (\varepsilon_p^d)(\%\Delta p) = (-0.5)(10\%) = -5\%$$

  - Remember the ratio formula above and its use is clearer

Empirical estimates of demand elasticities

- Elasticities are larger if substitutes are better
- Demand elasticities tend to be larger (more elastic) in long run relative to short run
  - E.g., demand for gasoline
- Elasticities tend to be larger when expenditures on good is larger fraction of buyers' income
  - luxury items (e.g., BMW's) tend to be less elastic

(See handouts for examples.)

Calculating elasticity

- Suppose instead that you know only two points on the demand curve and you desire an "average" elasticity of demand over the arc. You can calculate the arc elasticity by simply computing the relevant percentages.

  Note: the percentage change between two points depends upon which point is the initial point.
  - E.g., 9 to 10 is 11% increase; 10 to 9 is 10% decrease.
  - To eliminate this dependency, in the Becker and Miller sections, we will divide by the average of the two points rather than the initial point.
  - E.g., 9 to 10 is a (centered) 1/9.5=10.5% increase; 10 to 9 is a (centered) 10.5% decrease.

Calculating elasticity

- Given two points, (Q_1 ,p_1) and (Q_2 ,p_2), the arc elasticity of demand is therefore

$$\varepsilon_p^d(arc) = \frac{Q_1^d - Q_2^d}{\frac{1}{2}(Q_1^d + Q_2^d)} \Big/ \frac{p_1 - p_2}{\frac{1}{2}(p_1 + p_2)} = \frac{\Delta Q^d}{\Delta p} \frac{(p_1 + p_2)}{(Q_1^d + Q_2^d)}$$

  [NB: the textbook does not center the percentages.]
- Think of the arc elasticity as an average elasticity between two points on the demand curve.
- With more than two data points, one employs statistical methods to estimate the demand function and use it to calculate an elasticity.
  - **Aside:** Economists often assume demand has a constant elasticity form in the range of the data, in which case the regression delivers something like a multi-point arc elasticity for that data range. Technically, if you take natural logs (don't worry if you don't know what this is) of your price and quantity data and perform a linear regression, the estimated coefficient will be an estimated "arc" elasticity using multiple data points.

Calculating elasticity

- How is it calculated from a demand function at a point?
  - If you have the equation of the demand function (or its slope at a point) then you may use the point elasticity formula:

$$\varepsilon_p^d(point) = \frac{\partial Q^d}{\partial p}\frac{p}{Q^d}$$

  You need to know (a) the derivative of the demand function with respect to price at this point (or its slope), and (b) the ratio of price to output at this point.

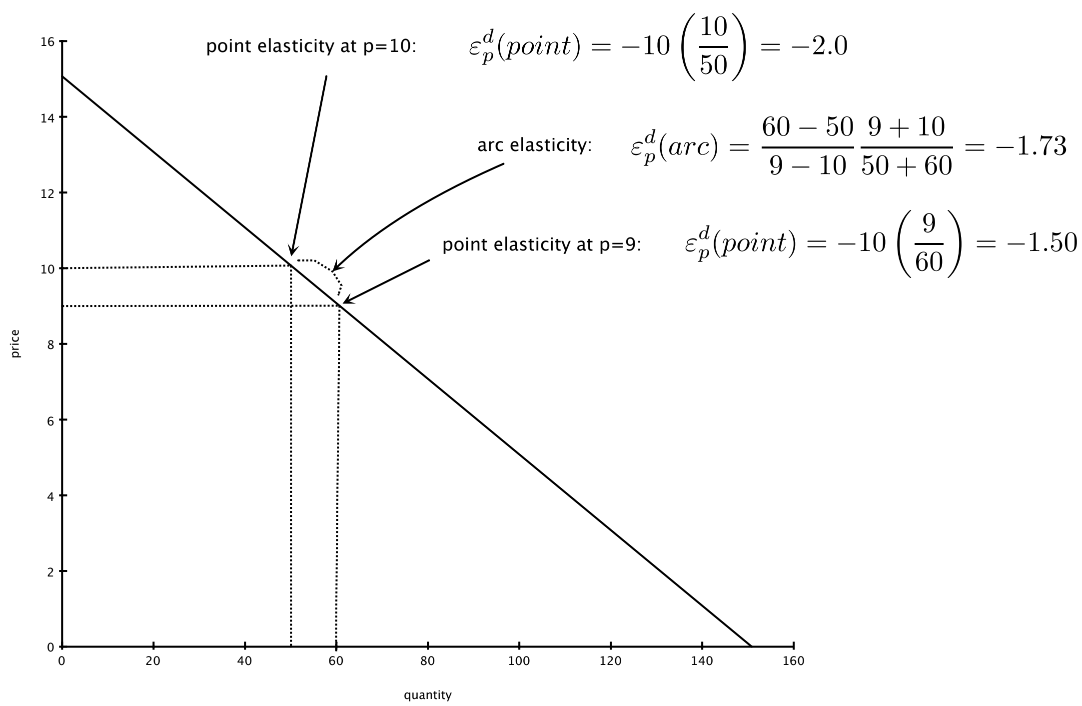
- Compare this formula with that of the arc elasticity

$$\varepsilon_p^d(arc) = \frac{\Delta Q^d}{\Delta p}\frac{\frac{1}{2}(p_1+p_2)}{\frac{1}{2}(Q_1^d+Q_2^d)} \quad \text{vs} \quad \varepsilon_p^d(point) = \frac{\partial Q^d}{\partial p}\frac{p}{Q^d}$$

  - **Aside:** In the limit as the two arc prices become closer to *p*, the arc elasticity converges to the point elasticity

Example of arc and point elasticities

- Suppose the demand function is $Q^d = 150 - 10p$



Remarks about demand elasticity

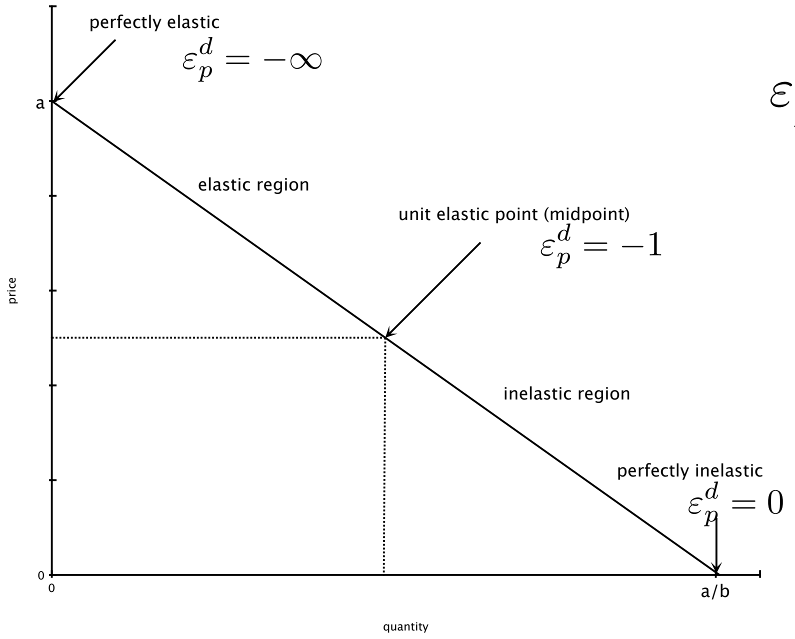
- Because demand curves slope downwards, (own-price) elasticities of demand are negative
- Larger elasticity (in absolute value) means buyers are “more price sensitive”
  - when $|\varepsilon_p^d| > 1$, demand is elastic, (price sensitive)
  - when $|\varepsilon_p^d| < 1$, demand is inelastic, (price insensitive)
  - when $|\varepsilon_p^d| = 1$, demand is unit elastic.

Special cases:

- Vertical demand curve: (perfectly inelastic)
$$\varepsilon_p^d = 0$$
- Horizontal demand curve: (perfectly elastic)
$$\varepsilon_p^d = -\infty$$
- Linear demand “curve”
- Iso-elastic (constant elasticity) demand curve

Linear demand curve

- Suppose inverse demand function is $p = a - bQ^d$

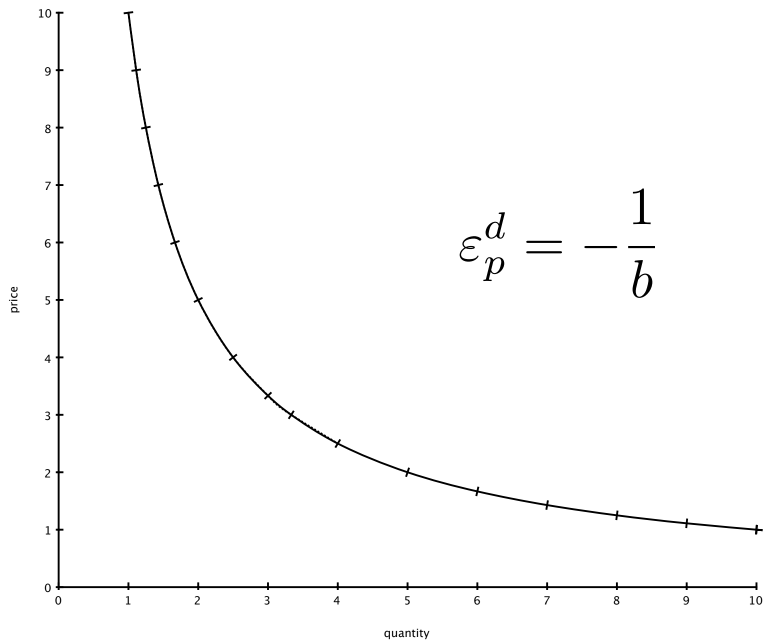


$$\varepsilon_p^d = -\frac{1}{b}\left(\frac{a - bQ^d}{Q^d}\right)$$

- Midpoint of linear demand curve is always the unit-elastic point

Iso-elastic demand curve

The inverse demand function is $p = aQ^{-b}$



$$\varepsilon_p^d = -\frac{1}{b}$$

NB: demand is linear in logs: $\ln p = \ln a - b \ln Q^d$

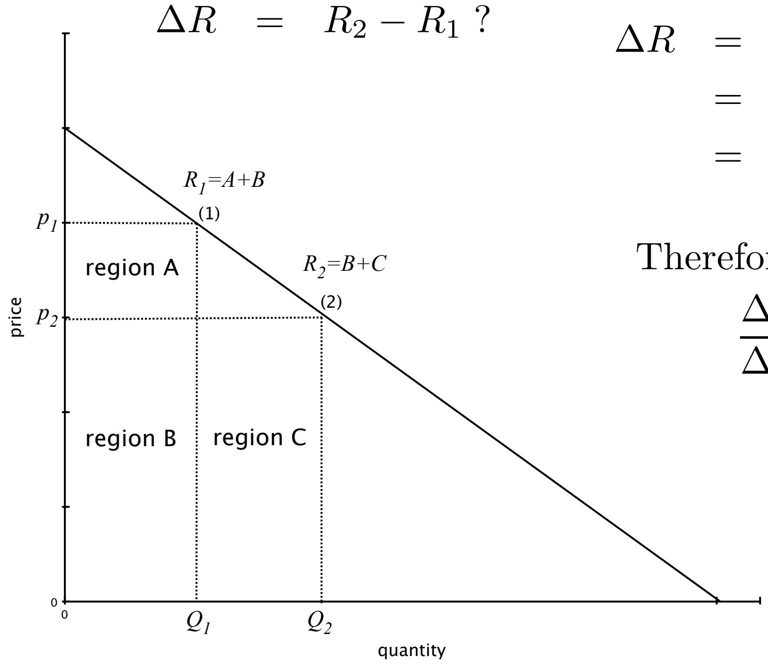
Revenue and demand elasticities

- Result:
  - **elastic** region:
    Q^d and market revenue, R, move together
    (alternatively, p and R move in opposition)
  - **inelastic** region:
    Q^d and market revenue, R, move in opposition
    (alternatively, p and R move together)
  - R is at its maximum where $\left|\varepsilon_p^d\right|$ = 1 (unit elastic)

- Note: market revenue = market expenditure

Logic for revenue-elasticity result

- $$\begin{aligned}\Delta p &= p_2 - p_1 < -0 \\ \Delta Q &= Q_2 - Q_1 > 0 \\ \Delta R &= R_2 - R_1 \ ?\end{aligned}$$



$$\begin{aligned}\Delta R &= B + C - (A+B) = C - A \\ &= p_2(Q_2 - Q_1) - Q_1(p_1 - p_2) \\ &= p_2 \Delta Q + Q_1 \Delta p\end{aligned}$$

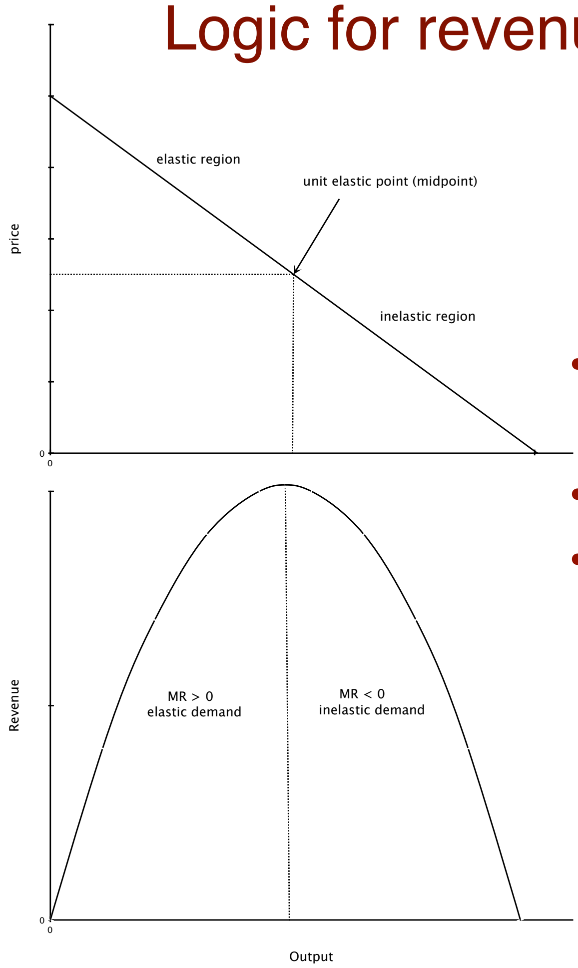
Therefore,

$$\begin{aligned}\frac{\Delta R}{\Delta Q} &= p_2 + Q_1\left(\frac{\Delta p}{\Delta Q}\right) \\ &= p_2\left(1 + \frac{\Delta p / p_2}{\Delta Q / Q_1}\right) \\ &\approx p_2\left(1 + \frac{1}{\varepsilon_p^d}\right)\end{aligned}$$

Logic for revenue-elasticity result

- To repeat: $$\frac{\Delta R}{\Delta Q} \approx p_2 \left(1 + \frac{1}{\varepsilon_p^d}\right)$$
- The left-hand side is marginal revenue (MR) for the market.
  - if demand is elastic, ε_p^d < -1, which implies $MR > 0$ ($MR > 0$ means that Q and R move together)
  - if demand is inelastic, $0 > \varepsilon_p^d > -1$, which implies $MR < 0$ ($MR < 0$ means that Q and R move in opposition)
  - and, of course, if $\varepsilon_p^d = -1$ (unit elastic demand), then $MR=0$, which must hold if revenue is at a maximum

Logic for revenue-elasticity result



- Top panel is demand curve; bottom panel is revenue curve;
- Horizontal axis (output) is aligned
- Useful diagram to remember!

Revenue-elasticity result with calculus

- The previous argument was an "approximate" proof of the result. It was only approximate because

$$\varepsilon_p^d(arc) \approx \frac{\Delta Q/Q_1}{\Delta p/p_2}$$

- The argument is exact if we use calculus to differentiate the *R(Q)=P(Q)Q*, and apply the point elasticity formula:

$$\begin{aligned} MR(Q) &= R'(Q) = P(Q) + P'(Q)Q \\ &= P(Q)\left(1 + P'(Q)\frac{Q}{P(Q)}\right) \\ &= P(Q)\left(1 + \left(\frac{\partial Q^d}{\partial p}\frac{P(Q)}{Q}\right)^{-1}\right) = P(Q)\left(1 + \frac{1}{\varepsilon_p^d}\right) \end{aligned}$$

where we have used the formula for the point elasticity.

Cross-price elasticities of demand

- A cross-price elasticity of demand measures

$$\varepsilon_{x^d,p_y} = \varepsilon_{p_y}^{x^d} = \frac{\%\Delta x^d}{\%\Delta p_y}$$

- Same idea as before with own-price elasticities
  - point cross-price elasticity of demand:

$$\varepsilon_{p_y}^{x^d} = \left(\frac{\partial x^d}{\partial p_y}\right)\left(\frac{p_y}{x^d}\right)$$

  - arc cross-price elasticity of demand:

$$\varepsilon_{p_y}^{x^d} = \left(\frac{x_1 - x_2}{p_{y,1} - p_{y,2}}\right)\left(\frac{p_{y,1} + p_{y,2}}{x_1 + x_2}\right)$$

Cross-price elasticities of demand

- Demand complements $\iff$ $\varepsilon_{p_y}^{x^d} < 0$
- Demand substitutes $\iff$ $\varepsilon_{p_y}^{x^d} > 0$

(See handout for real examples)

Income elasticities of demand

- An income elasticity of demand measures

$$\varepsilon_{Q^d,I} = \varepsilon_I^d = \frac{\%\Delta Q^d}{\%\Delta I}$$

- Same idea as before with own-price elasticities
  - point income elasticity of demand:

$$\varepsilon_I^d = \left(\frac{\partial Q^d}{\partial I}\right)\left(\frac{I}{Q^d}\right)$$

  - arc income elasticity of demand:

$$\varepsilon_I^d = \left(\frac{Q_1 - Q_2}{I_1 - I_2}\right)\left(\frac{I_1 + I_2}{Q_1 + Q_2}\right)$$

Income elasticities of demand

- Normal goods $\iff$ ε_I^d > 0
  - special case: luxury goods $\iff$ ε_I^d > 1
- Inferior goods $\iff$ ε_I^d < 0

(See handout for real examples)

Elasticity of supply

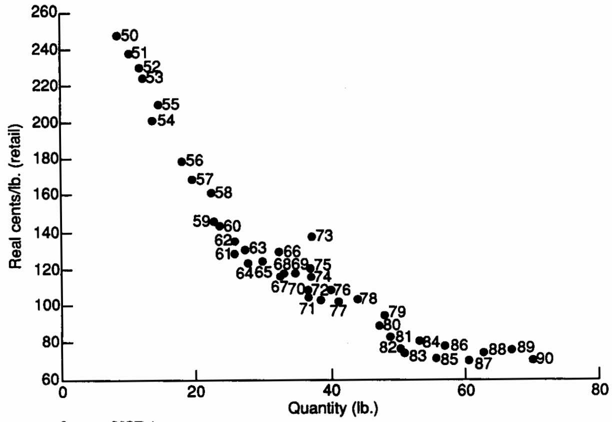
- Same idea as own-price elasticity of demand, but now with respect to the supply function (or curve):

$$\varepsilon_{Q^s,p} = \varepsilon_p^s = \frac{\%\Delta Q^s}{\%\Delta p}$$

- Supply elasticity is positive if supply slopes upward
- The point and arc variations of this elasticity are exactly what you expect (change "d" to "s" in demand formulae).
- Examples:
  - short-run supply elasticity for aluminum = 2.8
  - (See handout for further examples)

A remark about estimation of Supply and Demand curves

- If you have price and output data points over time, you may be able to estimate a demand or supply curve.
- Critically, you need some information about shifts in the curves to identify (or disentangle) demand from supply.
    - E.g., this data on Chicken sales does not necessarily yield a demand curve.



Source: USDA.

FIGURE 1 Broiler Price-Quantity Relationships, 1950–1990

Day 2:
Consumer choice: preferences & utility, constraints, optimal choice

Consumer choice - Basic Theory

- Topic 1: Consumer preferences (“ranking principle”)
  - indifference curves
  - utility functions
- Topic 2: Feasible choices (budget sets, etc.)
- Topic 3: Optimal choice (“choice principle”)
  - graphical approach (“no intersection” principle)
  - equating rates of return
  - arbitrage of relative price differences

Consumer preferences and the ranking principle

- We assume that our representative consumer is “rational” in two ways:
  - the **consumer can rank** any set of available bundles from best to worst, with the possibility that a rank may contain multiple bundles for which the consumer is indifferent
    - This is the **ranking principle**
  - the **consumer will choose** the best (according to this ranking) available bundle
    - This is the **optimal choice principle** (more on this later)
- The ranking principle is often deduced from a few axioms, namely that a consumer’s preferences are **complete** and **transitive**

Indifference curves

- We can graphically depict a consumer's preferences over various bundles of goods (at least in two dimensions) by constructing indifference curves
- An **indifference curve** is a collection of consumption bundles that generate the same level of "happiness" or "utility" or "economic satisfaction" for a consumer
  - I.e., any two points on the same indifference curve are equally ranked by the consumer
- In addition to the ranking principle, we assume the **more-is-better principle**: that is, take any bundle and increase some component by a small amount and the new bundle is strictly preferred by the consumer

An example of an indifference curve

- Suppose there are only two goods - soup and bread - and that each is valued
- We can ask a consumer to compare a bundle of
  - (0.3 liters of soup, 50 grams of bread) to (0.2 liters of soup, y grams of bread) and determine what value of y would make the consumer indifferent between the two
  - Because soup and bread are desirable and the second bundle has less soup, we know that it must have more bread to be indifferent. Thus, $y > 50$ grams. Suppose indifference happens at y=60 grams. Then

    (300ml soup, 50g bread) ~ (200ml soup,60g bread)

    and both these points lie on the same indifference curve.
- what amount of bread

Example of an indifference curve, cont.

- We have the following indifference from our consumer:

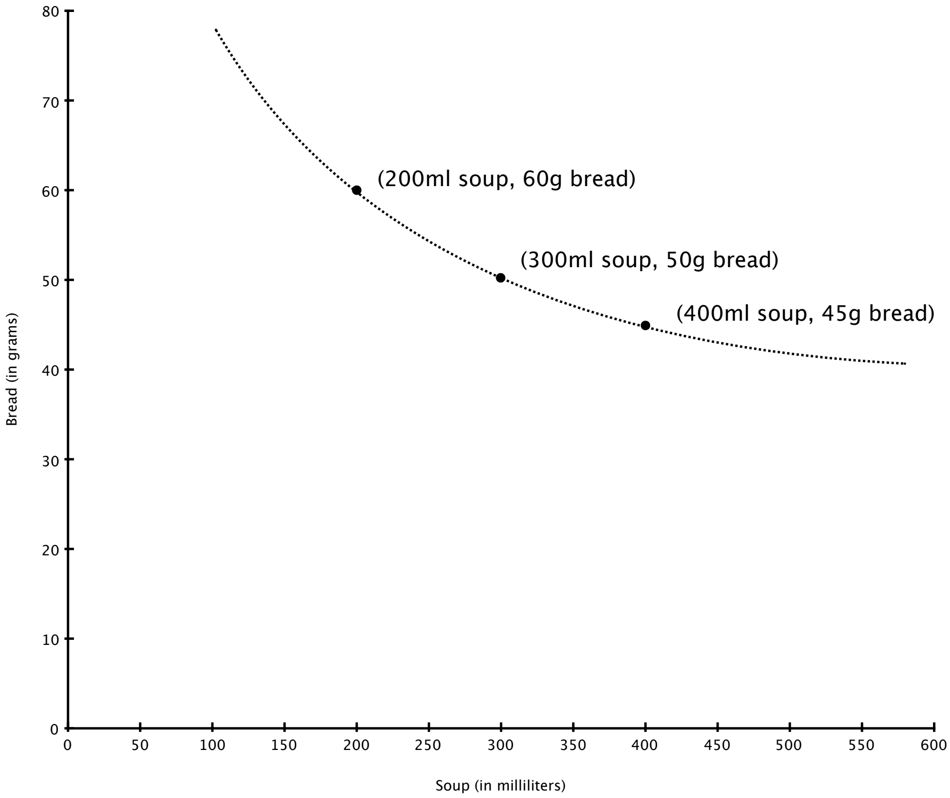
  (300ml soup, 50g bread) ~ (200ml soup,60g bread)

  Both these points lie on the same indifference curve.
- Note that increasing soup from 200ml to 300ml is worth forgoing 10g of bread to this consumer.
- We can repeat the analysis with 400ml of soup. Now, *y* must be less than 50g of bread. But with all that soup, bread may be very valuable relative to soup, so perhaps the indifference point is just *y*=45g of bread. That is, in exchange for an increase of 100ml of soup starting from 300ml, this consumer is only willing to forgo 5g of bread.

- Let's plot the three points.

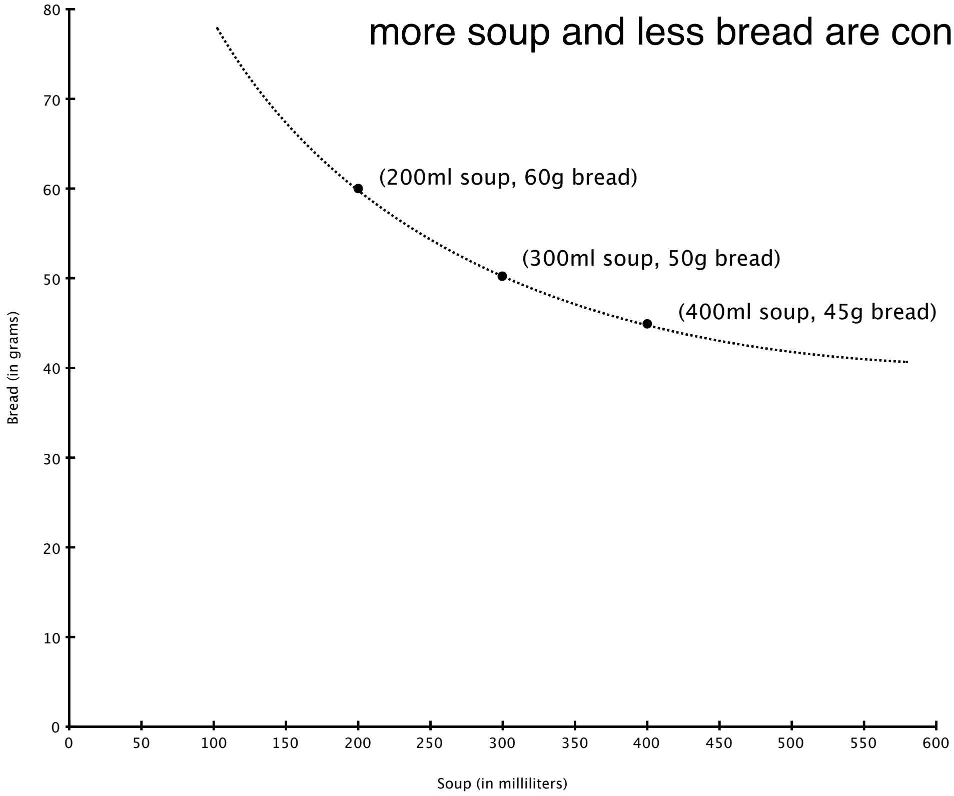
An indifference curve

- Dotted curve is the indifference curve



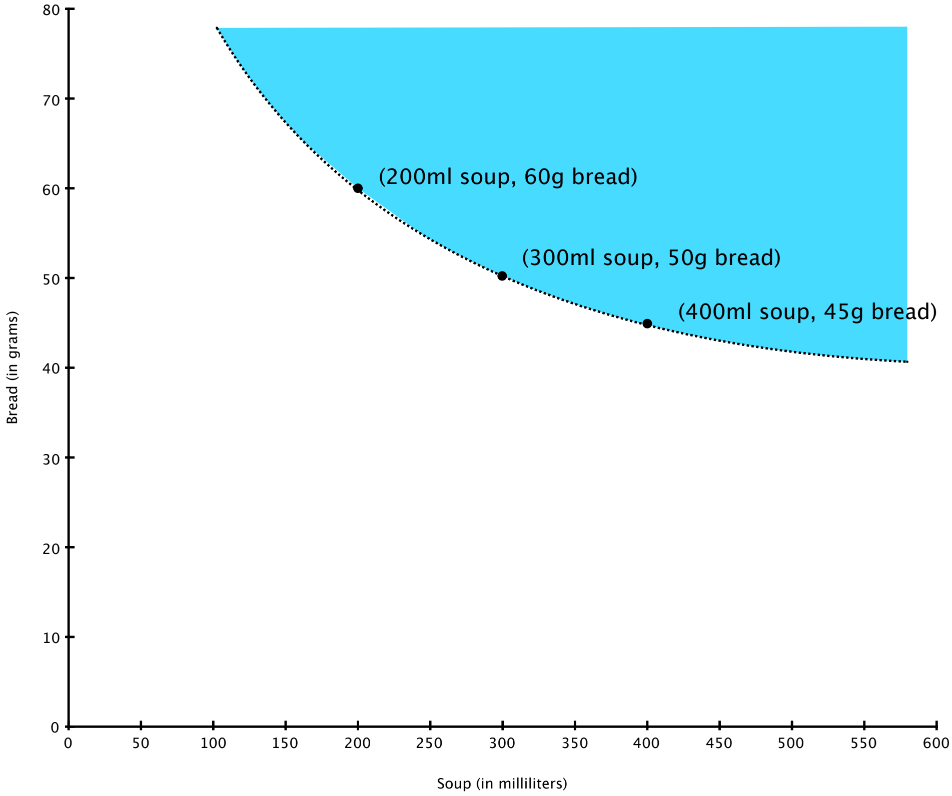
An indifference curve

- Curve becomes flatter as soup increases. This reflects the fact the consumer's relative value of soup to bread falls as more soup and less bread are consumed



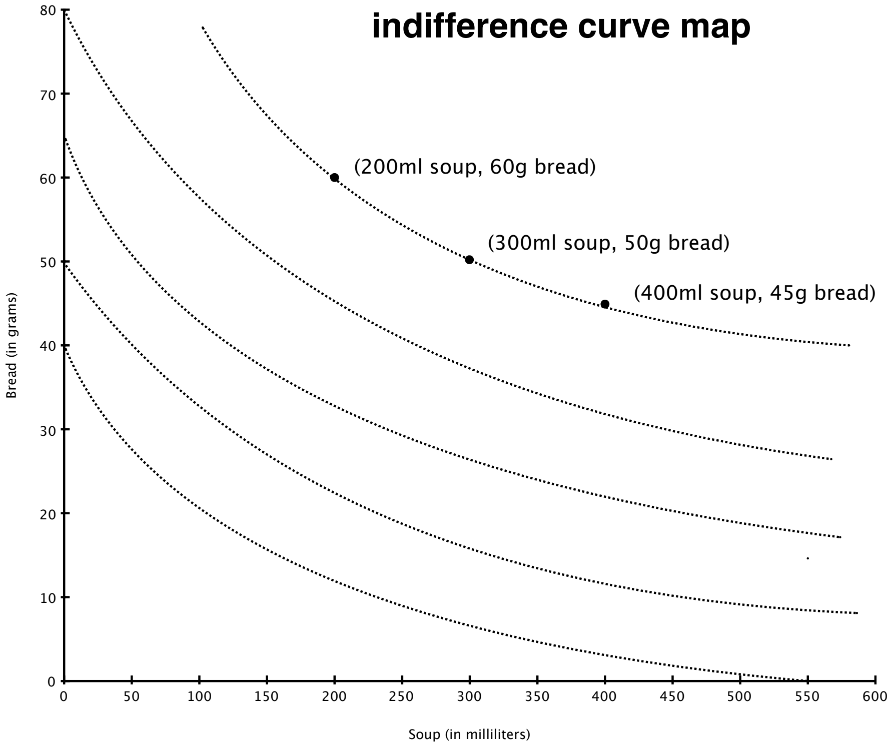
An indifference curve

- Note: all points to North-East of curve are strictly preferred



An indifference curve map

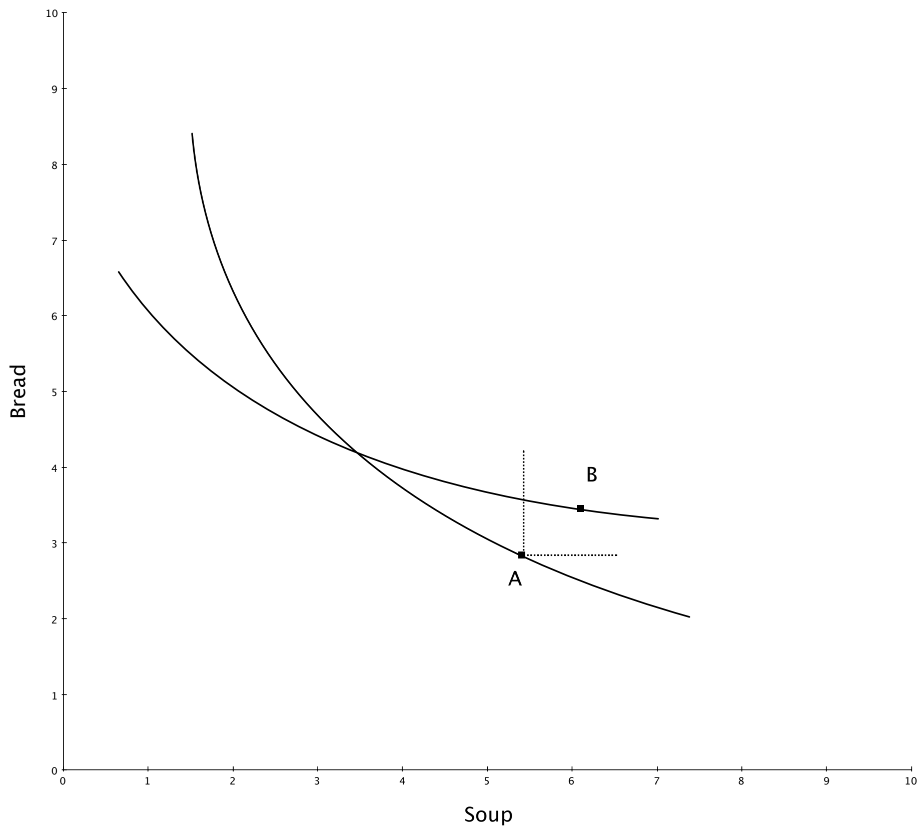
- We can repeat the analysis using different initial bundles to obtain additional indifference curve.
- A collection of indifference curves is called and **indifference curve map**



General properties

- In a world of only two goods, *x* and *y*, the ranking principle and the more-is-better principle imply that **indifference curves are downward sloping**. Why?
  - If an indifference curve were upward sloping, then for any two bundles on the curve, one bundle would have strictly more of both goods than the other. The more-is-better principle implies that the larger bundle is better, and thus cannot be ranked the same.
- **Indifference curves cannot cross**. Why?
  - If two curves intersect the same bundle, the ranking principle implies that all points on both curves are indifferent (i.e., there is only a single indifference "curve"). But any crossed curve has two bundles, one of which has more of every good; thus, they cannot be indifferent.

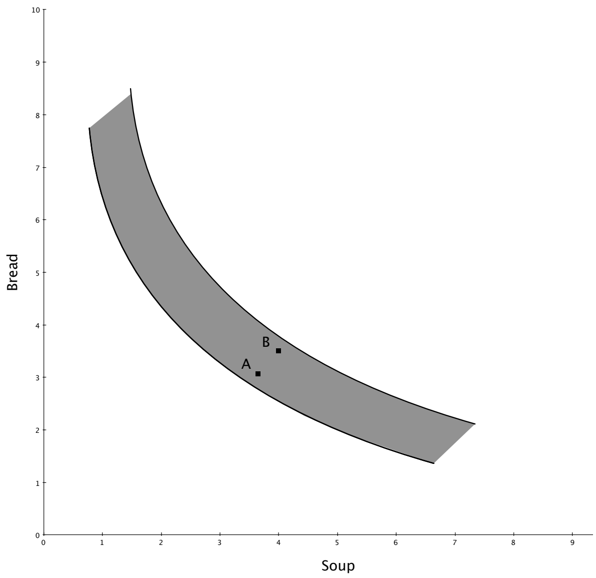
Figure 4.4: Indifference Curves Do Not Cross

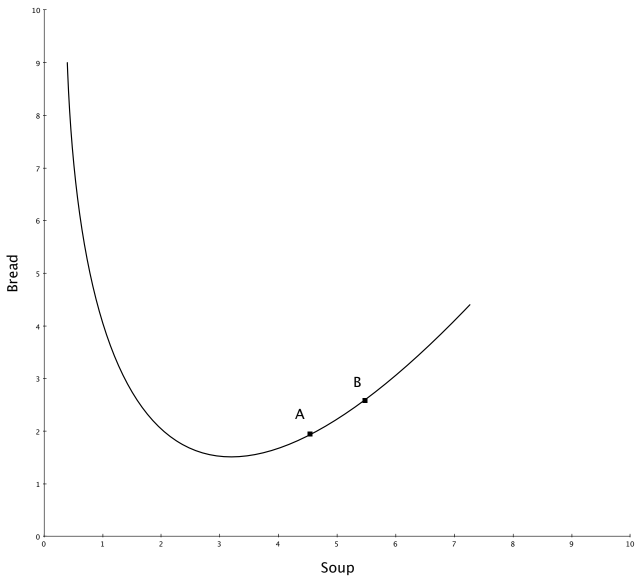


Properties, continued

- **Indifference curves are "thin"**. Why?
  - For the same reason that two indifference curves cannot intersect, each curve must be thin. Otherwise, a thick curve would contain two bundles for which one has strictly more of every good than the other. The points cannot be indifferent according to the more-is-better principle.
- If goods are continuously variable, then **indifference curves are continuous curves**.
  - That is, the all the points on an indifference curve can be drawn without lifting your pencil from the page. If goods must be consumed in discrete amounts (e.g., you can consumer 1 car or 2 cars, but not 1.3 cars), then indifference curves will be discontinuous (e.g., collection of separate points or curved segments).
  - We assume continuity in these lectures for convenience

Figure 4.2: Indifference Curves Ruled Out by the More-is-better Principle





Properties, continued

- **An indifference curve separates the space of all possible consumption bundles into three sets**
  - the **indifference set** which are the points on the indifference curve itself
  - the **better-than set** (the set of all bundles that are strictly better than those on the indifference curve)
  - the **worse-than set** (the set of all bundles that are strictly worse than those on the indifference curve)

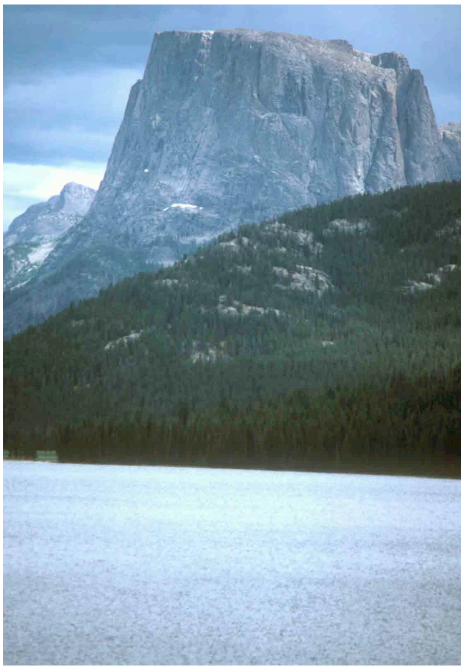
  We will return to these ideas when we investigate the consumer's optimal choice

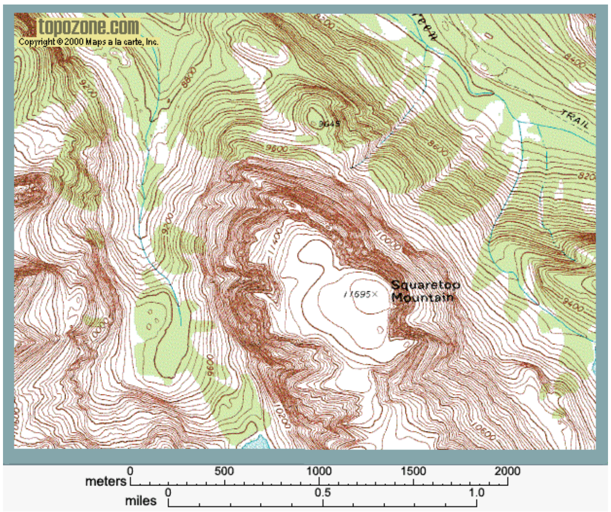
Properties, continued

- An **indifference curve map** presents a collection of indifference curves.
  - each curve represents a different level of utility or happiness
  - curves cannot cross (see above)
- Think of an indifference curve map like an **elevation** (or topographical) **map** for altitude. The indifference curve map gives us a sense of the terrain as it relates to happiness. Here, happiness is similar to altitude and the components in the consumption bundle relate to the geographical coordinates of the topographical map.
- NB: **more-is-better** implies "terrain" is always uphill

An example of an elevation map

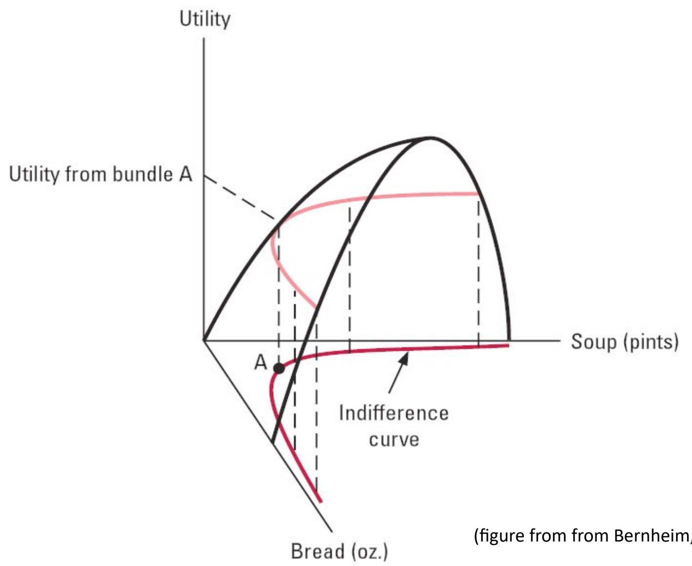
- Squaretop mountain





Properties, continued

- Indeed, if we had a way to measure happiness such as a utility function (more on this idea below), then the utility function provides us with a "hill" of sorts and the indifference curves are contour lines:



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Properties, continued

- Are indifference curves **convex** or **concave**?
  - First, we need to be clear about terms. For most of you, what you think of as a "convex" curve is what mathematicians and economists call a "concave" curve, and vice versa. Our previous example is a **convex** curve.
  - As a basic definition, we say a curve is **convex if it lies below all its chords** (a **chord** is a line segment that begins and ends on the curve). Thus, satellite dishes are convex and umbrellas when opened over your head are not. They are in fact concave. A curve is **concave if it lies above all its chords**.
  - Now think about what a **convex indifference curve** represents. They get flatter as you move down the curve (less *y*) and to the right (more *x*). Thus, a given reduction in *y* must be compensated by an increasing amount of *x*.

Properties, continued

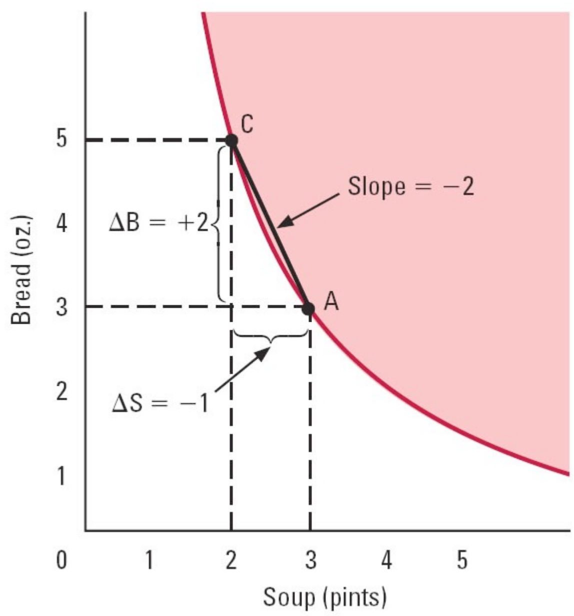
- Because a given reduction in *y* must be compensated by an increasing amount of *x*, the relative value of *x* to *y* is falling for the consumer.
- The slope of the indifference curve at a given bundle represents this relative value of *x* to *y*. We call this the **marginal rate of substitution of *x* with *y***. Formally,

$$MRS_{x,y} = -\left.\frac{\Delta y}{\Delta x}\right|_{\text{along an IC}}$$

- Hence, convex indifference curves have decreasing marginal rates of substitution
- **Economically**, this implies that the more one has of one good relative to the other, the less value it has relative to the other; preferences for diverse consumption are convex.
- What do concave preferences (increasing MRS) imply?

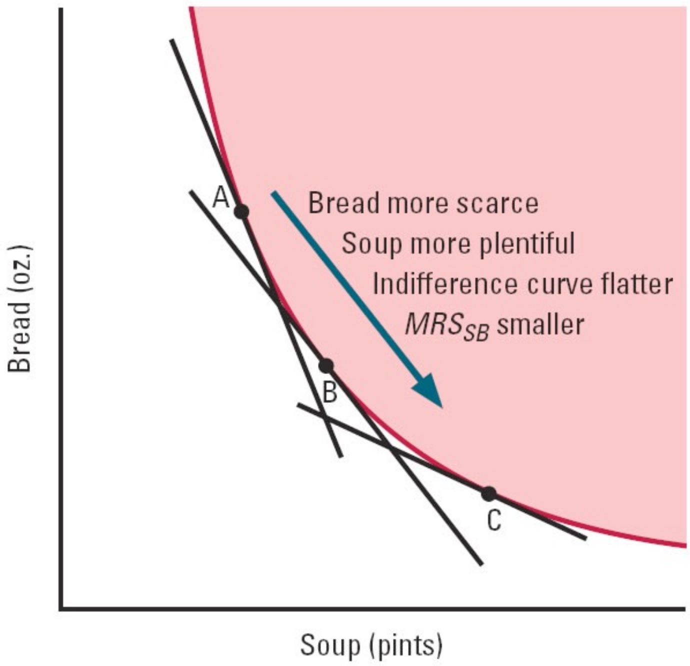
Figure 4.8: Rates of Substitution

- Look at move from bundle A to C
- Consumer loses 1 soup; gains 2 bread
- Willing to substitute for soup with bread at 2 ounces per pint



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Figure 4.11: MRS along an Indifference Curve



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Polar cases for convex indifference curves

- **Perfect substitutes:** two goods are perfect substitutes for a consumer if their marginal rate of substitution is constant and equal across all consumption bundles. Thus, the two goods substitute at a non-varying rate.
  - e.g., regular strength versus extra strength pain medicine
  - e.g., gasoline/petrol versus ethanol for some automobiles
- **Perfect complements:** two goods are perfect complements for a consumer if they generate value only when combined in a fixed (but not necessarily a 1-to-1) proportion.
  - e.g., left shoes and right shoes
  - e.g., dining room chairs and tables

Figure 4.12: Perfect Substitutes

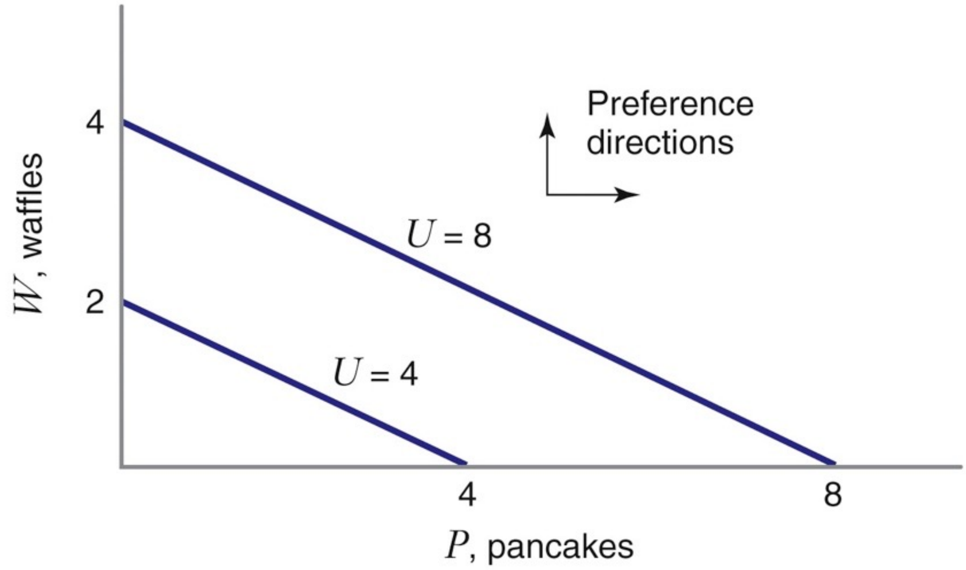


Figure 3.13
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Figure 4.13: Perfect Complements

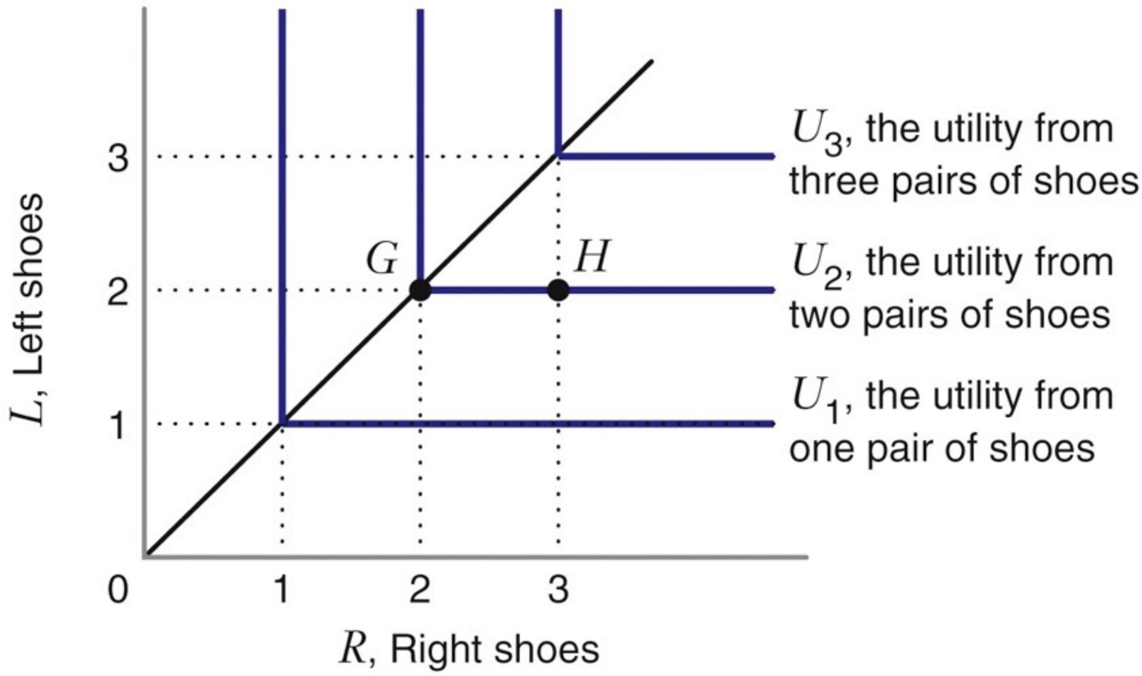

Figure 3.14
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Marginal rate of substitution

- The marginal rate of substitution can be thought of as a personal exchange rate of one good for the other.
- Because optimal choice involves tradeoffs, we will use the notion of *MRS* to address the marginal costs and benefits of moving from one choice to another
- It is useful to have an alternative construction of *MRS* that is based on utility rather than indifference curves. We turn to that construction now.

Characterizing *MRS* with a utility function

- A **utility function** is a function that maps from the coordinates of a consumption bundle to a numerical measure of utility or happiness.
- Thus, in a two good world, we have a function $u(x,y)$
  - e.g., $u(x,y) = xy + 2x$
- If we have many goods, say $(x_1, x_2, \ldots, x_n)$, then we write $u(x_1, x_2, \ldots, x_n)$ as the value of utility.
- A utility function **represents preferences** if for any two bundles (x,y) and (x',y') we have:

$$(x,y) \text{ is preferred to } (x',y') \iff u(x,y) > u(x',y')$$

- The **ranking principle** implies such a function exists.

Utility functions, continued

- Note that if *u(x,y)* represents preferences, then so does *f(u(x,y))* where *f(u)* is any strictly increasing function. Thus, a utility function is an **ordinal** idea. That is, it merely presents the consumer's ranking in numerical terms. Changing the measurement units for happiness doesn't change the ranking.
  - E.g., if $u(x,y) = xy + 2x$ represents preferences, then so does $u(x,y) = 2x\,y + 4x$ as well as $u(x,y)= x^2(y^2+ 4y + 4)$ (the square of the first function, assuming $xy+2x>0$).
- In the past, the **utilitarians** (most notably, Jeremy Bentham) insisted that **cardinal** utility functions could be used to evaluate social welfare. I.e., one could construct a function measuring happiness and apply it to different individuals. Such **interpersonal comparisons of utility** are generally frowned upon by modern economists (and philosophers).

Utility functions - why do we care?

- We are going to use a utility function representation (anyone will do) to characterize *MRS*.
- To this end, we need to define **marginal utility**:
  - the marginal utility of *x* is the numerical amount that utility increased from a small increase in *x*. If Δx is the increment in *x* and Δu is the change in utility, then the marginal utility of *x* is
$$MU_x = \frac{\Delta u}{\Delta x}$$
  - In calculus terms, we write
$$MU_x = \frac{\partial u(x,y)}{\partial x}$$
  - Similarly, we can define the **marginal utility of *y***:

$$MU_y = \frac{\Delta u}{\Delta y} \quad \text{or} \quad MU_y = \frac{\partial u(x,y)}{\partial y}$$

Utility functions - why do we care?

- Given *u(x,y)* represents preferences, the value of this function must be constant *along any indifference curve*.
- Now consider the point (*x,y*) on an indifference curve. Let *Δx* and *Δy* be movements from this point that remain on the indifference curve. That is, (*x+Δx,y+Δy*) is on the same indifference curve as (*x,y*).
- For small *Δx* and *Δy*, it follows that the net marginal change in utility from *Δx* and *Δy* is

$$\Delta u = MU_x \Delta x + MU_y \Delta y$$

- But if we stay on the same indifference curve, then *Δu=0*:

$$0 = MU_x \Delta x + MU_y \Delta y$$

- Algebra reveals $-\frac{\Delta y}{\Delta x} = \frac{MU_x}{MU_y}$, which is also the $MRS_{x,y}$

Utility functions - why do we care?

- To summarize, if *u(x,y)* represents a consumer's preferences, then the consumer's $MRS_{x,y}$ at *(x,y)* is characterized as either
  - the slope of the indifference curve at (*x,y*)

$$MRS_{x,y} = -\frac{\Delta y}{\Delta x}$$

  - the ratio of the marginal utilities of consumption:

$$MRS_{x,y} = \frac{MU_x}{MU_y}$$

- If you are comfortable with calculus, you'll notice that the chain rule implies any transformation of *u(x,y)* will leave $MRS_{x,y}$ unchanged. $MRS_{x,y}$ is derived from a consumer's ranking, not any specific cardinal utility function.
- We can think of $MRS_{x,y}$ therefore as a ratio of marginal valuations; higher ratio means *x* is (marginally) more valuable.

Utility functions and *MRS* -

- Suppose that there are more than two goods in the economy. Fortunately, the previous derivation of *MRS* for *x* and *y* easily extends to an arbitrary number of goods.
- Choose any two goods. Then the marginal rate of substitution for good x_i with good x_j is simply

$$MRS_{x_i,x_j} = \frac{MU_{x_i}}{MU_{x_j}}$$

- Below, we will see that this affords us a nice generalization of a consumer's optimal choice for an arbitrary number of goods.

Feasible choices: budget constraints, etc.

- Previously, we have characterized consumer preferences both graphically (indifference curves) and mathematically (using marginal utility ratios)
    - we can characterize a consumer's choice between any two bundles, even for cases in which the consumer cannot afford either (e.g., we characterize whether a poor person prefers a Mercedes or a BMW)
- Now we will characterize a consumer's set of available choices both graphically (budget lines) and mathematically (using algebra and price ratios).
    - in both cases, we will describe the available consumption set even putting aside issues of preferences (e.g., we describe what combinations of beef and pork a vegetarian can afford to buy).
- In both cases, preferences and constraints will be illustrated in the same graphical space, with each axis measuring a particular good. (E.g., soup and bread on the axes.)

Overview of constraints

- The set of available consumption bundles (also known as the **constraint set**) are those that the consumer can obtain while satisfying budget constraints (as well as possible constraints on time, legal restrictions, etc.).
- In general, the constraint set will be a strict subset of the universe of possible consumption bundles.
- In a 2-good world, the constraint set will be a strict subset of the positive quadrant.
  - When the only constraint is budgetary, the constraint set is called a **budget set**, and it is defined by a right triangle.
  - We will start with this simple geometric case.

A consumer's budget constraint

- Suppose that a consumer has I dollars to spend on two different goods, x and y. The price of x is p_x dollars per unit and the price of y is p_y dollars per unit. The consumer's total expenditures for a purchase of (x,y) is therefore

$$p_x x + p_y y.$$

- A consumer's **budget constraint** is the requirement

$$p_x x + p_y y \leq I.$$

- The **budget line** is is the set of all (x,y) that exactly spends the consumer's income, I:

$$p_x x + p_y y = I.$$

- The budget line is a boundary of the budget set.

A consumer's budget constraint

- We now want to depict the budget constraint set graphically.
- The budget line

$$p_x x + p_y y = I$$

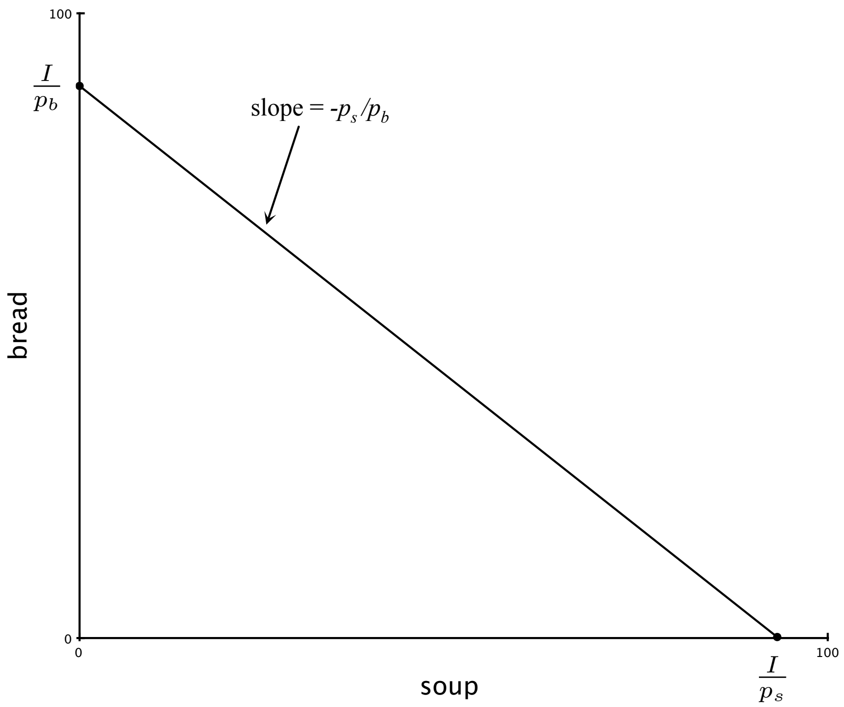
can be solved for *y* to obtain instead

$$y = \frac{I}{p_y} - \frac{p_x}{p_y} x$$

- Thus the vertical intercept is at $\frac{I}{p_y}$. This is the amount of *y* that can be afforded if the entire income is spent only on *y* and not *x*. The slope of the budget line is $-\frac{p_x}{p_y}$.

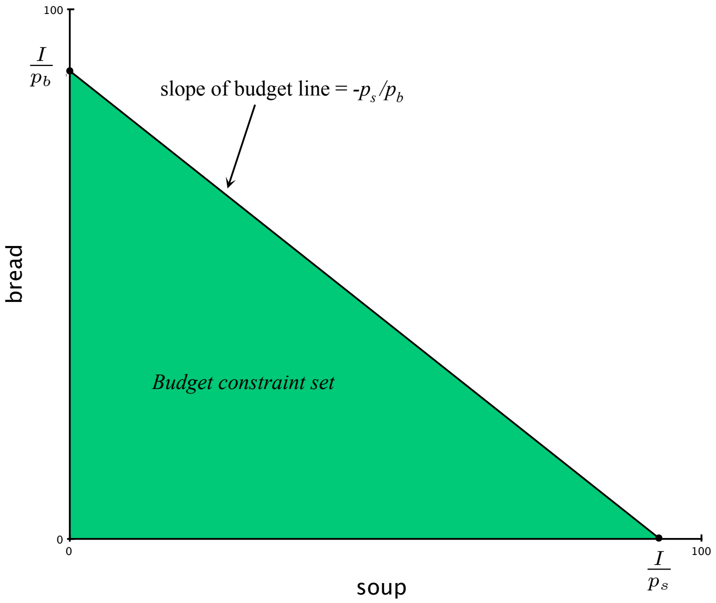
Graph of budget line

- Let's return to our setting of soup, *s*, and bread, *b*.
- The budget line is



Budget constraint

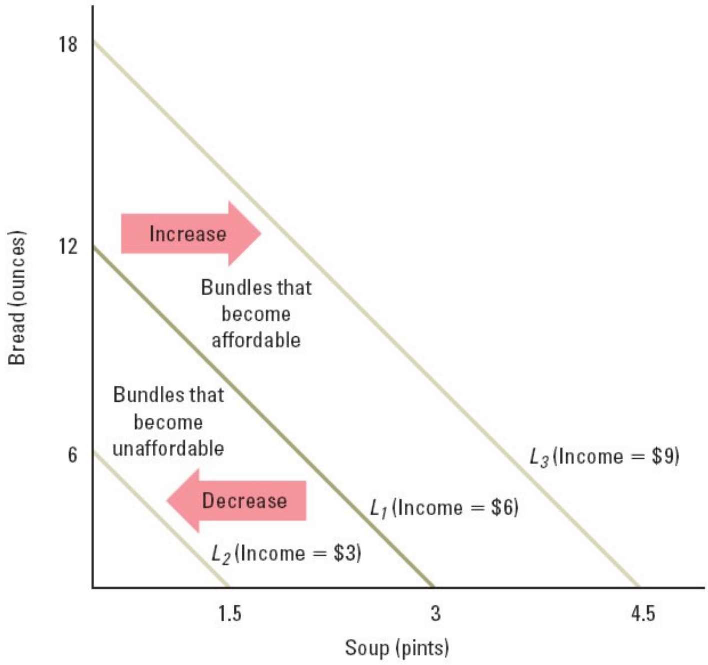
- The budget constraint set is the shaded region contained in the triangle defined by the budget line.
- The budget line is the part of the budget constraint set that exactly spends all income.



Impact of income and price changes on budget lines

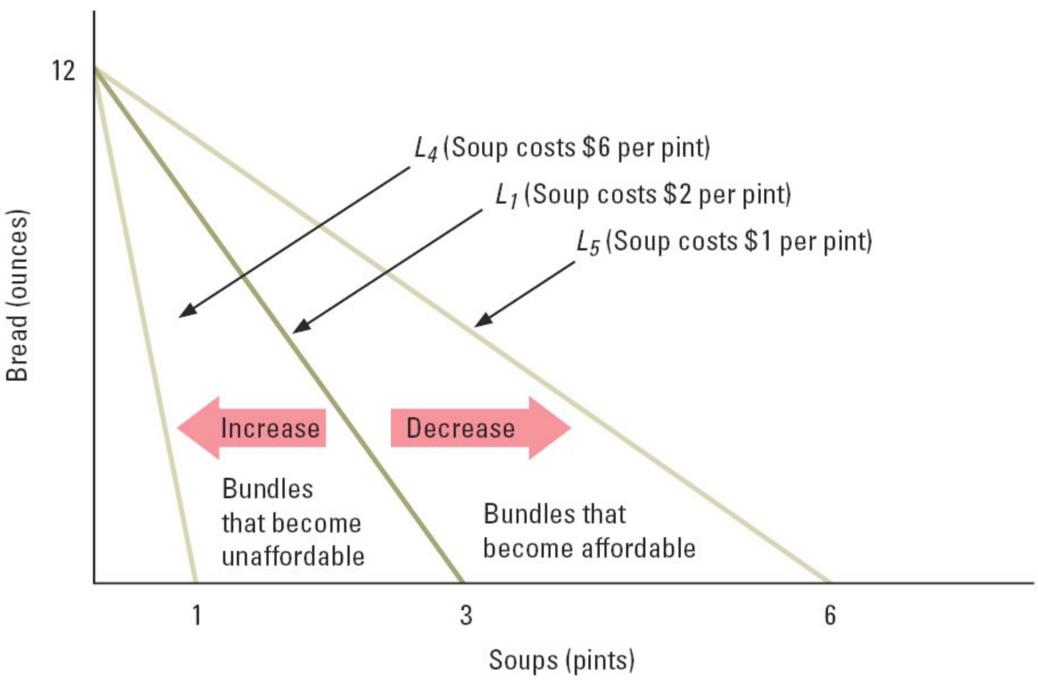
- A variation in income changes the intercepts but not the slope of the budget line. Hence, income changes shift the budget line in parallel fashion
  - income increase shifts budget line outward; income decrease shifts budget line inwards
- Change in one price changes the intercept of that good without affecting the intercept of the other. The budget line pivots on the intercept of the other good and the slope changes accordingly.
  - e.g., if the price of *x* decreases, then the budget line pivots on the y intercept and swings outward along the *x* axis. The budget set grows larger as more *x* can be bought.
  - Inward for a price increase

Effects of Changes in Income on the Budget Line



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Effects of a Change in the Price of Soup



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Budget Line properties

- Budget line is the boundary of the budget set that represents bundles that exhaust all income
- x-intercept is I/p_x and the y-intercept is I/p_y
  - points represent most that can be afforded of any good
- Budget line slope is $-p_x/p_y$ (a ratio of prices)
  - this is an exchange ratio like $MRS_{x,y}$
  - measured in units of y per x (just like price ratio)
- Income changes generate parallel shifts
- Price changes rotate the line, pivoting through the intercept point of the good with unchanged price

Generalizations of budget lines

- More than 2 goods
- Multiple (overlapping) constraints
  - time constraints
  - legal constraints
- Nonlinear budget lines (e.g., quantity discounts)
- In all cases, any point on the boundary of the constraint set has a slope defined by a ratio of "prices" (an exchange rate)

Generalization: More than two goods

- With 3 goods, the budget line becomes a budget plane. With more than 3 goods, the plane is referred to as a **hyperplane**, but it is difficult to visualize.
- Algebraically, we have the constraint set

$$\sum_{i=1}^{n} p_i x_i \leq I$$

- The associated budget hyperplane (i.e., the set of bundles that exactly spends all income) is

$$\sum_{i=1}^{n} p_i x_i = I$$

- For any pair of goods there is a price ratio as before.

More realistic set of goods

Consumer Expenditures, US, 2001

Yearly After Tax Income: $42,362

Yearly Total Expenditures: $40,900

Allocation of Spending

Food	$5,904
Housing	$12,248
Transportation	$8,672
Health care	$2,239
Entertainment	$1,958

Note: savings here is $42,362-40,900=$1462/year. This can be thought of as expenditures on “future consumption”. Thus, there are six relevant goods categories.

Generalization: Multiple constraints

- Time and money constraints
  - suppose that *x* is "meals at restaurants / month" and *y* is number of evenings spent at "pubs/month"
  - The price of eating out at a restaurant is $p_x = £20$/meal and the price of going to a pub is $p_y = £10$/evening.
  - $I=£200$ is devoted to these activities each month; thus,

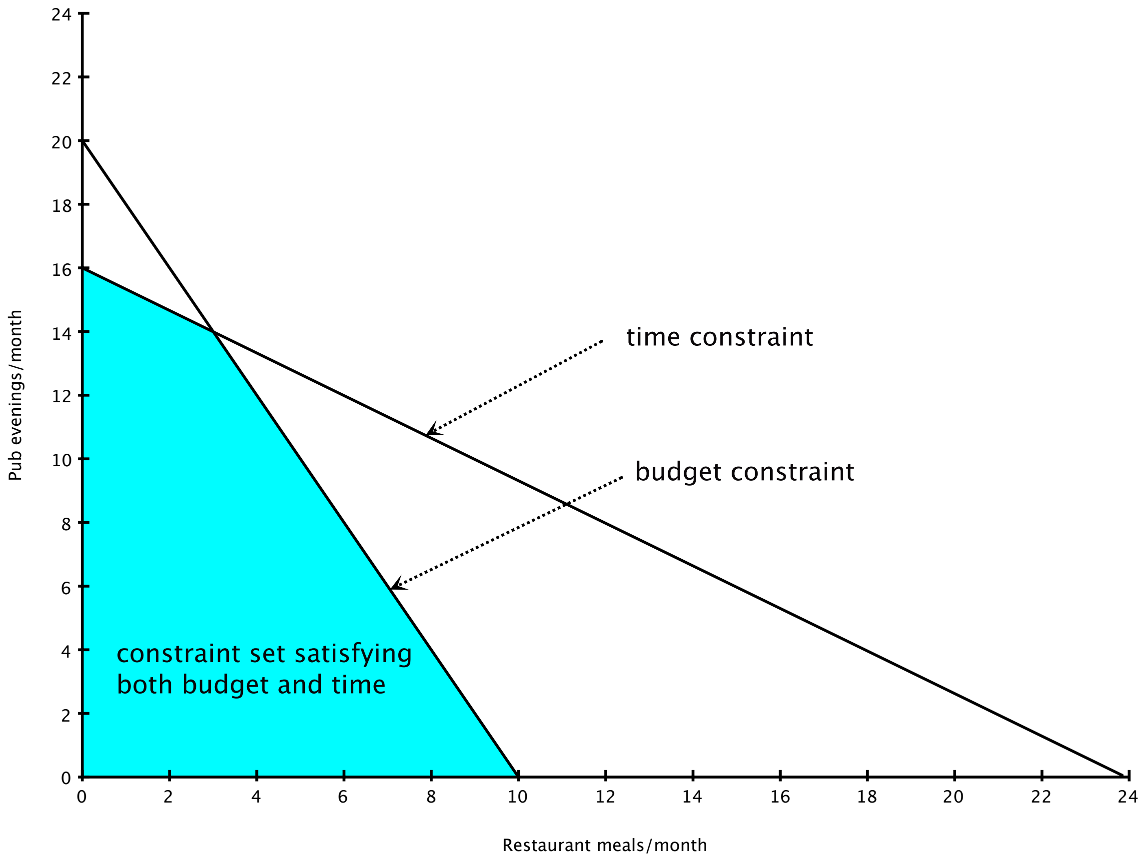
$$200 \geq 20x + 10y$$

  - Suppose each activity has a "price" in time as well. Restaurant dining uses 2 hours of your time, $t_x = 2$h/meal; going to a pub uses 3 hours, so $t_y = 3$h/evening. If only 48 hours are available each month, $T=48$, we have as a time constraint

$$48 \geq 2x + 3y$$

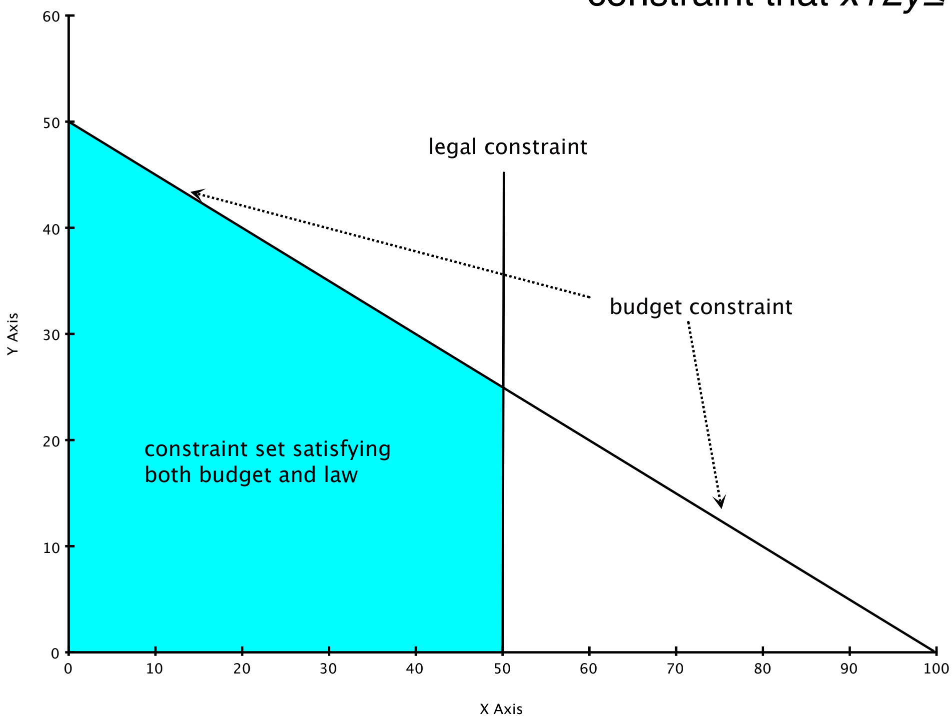
  - Bundles in the constraint set satisfy ***both*** constraints.

Generalization: Multiple constraints



Generalization: Multiple constraints

- Legal restrictions on consumption
  - e.g., suppose that legally $x \leq 50$ in addition to the budget constraint that $x+2y \leq 100$.



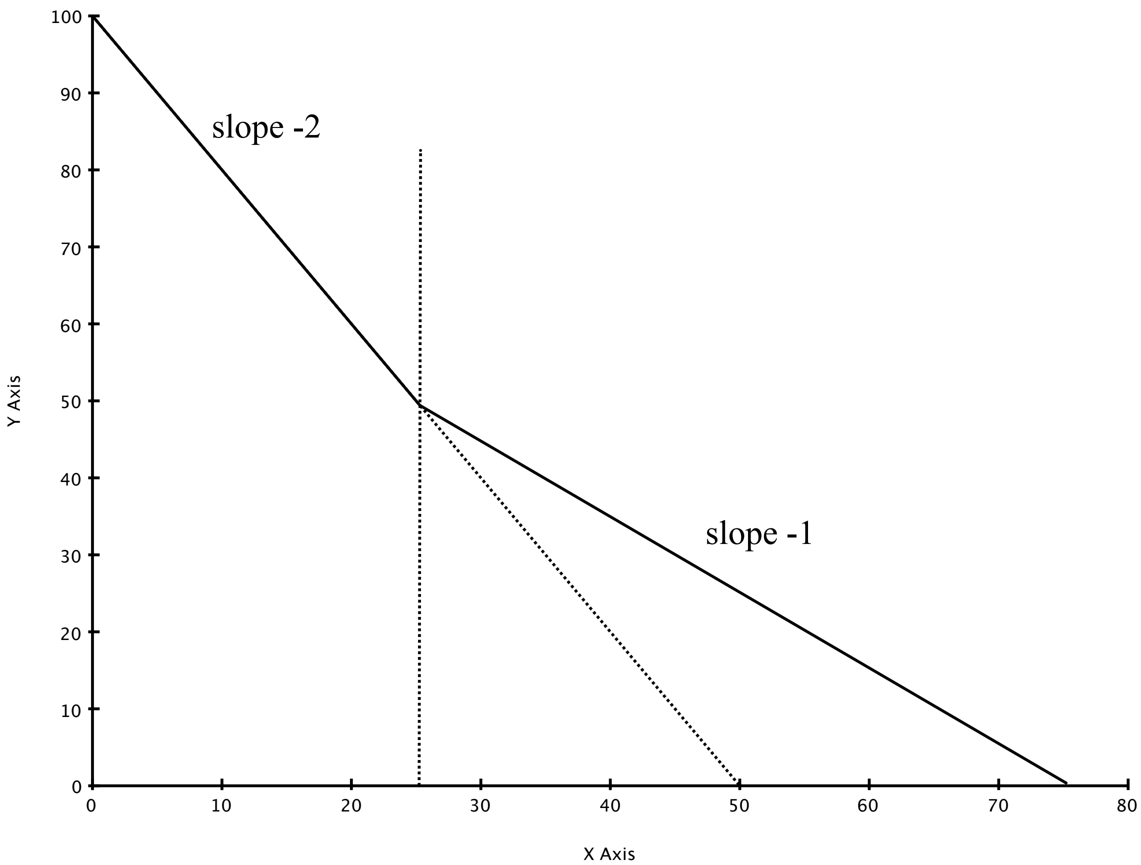
General character of any constraint set

- The boundary of the constraint set will consist of points that exactly satisfy at least one constraint
  - in this case, we say that this constraint is **binding** or **active**
- The slope of a binding constraint will be a ratio of "prices" -- an exchange rate of goods with respect to the active constraint
- At any boundary point which simultaneously satisfies more than one constraint there will generally be different slopes in different directions (e.g., a kink on the frontier of the budget set)
- **Bottom line:** the tradeoff on a constraint set is given by some implicit or explicit exchange rate.

Generalization: Nonlinear budget lines

- Sometimes the unit price one pays depends upon how much is purchased. Consider the following example:
  - Suppose that $I=100$, $p_y = 1$, but p_x has two possible values. For the first 25 units of x, $p_x = 2$, but for each unit greater than 25, the marginal price is reduced to $p_x = 1$. Thus, 26 units of x costs $2(25)+1=51$ dollars and ***not*** 26 dollars. (Not all quantity discounts look like this. Other possibilities?)
  - It is helpful to draw a few points to start. If the consumer spends all his money on y, then $y=100$. If the consumer spends all of his money on x, then $x=75$. Yes? So we know that these are the axis intercepts.
  - At $x=25$, the consumer can afford $y=50$. Another point.
  - If the consumer chooses an $x \leq 25$, then the budget line has a slope of -2. For $x>25$, the slope is -1. Why?

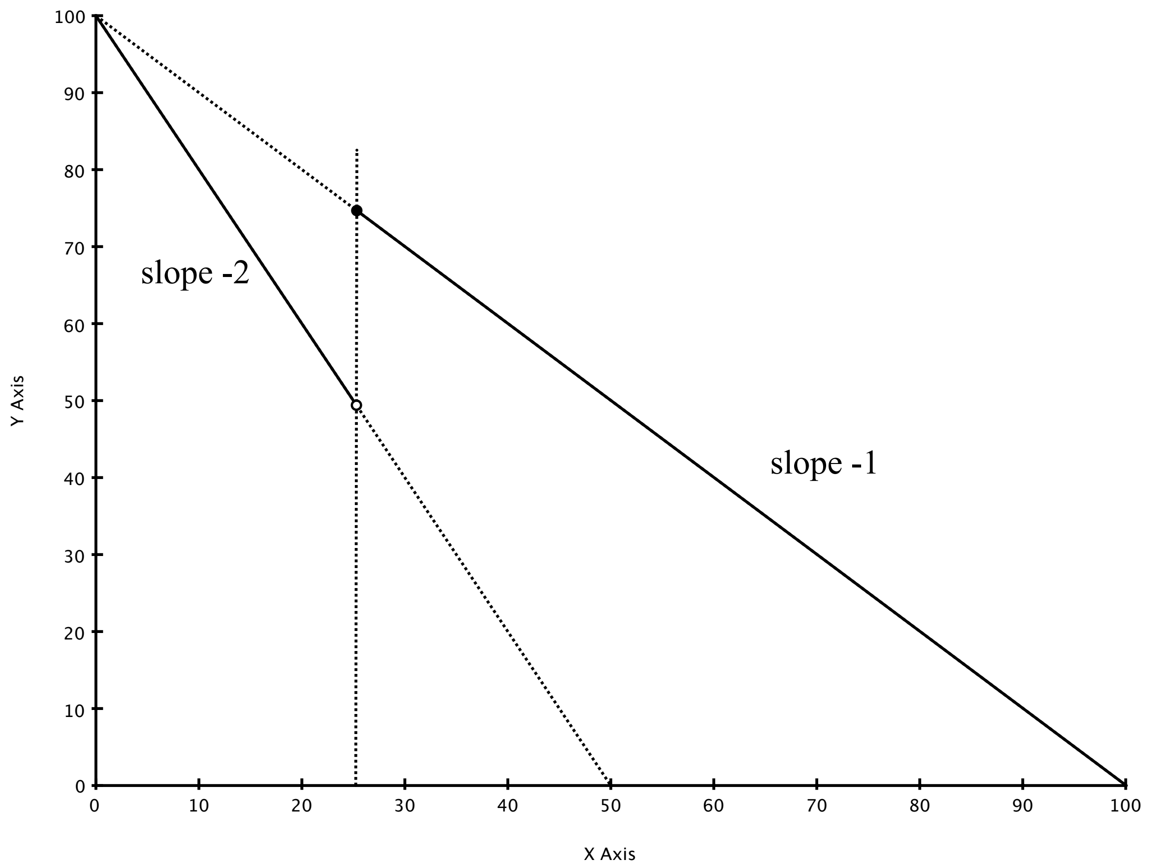
Generalization: Nonlinear budget lines



Variation on quantity discounts

- Sometimes the discount is given for all units once a certain purchase size is reached. In this case, suppose that the price for all *x* drops from 2 to 1 once purchases at least 25 units
  - hence, 26 units of *x* costs 26 dollars now, not 51.

Generalization: Nonlinear budget lines



Optimal Choice

- So far, we have established …
  - the **ranking principle** allows us to characterize a consumer's preferences with either indifference curves or with a utility function that generates the same $MRS_{x,y}$.
  - the constraint set (available bundles) is geometrically defined by a boundary (e.g., budget line) with slope given by a ratio of "prices"
- Now we apply the **choice principle**: a consumer chooses the best bundle (given his ranking) from the constraint set
- Putting this all together, there are two characteristics of the consumer's optimal choice
  - An optimal **choice must lie on the boundary** of the constraint set (because more-is-better assumption)
  - An optimal choice on the boundary satisfies the **no-overlap** condition (also known as "tangency condition")

  - A bundle on the highest indifference curve that touches the budget line

Rule 1: Optimal choice lies on boundary

- Because more is better and all possible uses of income are included as goods, it is never optimal to spend less than *I*.
  - **Note:** the key here is that everything you can spend money on is defined as some good. Thus, saving for the future is the good "future consumption" or "self-insurance against bad times." Similarly, leaving money to your children is a "bequest" good. In this sense, "spending all of your money" is more appropriately thought of as "allocating all of your income to some purpose."
- This is a requirement that resources are not thrown away or wasted. **All resources are fully allocated.** We will refer to Rule 1 as the **no-waste condition** in some instances. Gary Becker takes this idea further (he called it the **scarcity principle**) and derives some interesting implications. (See Becker, *Economic Theory*, for more on this.)

Implications of full allocation (no waste)

- Let the consumer's optimal demands be given by consumer demand functions, one for each good, *i=1,…,n*

$$x_i = D_i(p_1, \ldots, p_n, I)$$

Because income is scarce, it is exhausted across these *n* goods and so

$$\sum_{i=1}^{n} p_i D_i(p_1, \ldots, p_n, I) = I$$

- Because this relationship must be true *for all* prices and income, it is an identity and we can differentiate it with respect to any price or income and the result is still true!

No-waste implications, continued

- Implications for income elasticities of demand:
  - First, differentiate with respect to *I.*

$$\sum_{i=1}^{n} p_i \frac{\partial D_i(\cdot)}{\partial I} = 1$$

  - Multiply each term by $\dfrac{D_i}{I}\dfrac{I}{D_i}$ and rearrange a bit.

$$\sum_{i=1}^{n} \frac{p_i D_i}{I} \frac{\partial D_i(\cdot)}{\partial I} \frac{I}{D_i} = 1$$

  - Define the budget share of good *i* as $s_i = \dfrac{p_i D_i}{I}$. Using the formula for income elasticities, we have

$$\sum_{i=1}^{n} s_i \varepsilon_I^{d,i} = 1$$

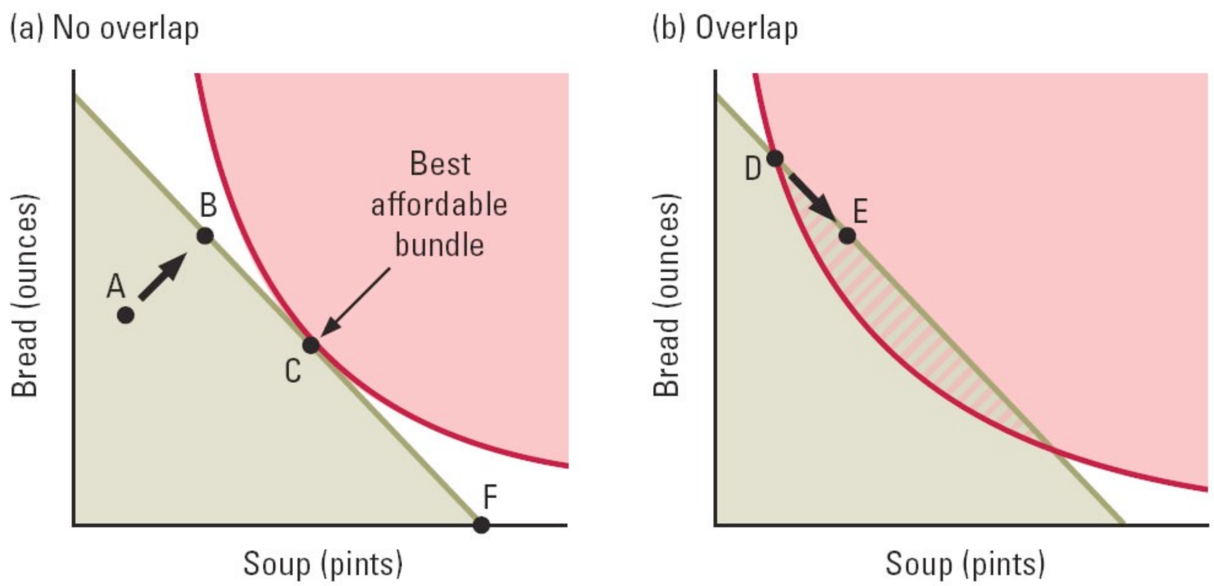
  - **The weighted average of income elasticities equals 1**.

Rule 2: No-overlap / “tangency” condition

- Rule 1 establishes that the optimal choice lies on the boundary of the constraint set. Rule 2 indicates which point on the boundary is best.
- **No-overlap condition: the indifference curve through the optimal boundary point is such that its preferred points do not overlap with the constraint set.**
  - If this were not the case, then there would be a preferred point that is in the constraint set which would violate the principle of optimal choice.
  - If the boundary of the constraint set and the indifference curves are both smooth at the optimal consumption bundle, then the no-gap principle requires that both curves are tangent. Hence, sometimes called the **tangency condition**:

$$MRS_{x,y} = \frac{p_x}{p_y}$$

Choosing Among Affordable Bundles



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Rule 2: No-overlap / "tangency" condition

- This condition is so fundamental to economics that we will examine it from a few angles.
- Suppose that the indifference curves are smooth, the constraint set is defined by a single budget line, and that the optimal bundle is an **interior point** (positive amounts of both goods, as opposed to a **corner point**).
- Our no-overlap condition is simply the tangency condition:

$$MRS_{x,y} = \frac{p_x}{p_y}$$

- **Graphically:** choose any point on the constraint boundary and look at the $MRS_{x,y}$ at that point. If

$$MRS_{x,y} > \frac{p_x}{p_y}$$

then the indifference curve is steeper than the budget line. A movement down the budget line (more *x*, less *y*) is preferred.

Rule 2: No-overlap / "tangency" condition

- **Graphically:** if instead $$MRS_{x,y} < \frac{p_x}{p_y}$$

then the indifference curve is flatter than the budget line. A movement up the budget line (to less *x*, more *y*) is preferred.

- In sum, a comparison of *MRS* and the price ratio indicates in which direction an improvement can be made.
- **Capital budgeting; rates of return.** We know that we can characterize *MRS* as the ratio of marginal utilities from some representative utility function. Hence, the tangency condition becomes

$$\frac{MU_x}{MU_y} = \frac{p_x}{p_y} \iff \frac{MU_x}{p_x} = \frac{MU_y}{p_y}$$

  - Notice that the right hand expression states that the **marginal utility per dollar spent is equal across goods**

Rule 2: No-overlap / "tangency" condition

- We can reinterpret the graphical rules using these rates of marginal utility per dollar:
  - If $\frac{MU_x}{p_x} > \frac{MU_y}{p_y}$, then buy **more** *x* and **less** *y*;
  - If $\frac{MU_x}{p_x} < \frac{MU_y}{p_y}$, then buy **less** *x* and **more** *y*;
- **Arbitrage.** As a general principle in economics, if there exists any pair of goods such that the relative price ratio in market A is different from the relative price ratio in market B, then there is an **arbitrage opportunity** and money can be made taking advantage of the discrepancy in exchange rates.
  - An example will make this idea a bit clearer:

Rule 2: No-overlap / "tangency" condition

- An example of "Arbitrage": Suppose that in Market A the prices (in currency A, say dollars) are

$$p_x^A = 1, \ p_y^A = 2$$

while the prices in Market B (in currency B, say £) are

$$p_x^B = 3, \ p_y^B = 3$$

- Suppose also that the currencies cannot be traded. Consider the following arbitrage strategy: spend 1 dollar to buy a unit of *x* in market A and export it to market B. In market B, this unit of *x* can be traded one-for-one with a unit of *y*. Obtain one unit of *y* and import it into market A. Now exchange the unit of *y* for two dollars. You have doubled your investment.
- This works because the price ratios in the countries are different. You double your money because the ratio of price ratios is 1:2.

Rule 2: No-overlap / "tangency" condition

- Now, back to consumer choice. Think of market A as the real market for the two goods.
- Think of market B as market with currency denominated in units of utility. The price ratio is therefore simply *MRS*.
- If *MRS* is not equal to the price ratio, then there is an arbitrage opportunity between the real market and the fictional utility market in the consumer's mind. A strange metaphor, to be sure, but it is insightful.

General no-overlap / "tangency" condition

- Everything we have said extends to **more than 2 goods**. Between any pair of goods, we will have a no-overlap or tangency condition. In the simple case with smooth indifference curves and budget lines with an interior solution, we have for any two goods, say x_i and x_j we have:

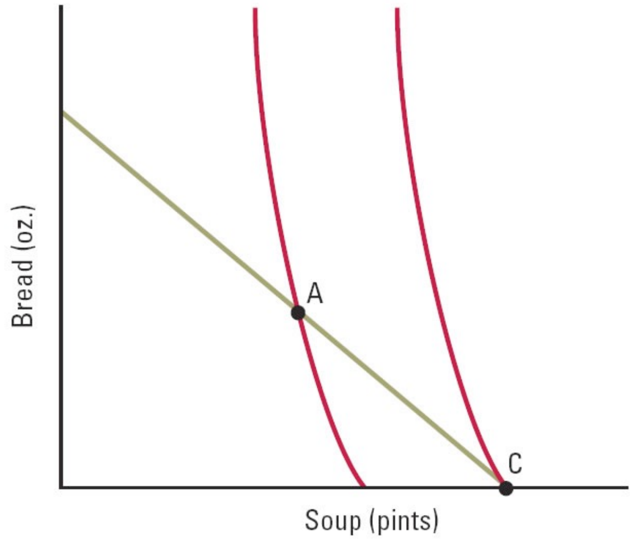
$$MRS_{i,j} = \frac{MU_i}{MU_j} = \frac{p_i}{p_j}$$

- When the solution is **not interior** (at least one good is not consumed, say $x_k=0$), then it must be that the *MRS* for that good with respect to any other that is consumed:

$$MRS_{k,i} = \frac{MU_k}{MU_i} \leq \frac{p_k}{p_i} \quad \text{or} \quad \frac{MU_k}{p_k} \leq \frac{MU_i}{p_i}$$

A Boundary Solution

- Bundle C is the best affordable bundle
- C is also a boundary solution



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

NB: the previous formula should be carefully applied. Because *b=0* at the optimal point, *C*, you want to use $MRS_{b,s}$ which is flatter than the budget line.

General no-overlap / "tangency" condition

- If instead the optimal bundle is interior, but either the constraint boundary or the indifference curve are **kinked**, then a similar condition is met for a slight increase above x_k^*:

$$MRS_{k,i} = \frac{MU_k}{MU_i} \leq \frac{p_k}{p_i} \quad \text{or} \quad \frac{MU_k}{p_k} \leq \frac{MU_i}{p_i} \quad \text{for } x_k > x_k^*$$

- For a slight decrease below x_k^*, the reverse is true:

$$MRS_{k,i} = \frac{MU_k}{MU_i} \geq \frac{p_k}{p_i} \quad \text{or} \quad \frac{MU_k}{p_k} \geq \frac{MU_i}{p_i} \quad \text{for } x_k < x_k^*$$

Deriving a demand function

- There is very little reason to know how to do this, but it may be conceptually interesting to some of you.
- Suppose that you have a smooth utility function that represents a consumer's preferences. You can derive the consumer's demand function in 2 steps:
  - Using calculus and $u(x,y)$, obtain an algebraic expression for $MRS_{x,y}$ which will be a function of x and y.
  - Given the optimal bundle is interior, it satisfies our 2 rules:
    - No waste: $p_x x + p_y y = I$
    - Tangency: $MRS_{x,y} = p_x / p_y$
  - These two rules give 2 equations for 2 unknowns (x and y).
  - Solving this system yields demand functions for x and y.

All other goods ...

- As a graphical simplification, we of will often take the good that we are most interested in, say good x, and construct good y to be an optimal bundle of **all other goods (AOG)**. Thus, we artificially create a 2-good economy.
  - This is not without some problems, as the composition of the AOG bundle may depend on the choice of x. If these **composition effects** are not too big (they often are negligible), this simplification can be useful.
  - The *AOG* "trick" is due to Hicks, and so is sometimes referred to as a **Hicksian** good.
  - Because the unit of measurement of the AOG bundle is arbitrary, we can choose units such that the price of the bundle is \$1/unit. Thus, we can just think of y as left over money not spent on x.

Day 3:
Income and substitution effects, consumer surplus, choice under uncertainty, incomplete information

Consumer Choice - Advanced topics

- Topic 1: Graphical analysis of demand relationships.
  - The connection between indifference curves and individual demand curves
- Topic 2: Cost-of-living allowances
- Topic 3: Choice under uncertainty
  - expected utility theory
  - moral hazard
  - adverse selection
  - diversification

Topic 1: Graphical analysis of demand relationships

- We are interested in determining the response of a consumer's optimal choice with respect to own-price changes, cross-price changes and income changes.
- Sometimes this analysis is referred to as **comparative statics.**
- We will proceed with the simple case of two goods and a simple budget constraint.
- Changes in a price or income shifts the budget line and traces out a series of optimal bundles (for the given economic data).
- These optimal consumption graphs are referred to as **consumption curves**. When we vary price, we have a **price consumption curve (PCC)**; when we vary income, we call it an **income consumption curve (ICC).**
- From the PCC or ICC we can derive various demand relations.

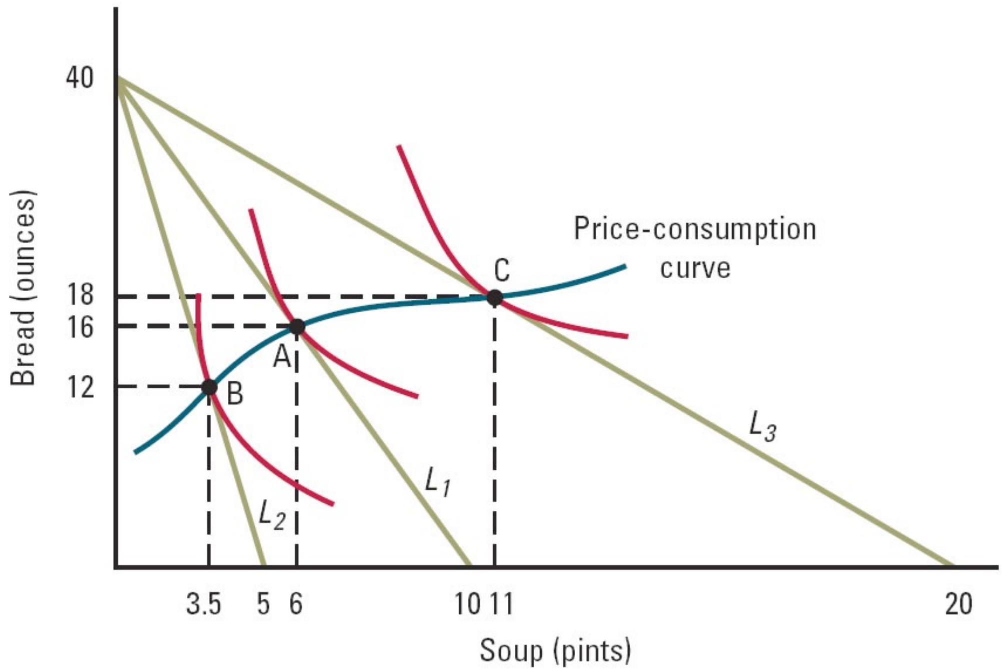
Price consumption curves

- We begin with an analysis of a consumer's optimal choice as we change the price of one good, holding income and the prices of all other goods fixed.
- As we vary the price of *x*, for example, we trace out all the corresponding optimal combinations of *x* and *y*. This is the **price consumption curve (PCC)** for good *x.*
- If the **PCC is increasing**, then reducing the price of good *x* increases the demand for *x and the demand for y*. Thus, the two goods are **demand complements**.
  - E.g., Left shoes and Right shoes (perfect complements)
  - E.g., Soup and Bread (figure 5.11)
- If the **PCC is decreasing**, then reducing the price of good *x* increases the demand for *x but reduces the demand for y*. Thus, the two goods are **demand substitutes**.

Complements and Substitutes, redux

- Note that two goods that are **perfect complements** (a relationship determined entirely by preferences), then they must be **demand complements** (a relationship that generally depends on preferences, prices and income).
  - This is evident by noting that the PCC is always upward sloping. Draw a few pictures to convince yourself.
- If two goods are **perfect substitutes**, then the goods can never be demand complements. Either the change in the price of one good has no effect on the demand of the other, or the goods are demand substitutes.

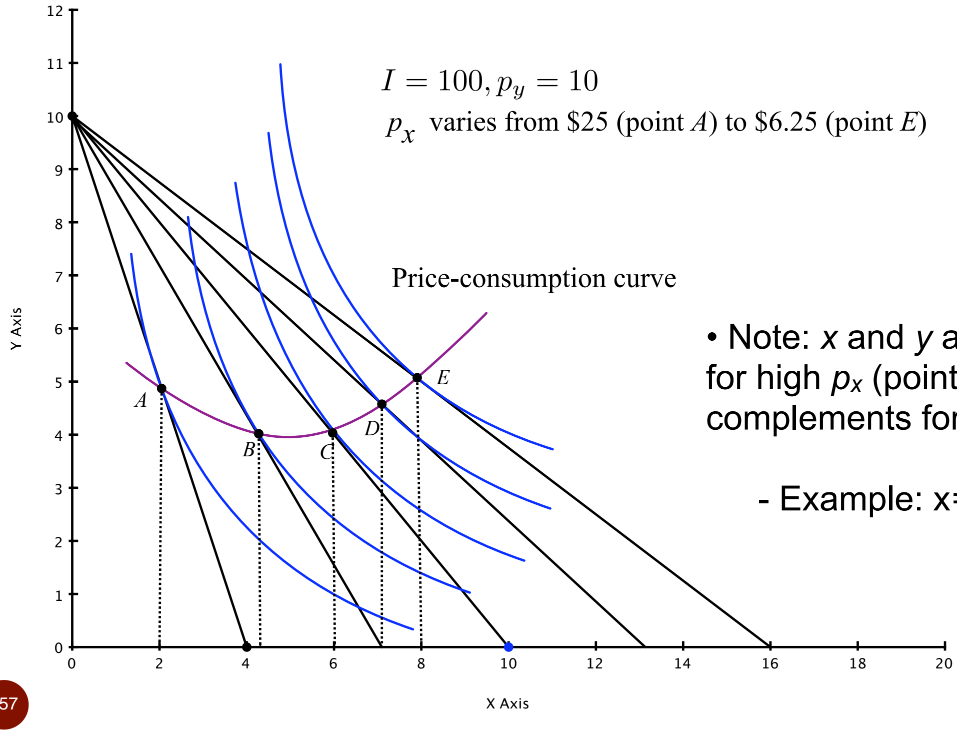
Effect of a Change in the Price of Soup on Consumption



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Note: Soup and bread are demand complements because PCC is upward sloping

Graphical analysis of demand relationships



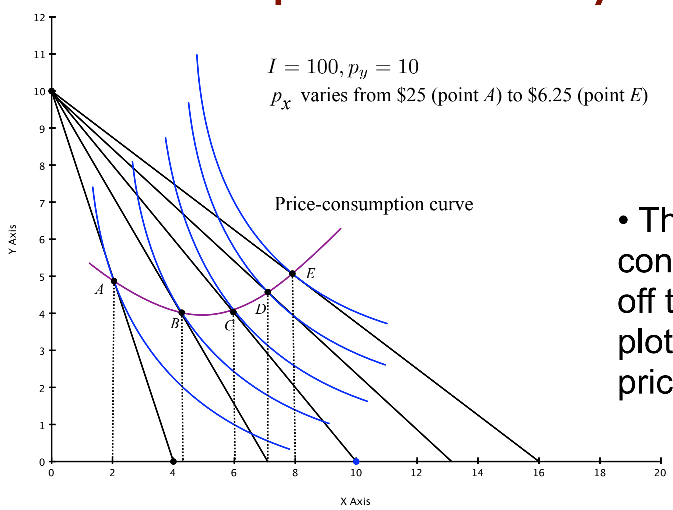
• Note: *x* and *y* are demand substitutes for high p_x (points *A*, *B*) and demand complements for low p_x (points *C, D, E*)

- Example: x=housing, *y=AOG*

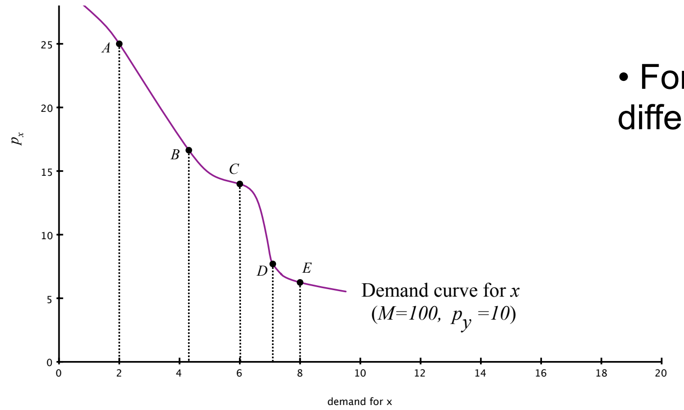
Graphical derivation of consumer's demand curve

- The PCC for good *x* contains all the information needed to derive the consumer's demand curve for *x*
- Recall: the consumer's demand curve for *x* is based on some fixed background values of *I* and p_y.
  - For different values of *I* and p_y we obtain different demand curves.

Graphical analysis of demand



• The demand curve for good x is constructed by taking the values of x off the horizontal axis above and plotting them against the associated prices for x for each budget line.

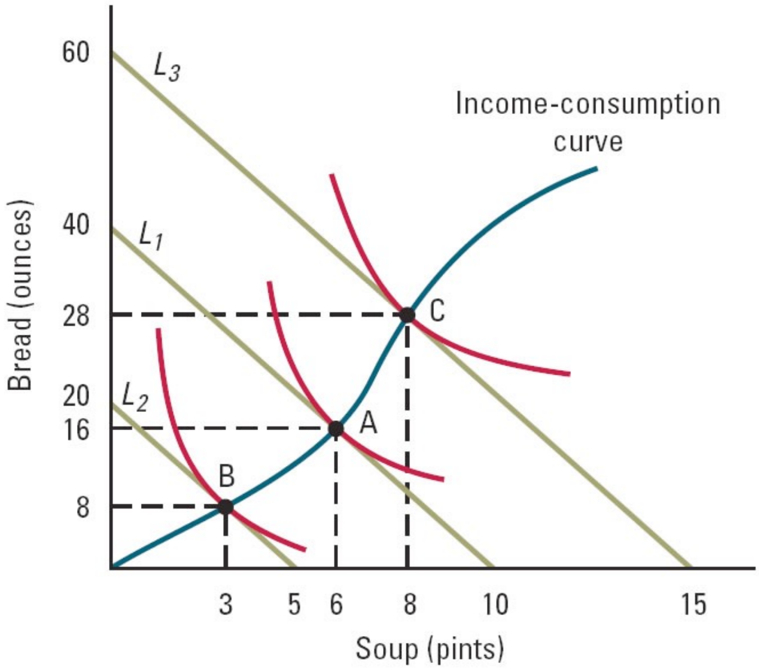


• For different I and p_y, we have different demand curve (shifted)

Income consumption curves

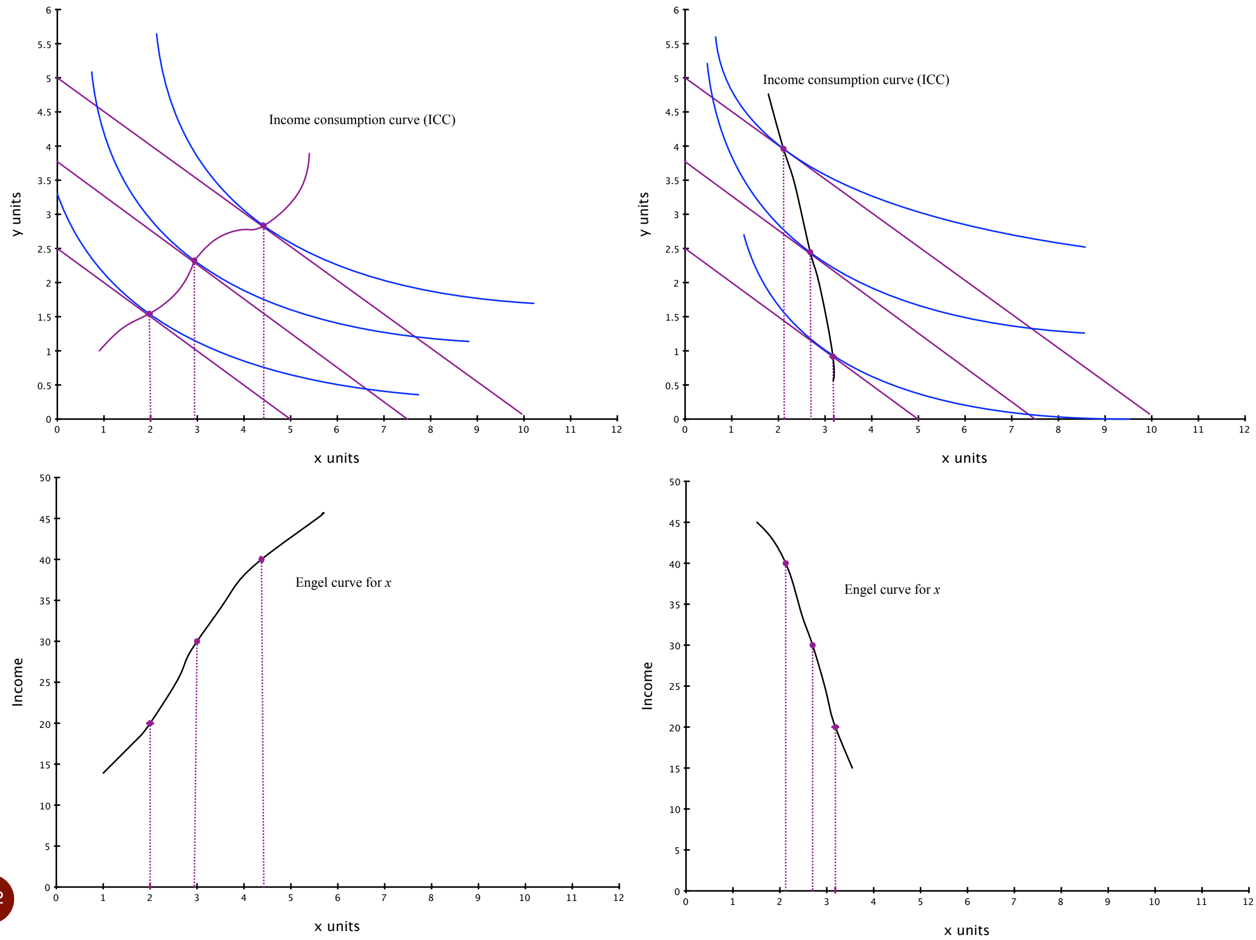
- Now consider a consumer's optimal choice as we change the consumer's income, holding all prices prices fixed.
- As we vary I, we trace out all the corresponding optimal combinations of x and y. This is the **income consumption curve (ICC)**.
- Suppose that there are only **2 goods**. If the **ICC is increasing**, then both goods are normal (their demands increase with income). If the **ICC is decreasing,** then one good is normal and the other is inferior. As income rises, one good's demand increases and the other falls, and hence there is a negative relationship between the two.
  - Both goods cannot be inferior in a 2-good economy. Why?
  - No good can be inferior for all income levels. Why?
- More generally in an economy with more than two goods, at least one good must be normal.

Effect of a Change in Income on Consumption

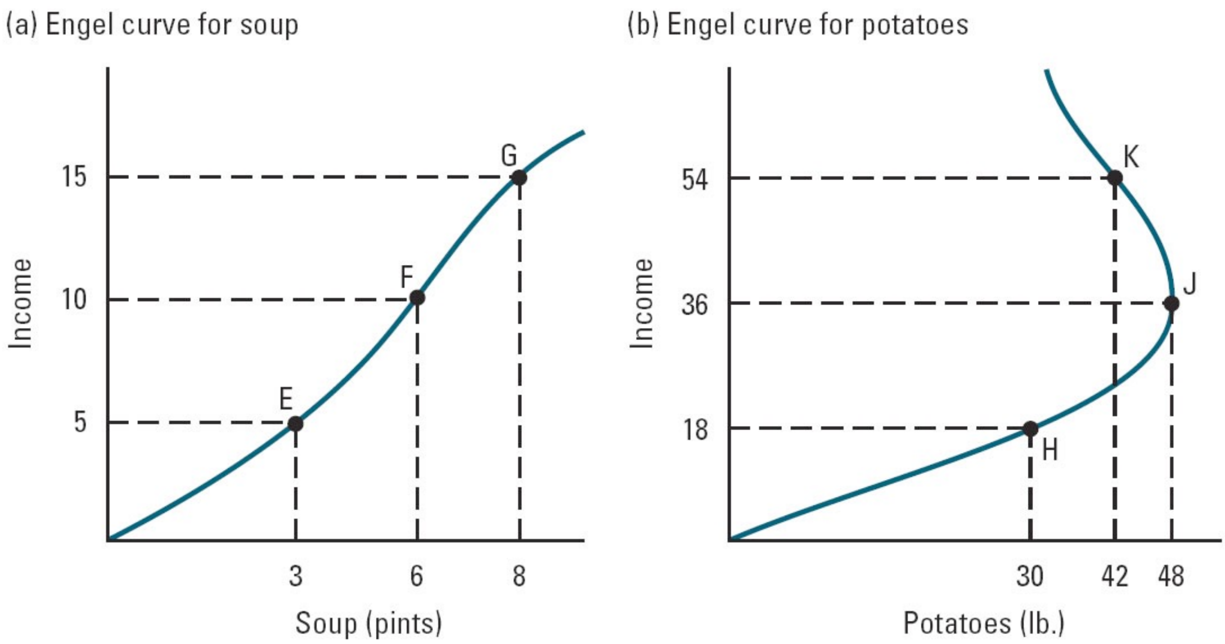


(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Normal and Inferior Goods



Engel Curves for Soup and Potatoes



Potatoes illustrates a good that begins normal and becomes inferior at higher incomes.

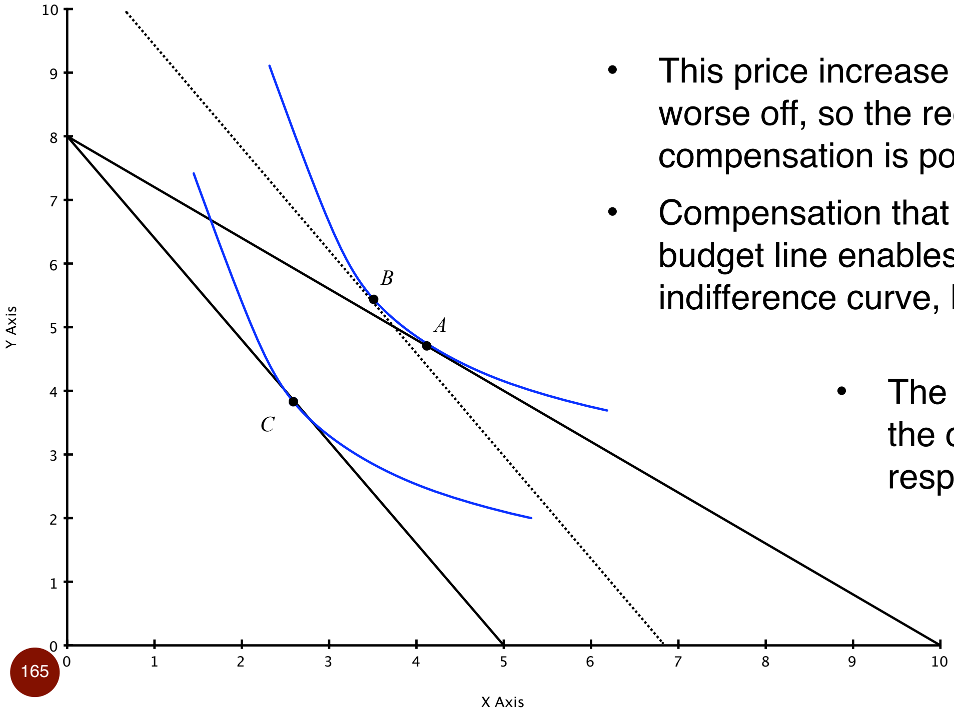
(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Topic 2: Cost of living allowances and substitution boas

- Suppose that a price changes (or several prices change together). In addition, suppose that our individual consumer is **compensated with income** so as to leave them equally well off as before:
  - if the price change would cause less happiness (i.e., prices increase), the consumer will be given a positive income to compensate
  - if the price change would increase happiness, then the consumer has income taken away to compensate
- In either case, you can think of this income compensation as a **perfect cost-of-living-adjustment (COLA)**. That is, with a perfect COLA, the consumer is indifferent to the price change.
- Nonetheless, the consumption bundle of the consumer will likely change under the new prices. The new optimal bundle is the **compensated demand response** to the price change.

Compensated income and demand

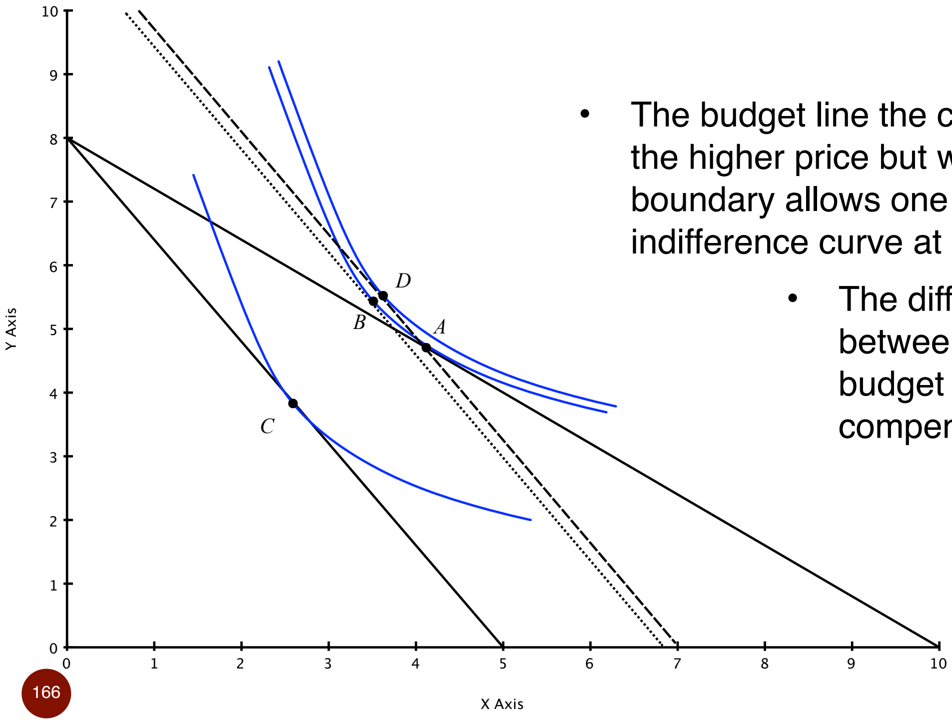
- Suppose the consumer starts out at point *A*, but a doubling of the price of *x* occurs and (absent any income compensation) the consumer would choose point *C*.



- This price increase makes the consumer worse off, so the required income compensation is positive.
- Compensation that generates the dotted budget line enables him to reach the original indifference curve, but at the higher price.
  - The movement from A to B is the compensated demand response.

An aside about COLA's

- Notice that if you gave the consumer sufficient income to afford the original bundle at point *A,* you would be **over-compensating** for the price increase.



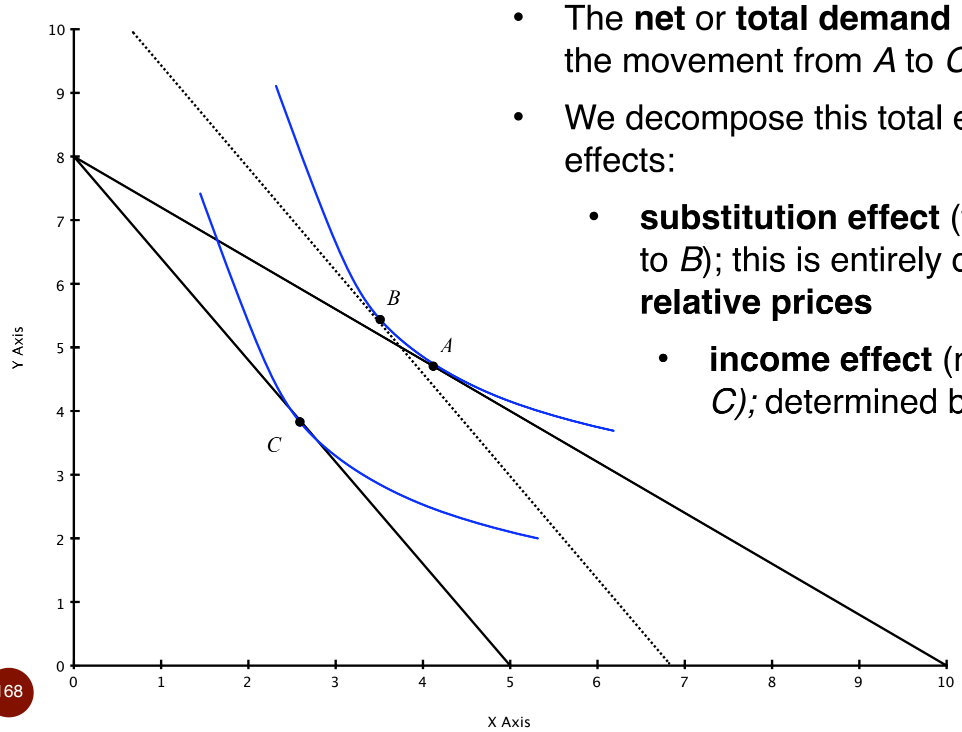
- The budget line the consumer would face at the higher price but with point A on the boundary allows one to reach a higher indifference curve at point D.
  - The difference in incomes between the dotted and dashed budget lines is the over compensation.

An aside about COLA's, continued

- This over-compensation is also the cause of a **substitution bias** in consumer-price indices.
  - The Boskin (1996) commission, for example, concluded that about 1.1% points of annual US inflation is actually not inflation but quality bias (0.7%) and substitution bias (0.4%). Recent studies (Lebrow, Rudd (2003)) find a lower quality bias (0.53%) but similar substitution bias (0.35%)
  - Need to have index weights vary as relative prices change, etc., to correct for substitution bias.

Optional: Income and Substitution effects

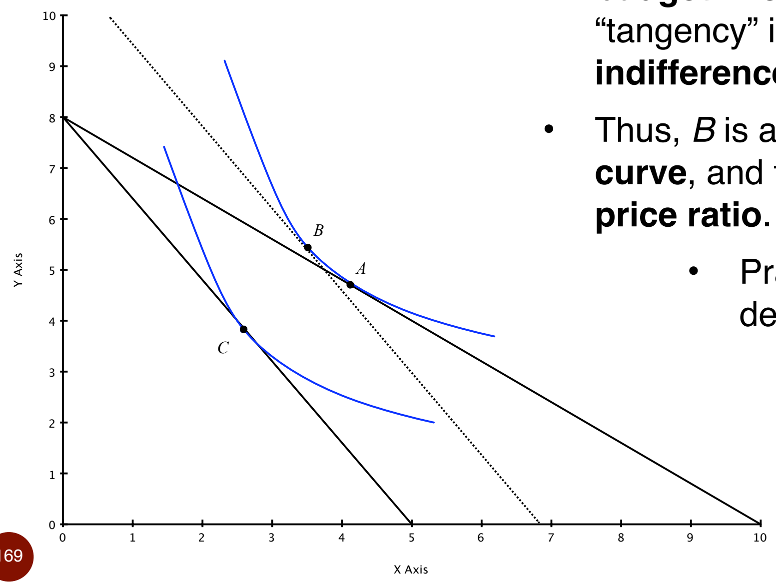
- Consider again a single price change, hold other prices and income fixed.



- The **net** or **total demand change** is given by the movement from *A* to *C*.
- We decompose this total effect into 2 separate effects:
  - **substitution effect** (the movement from *A* to *B*); this is entirely determined by **relative prices**
    - **income effect** (movement from *B* to *C);* determined by purchasing power

Optional: Income and Substitution effects

- When drawing these diagrams, always label the initial point as *A* and the final point as *C*.



- Point *B* is determined by taking the **new budget line** and shifting parallel until "tangency" is found on the **original indifference curve**.
- Thus, *B* is always on the **original indifference curve**, and the *MRS* at *B* is equal to the **new price ratio**.
  - Practice finding *B* using a price decrease, or some other changes.

Optional: Income and Substitution effects

- We can also apply this methodology for finding a decomposition point, *B*, for more complicated cases in which more than one price changes.
  - Step 1: indicate points *A* and *C* given the price changes.
  - Step 2: parallel shift the **new budget line** until it is tangent to the **original indifference curve**. Mark this as point *B*.

Optional: Income and Substitution effects

- So what is the point of this decomposition? What do the effects mean?
- **Substitution effect**. Perhaps this should have been called the **relative-price effect**. It tells us how a someone will adjust their consumption due entirely to relative prices
  - it is as if the consumer was given a perfect COLA so that no change in purchasing power arises
- **Income effect.** A price change also has an effect *as if* income has changed. Should probably be called the **purchasing-power effect** because it may arise even when income does not change.

Optional: Income and Substitution effects

- **Direction of the substitution effect**. In the simple setting of two goods, you need only consider what happens to the price ratio:
  - If the price change renders x more expensive relative to y, then the substitution effect is for less x and more y
  - If the price change renders x less expensive relative to y, then the substitution effect is for more x and less y
  - **Negative relationship.** Notice that x and y move in opposite directions going from A to B. This is because indifference curves are downward sloping and both A and B are on the same curve.
- With more than two goods, if a good's relative price falls with respect to every other good, it will be demanded more, and vice versa.

Optional: Income and Substitution effects

- **Direction of the income effect**. We need to know two things to sign the direction of the income effect.
  - First, does the uncompensated price change lead to a better or worse outcome for the consumer (is *C* on a higher or lower indifference curve than *A?*). That is, **what happens to purchasing power?**
    - If consumer is better off (e.g., some price falls), we say that purchasing power has increased due to the price change.
    - If consumer is worse off (e.g., some price rises), we say that purchasing power has decreased due to the price change.
  - Second, **is the good a normal or an inferior good?**
- If purchasing power falls and the good is normal, then the income effect is negative; demand is reduced. If purchasing power falls and the good is inferior, then the income effect is positive and demand increases. What about the other cases?

A few quick examples/questions:

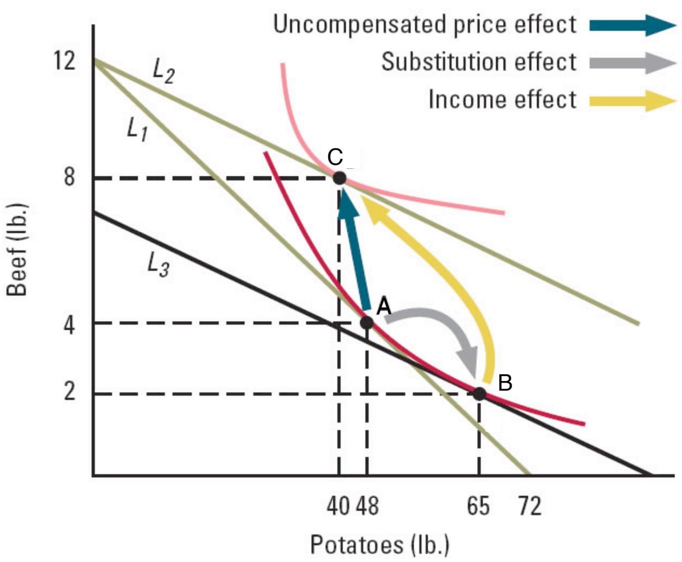
- Perfect Cost-of-Living Adjustments:
- Suppose that the price of left shoes goes up, but nothing else changes. What is the magnitude of the substitution effect between left and right shoes?
- Suppose that a consumer's income increases but no price changes. What is the size of the substitution effect between x and y?
- Consider an increase in the price of good x and its impact on the demand for good x (assuming there are only 2 goods).
  - the substitution effect implies that the demand for x should decrease
  - because the price has increased, purchasing power has decreased; if x is a normal good, then the income effect implies that the demand for x should decrease
  - thus, when x is normal, demand must be negatively sloped

A few quick examples/questions:

- Suppose that the price of x increases and x is an inferior good.
  - the substitution effect implies that the demand for x should decrease (same as before)
  - because the price increases, purchasing power decreases
  - because x is inferior, reduced purchasing power implies that the income effect on x is an increase in demand
  - the two effects operate in different directions
  - if the substitution effect for x is larger in magnitude than the income effect, the demand curve for x is downward sloping
  - if the substitution effect for x is smaller in magnitude than the income effect, the demand curve for x is rising in price! This is very rare and is known as a **Giffen good**
    - (e.g., Irish potato famine in 1845, rats and quinine, etc.)

Giffen Good

- Giffen goods are inferior, and the amount purchased moves in the same direction as the price
- Income effect is larger than the substitution effect



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

A few quick examples/questions:

- **Hard question**: Suppose that Jake consumes only two goods, beer and chips. When the price of beer falls, Jake consumes more chips. Is chips a normal or inferior good? Can we tell?
- Variation of above question. Suppose that the price of good y increases and the demand for both *x* and *y* falls (on net). What can we say about the directions and magnitudes of the substitution and income effects for *x*?
  - The substitution effect is for the demand of *x* to increase because it has become relatively cheaper than *y*.
  - If a price increases, then purchasing power falls. If *x* is an inferior good, this would imply that the income effect on demand for *x* is positive; if *x* is normal, then the income effect on demand is negative
  - Because the net/total effect is negative, *x* must be normal.

Income and substitution effects: Applications

- Price indices and substitution bias
- Labor supply
  - Think of two goods: leisure (the inverse of labor) and AOG. When the wage rises, so does the implicit price of leisure. The substitution effect is to work more and consume less leisure; and to increase consumption of AOG. The income effect is to feel wealthier (greater purchasing power). Because leisure is empirically a normal good, the income effect of a wage increase is to work less. Could go either way on net. Typically, labor supply curves are backward bending for this reason.
  - Note: this is not an example of a Giffen good because changing the wage does two things - it changes the price of leisure and it directly changes the value of a worker's labor endowment. Similar in spirit, however.

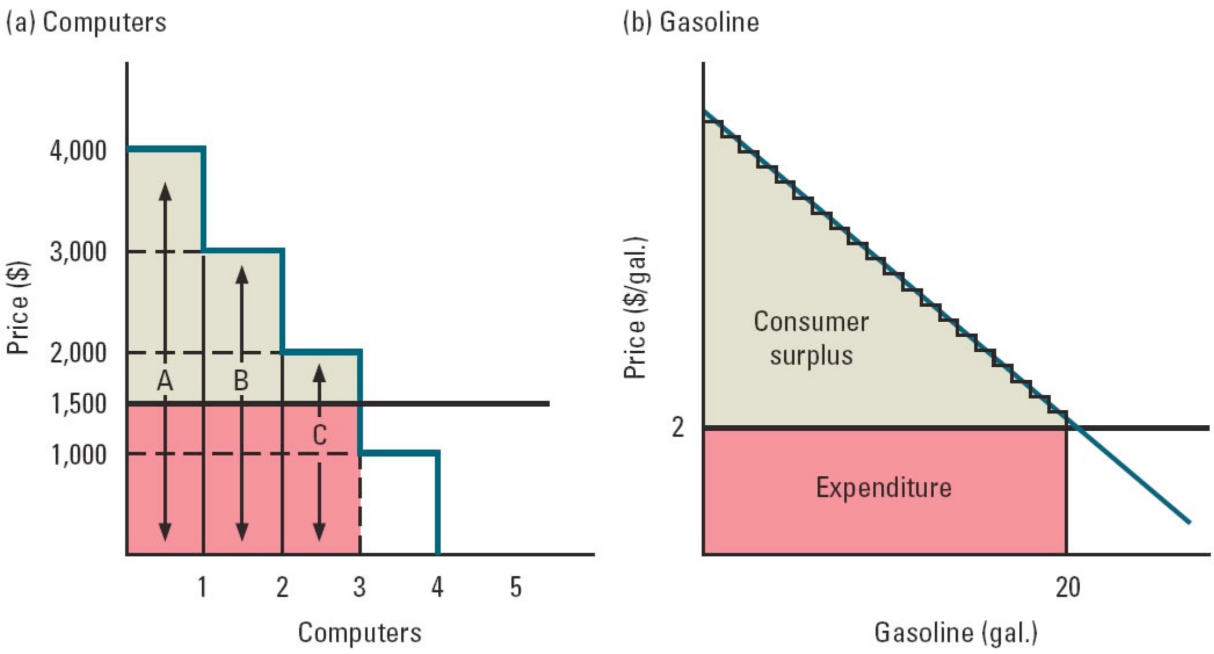
Income and substitution effects: Applications

- Borrowing and saving, and the effect of a change in the interest rate. Thus, if you are a net saver, an increase in the interest rate induces you to save more (the substitution effect) but your greater purchasing power from interest income induces you to spend more (the income effect) which reduce savings. You'll see more on this in macroeconomics.
- Quality of automobiles in Singapore relative to the United States
- Quality of oranges sold in Chicago relative to Florida

Consumer Surplus

- Consumer surplus is the monetary "profit" a consumer obtains from the market
  - recall our trading experiment in class
- We will define a consumer's surplus as the area below the consumer's demand curve and above the market price
- The total consumer surplus (across all consumers) is the area below the market demand curve and above the market price
- Consumer surplus is not exactly the same amount of money as the compensating income needed for losing access to the market. The difference is due to possible income effects. We will ignore this subtle point for most of the course.
- To calculate the *CS* for a linear demand function, *D(p)=a-bp*, remember that the area of a triangle is *(1/2)(base)(height)*. Thus, at a market price of p: CS=(1/2) (a-bp)(a/b - p).
  - note: (a/b) is the "choke" price for this market.

Consumer Surplus



(figure from from Bernheim, Whinston, *Microeconomics, 1st ed, 2008)*

Topic 3: Choice under uncertainty

- Main topic: We consider bundles that are uncertain
  - First, we broaden our definition of consumption bundle to include risk -- this requires new notation
  - Second, we develop the Expected Utility theorem and its implications for optimal choice
- Additional considerations with uncertainty:
  - Moral hazard and incentives
  - Adverse selection
  - Diversification and risk-adjusting returns

Uncertainty in consumption bundles

- With certainty, a consumption bundle is characterized as a vector or numerical list of quantities, where each number corresponds to the amount consumed of a particular good.
- Suppose instead that the bundle has uncertain payoffs for each good.
  - E.g., two goods: you order a “large” bowl of soup and a “regular” salad at a new restaurant, but you are uncertain about how much soup or salad you will actually get. There is a range of acceptable sizes, some more likely than others, but there is still some uncertainty until you are served.
  - E.g., one good: you buy into a mutual fund for your retirement savings. You are not sure what it will be worth when you retire, but you understand the possible outcomes and their likelihoods.

Uncertainty in consumption bundles

- To simplify the analysis, we will assume that consumers care only about one good -- money. Of course, for a given amount of money, they will optimally allocate it across the available goods according to our rules (no-waste/no-overlap).
- Such an uncertain consumption bundle is like the second example -- it is an asset that pays out various amounts of money (perhaps negative, a loss) with a various probabilities
- Notation:
  - there are n possible outcomes, denoted i=1,…,n; sometimes these are referred to as states or events
  - outcomes are mutually exclusive (one and only one occurs)
  - for each outcome, there is a payout, x_i, positive or negative
  - each outcome has a probability of ϕ_i, of occurring, $0 \leq \phi_i \leq 1$
  - the sum of the probabilities across all outcomes is 1.

Uncertainty, continued

- All together, we describe a random variable or asset by
$$\tilde{x} = \{x_1, \ldots, x_n, \phi_1, \ldots, \phi_n\}$$
The tilde ~ over the *x* indicates that the value of *x* is random.
- The requirement that the probabilities sum to 1 is stated as $\sum_{i=1}^{n} \phi_i = 1$
- A few statistics:
  - the expected value of $\tilde{x}$ (sometimes called the mean or weighted average) is given by the formula
  $$E\left[\tilde{x}\right] = \sum_{i=1}^{n} \phi_i x_i$$
  If each event is equally likely, then $\phi_i = \frac{1}{n}$ and the expected value is simply average value.

    Sometimes μ or $\overline{x}$ is used instead of $E\left[\tilde{x}\right]$

Uncertainty, continued

- An asset with no risk is called a sure-thing or riskless asset. It is described like any other asset but has a simpler form:
  - $\tilde{x}_{st} = \{x_1, \phi_1 = 1\}$
  - The expected value of a sure-thing satisfies the expectation formula, but it is trivial to calculate:
  $$E\left[\tilde{x}_{st}\right] = \sum_{i=1}^{1} \phi_i x_i = 1 \cdot x_1 = x_1$$
  - You should verify that a sure-thing also has zero variance.
  - The point of all of this is that we can still apply our formulas for risk to an asset with no risk.

Uncertainty, continued

- A few statistics:
  - the variance of $\tilde{x}$ is given by the formula

$$V\left[\tilde{x}\right] = \sum_{i=1}^{n} \phi_i \left(x_i - E[\tilde{x}]\right)^2$$

    Sometimes, variance is denoted $\sigma^2 = V\left[\tilde{x}\right]$. Hence, you may see in statistics class the formula

$$\sigma^2 = \sum_{i=1}^{n} \phi_i (x_i - \mu)^2$$

  - Variance is a measure of dispersion, of course. It is often used a proxy for the riskiness of an asset.

Expectations of a function of a random variable & Jensen's inequality

- Just as we can take the expectations of a random variable, we can also take the expectations of a function, $f(x)$, of a random variable
- To do this, we first convert each possible outcome, x_i, to the function evaluated at this outcome, $y_i = f(x_i)$. We then take the expectation of this new random variable:

$$\mathrm{E}[f(\tilde{x})] = \sum_{i=1}^{n} \phi_i f(x_i).$$

- It is very important to note that the "expectation of a function of a random variable" is generally NOT the same as the "function of the expectation of a random variable." Indeed, if f is not linear, then

$$\mathrm{E}[f(\tilde{x})] = \sum_{i=1}^{n} \phi_i f(x_i) \neq f\left(\mathrm{E}[\tilde{x}]\right) = f\left(\sum_{i=1}^{n} \phi_i x_i\right) = f(\overline{x}).$$

An example:

- An example where $\mathrm{E}[f(\tilde{x})] \neq f\left(\mathrm{E}[\tilde{x}]\right).$

$$n = 2, \phi_1 = \phi_2 = \frac{1}{2}, x_1 = 0, x_2 = 2, \text{ and } f(x) = x^2 :$$

$$E[\tilde{x}] = 1, f(E[\tilde{x}]) = 1, \text{ but } E[f(\tilde{x})] = \frac{1}{2}0^2 + \frac{1}{2}2^2 = 2.$$

Jensen's inequality

- There is a useful geometric relationship called "Jensen's inequality" which provides insights into risk aversion, option pricing, etc.
- Jensen's inequality holds that if *f*(*x*) is a **concave function** (i.e., it lies above all of its chords), then for any random variable

$$\mathrm{E}[f(\tilde{x})] = \sum_{i=1}^{n} \phi_i f(x_i) < f\left(\mathrm{E}[\tilde{x}]\right) = f\left(\sum_{i=1}^{n} \phi_i x_i\right).$$

- If *f*(*x*) is a **convex function** (i.e., it lies below all of its chords), then the inequality is reversed:

$$\mathrm{E}[f(\tilde{x})] = \sum_{i=1}^{n} \phi_i f(x_i) > f\left(\mathrm{E}[\tilde{x}]\right) = f\left(\sum_{i=1}^{n} \phi_i x_i\right).$$

- Lastly, in the case in which *f*(*x*) is linear, we have an equality:

$$\mathrm{E}[f(\tilde{x})] = f\left(\mathrm{E}[\tilde{x}]\right).$$

Preferences for risky assets

- As before, we assume the ranking principle applies -- that is, an individual can rank all assets from best to worse with possible indifference.
- We say that someone is **“risk averse”** if a riskless asset (sure thing) is always ranked strictly above any risky asset with the same expected value.
  - Note: this comparison requires that both the sure thing and the risky asset have the same expected values. A risk averse person may very well prefer a risky asset with a higher expected payoff to a sure-thing with a low payoff.
- Similarly, we say that an individual is **“risk loving”** if the ranking is reversed: a risky asset is always preferred to a riskless asset with the same expected payoff.
- A **“risk neutral”** person cares only about expected payoff. Hence, they are indifferent between the assets above.

Digression on risk loving

- Most individuals are NOT risk loving. Typically, when we think about examples of risk-loving, there is something else involved as well
  - Gambling, casinos, lottery tickets, betting on sports, etc.: In most cases, these activities are as much about entertainment value as about sound financial choices. Typically, gamblers care about the financial aspects of their bets but also obtain some additional utility from betting.
  - Irrationality. People are notoriously bad at figuring out expected values for events with small probabilities. They instead rely on heuristics that are not optimal and lead to behavior that sometimes looks like risk loving. Sometimes people just get the probabilities wrong. (E.g., “I’ve lost at roulette 10 times in a row now, so I am more likely to win on the next spin.” This is irrational. Why?)

Digression on risk loving

- Entrepreneurial spirit. Typically, successful entrepreneurs have a low cost to risk, but they are still risk averse unless they are given perverse incentives.

- Indeed, if you could find two risk loving people, you could put them in a room with access to their entire wealth and a pair of dice and they could make themselves incredibly happy for hours on end. Since we don't see this so much, risk lovers must be rare.
- That said, sometimes the economic incentives facing otherwise risk-averse people will make them behave as if they are risk loving.
  - E.g., if you own a call option on a stock, you will have risk-loving preferences over the stock return.

    We will discuss this more below.

Representing preferences for risk with utility

- Given that the ranking principle is satisfied for risky assets, it follows that there exists a utility function, $U(\tilde{x})$, for each consumer which represents his preferences.
  - That is, for two portfolios -- $\tilde{x}^a, \tilde{x}^b$ -- the consumer strictly prefers portfolio *a* to *b* (which we denote by $\tilde{x}^a \succ \tilde{x}^b$ using a script greater-than symbol) if and only if utility is higher for a than for b:

$$\tilde{x}^a \succ \tilde{x}^b \iff U(\tilde{x}^a) > U(\tilde{x}^b)$$

- Unfortunately, our new utility function is quite complicated.

Representing preferences for risk with utility

- The utility function over portfolios is quite complicated:
  - $U(\tilde{x})$ depends upon $x_1, \ldots, x_n$ and $\phi_1, \ldots, \phi_n$
  - The first components are the monetary payments of the assets. The utility function should be increasing in each of those values just as was the case for *u(x,y)* from before.
  - The impact in a change in probability for some event is more subtle. Because all the probabilities must sum to 1, an increase in one event must be offset with decreases elsewhere. Thus, it is not obvious how $U(\tilde{x})$ depends upon $\phi_1, \ldots, \phi_n$.
- Fortunately, we know that the utility function is an ordinal concept, and that any strictly increasing transformation of $U(\tilde{x})$ will also represent the consumer's preferences. Remarkably, there is a transformation which is simple and intuitive.

Expected utility representation

- We need to add an important assumption to our ranking principle called the Independence of Irrelevant Alternatives (IIA). It is easy to construct examples where people make choices that are inconsistent with the IIA assumption, but most people, once it is carefully explained, agree that they should follow the IIA principle to be "rational."
- We'll return to the criticisms of IIA later.
- With the addition of IIA, we can state a great simplification of the utility function representation for preferences over risk. It is known as the Expected Utility Representation Theorem or sometimes the expected utility property.
- The Theorem is due to John von Neumann, a famous Hungarian mathematician and the father of game theory. Sometimes the expected utility result is referred to as von Neumann-Morganstern utility (VNM).

Expected utility representation

- Theorem: If a consumer's preferences satisfy the ranking principle and IIA, then there exists a utility-of-money function, *u(x)*, which gives the "happiness" associated with *x* units of money and where the expected utility represents preferences:

$$U(\tilde{x}) = E[u(\tilde{x})] = \sum_{i=1}^{n} \phi_i u(x_i)$$

- The utility-of-money function is sometimes called a Bernoulli function (after 1 of 3 famous Bernoulli mathematicians).
- Note: Do not confuse the utility function over assets, $U(\tilde{x})$,(which gives the utility of a risky portfolio) with the utility-of-money function,$u(x_i)$, which gives the utility for the payoff x_i.

Simple example using EU theory

- Suppose that an individual's benefit function for money is $u(x) = \sqrt{x}$
- This person has an initial wealth of *W=100* and can choose one of three options.
  - Do nothing. The results in a portfolio that pays 100 with certainty.
  - Buy asset *A*. Asset *A* costs \$36 and with probability 1/2 pays off \$80, and with probability 1/2 is worthless.
  - Buy asset B. Asset B costs \$100. With probability 1/2, it pays off \$196; with probability 1/2 it pays off \$49.
- Which of the three choices yields the highest expected utility?
  - Do nothing yields $EU = u(100) = \sqrt{100} = 10.$
  - Buy asset *A*:

$$EU_a = \tfrac{1}{2}u(100-36+80) + \tfrac{1}{2}u(100-36+0) = 0.5\sqrt{144} + 0.5\sqrt{64} = 10.$$

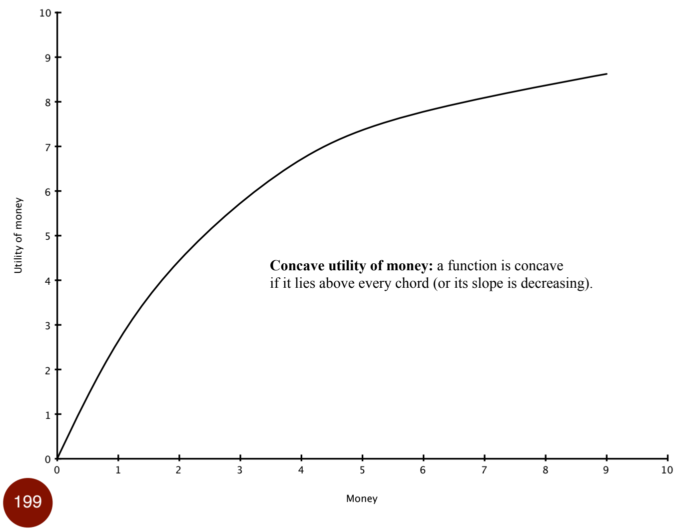
  - Buy asset *B:*

$$EU_b = \tfrac{1}{2}u(196) + \tfrac{1}{2}u(49) = 0.5\sqrt{196} + 0.5\sqrt{49} = 10.5.$$

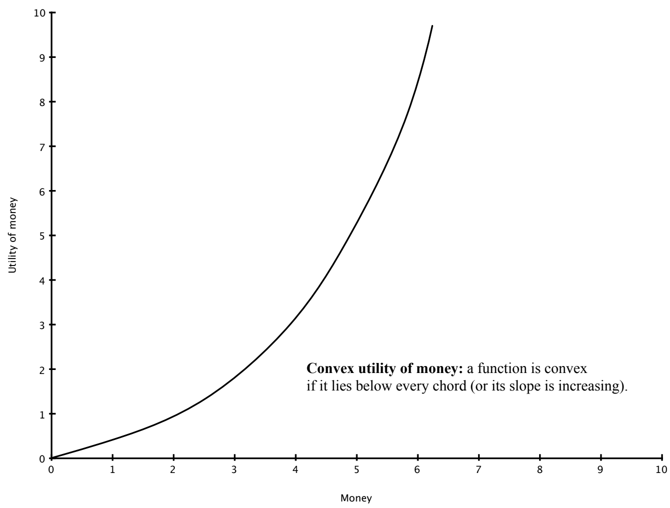
Implications of EU result

- An immediate implication is the result is that a consumer's preferences for risk are entirely determined by the utility-of-money function. Thus, a risk averse person should have a different looking *u(x)* function than a risk loving person.
- Here are two examples (both are increasing in money):

concave $u(x)$:



convex $u(x)$:



Implications of EU result

- The concave utility-of-money function indicates that more money is good, but that earning another million £ is not as satisfying as was the first million £. We say that there is diminishing marginal utility of money.
- Similarly, a convex utility-of-money function indicates that more money is good, and that earning another million £ is even better than was the first million £. We say that there is increasing marginal utility of money.
- While we are at it, note that a linear utility-of-money function has a constant marginal utility of money.
- We are now prepared to state the most important implication of the EU representation ...

Implications of EU result

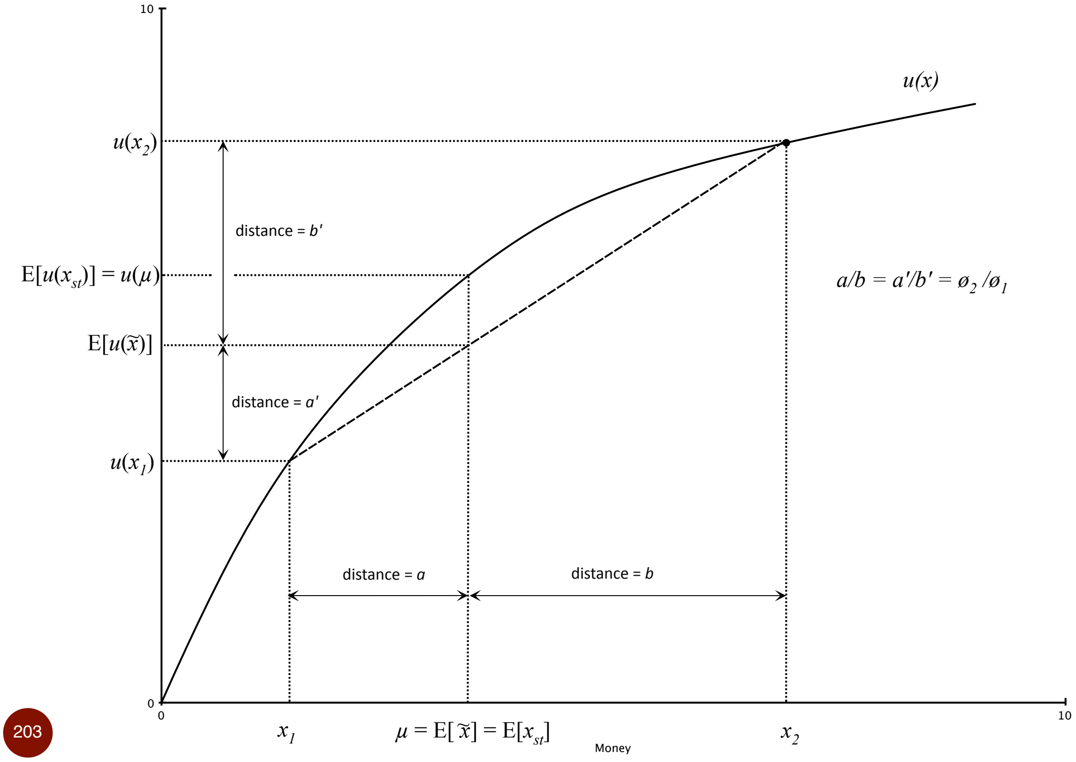
$u(x)$ is concave	$\Longleftrightarrow$	diminishing MU of money	$\Longleftrightarrow$	risk averse consumer
$u(x)$ is linear	$\Longleftrightarrow$	constant MU of money	$\Longleftrightarrow$	risk neutral consumer
$u(x)$ is convex	$\Longleftrightarrow$	increasing MU of money	$\Longleftrightarrow$	risk loving consumer

Connection between risk aversion and diminishing MU of money

- Intuitively, consider a choice between obtaining 1 million £ for sure or a 50-50 gamble of 0 or 2 million £.
  - This is an example of an actuarial fair bet (i.e., the expected values are the same between the choices).
  - Nonetheless, most of use would choose the sure thing (and therefore we are risk averse). Why?
  - First, note that this choice is equivalent to having 1 million £ in your bank account and considering a 50-50 bet for winning 1 million or losing 1 million -- a fair bet.
  - If you take this bet, there is a 50 percent chance of an increase in income to 2 million, and a 50 percent chance of a reduction to zero. In utility terms, if the first million is worth more than the second million, then an even bet in money terms is definitely not even in utility terms.

Risk aversion and concave utility for money

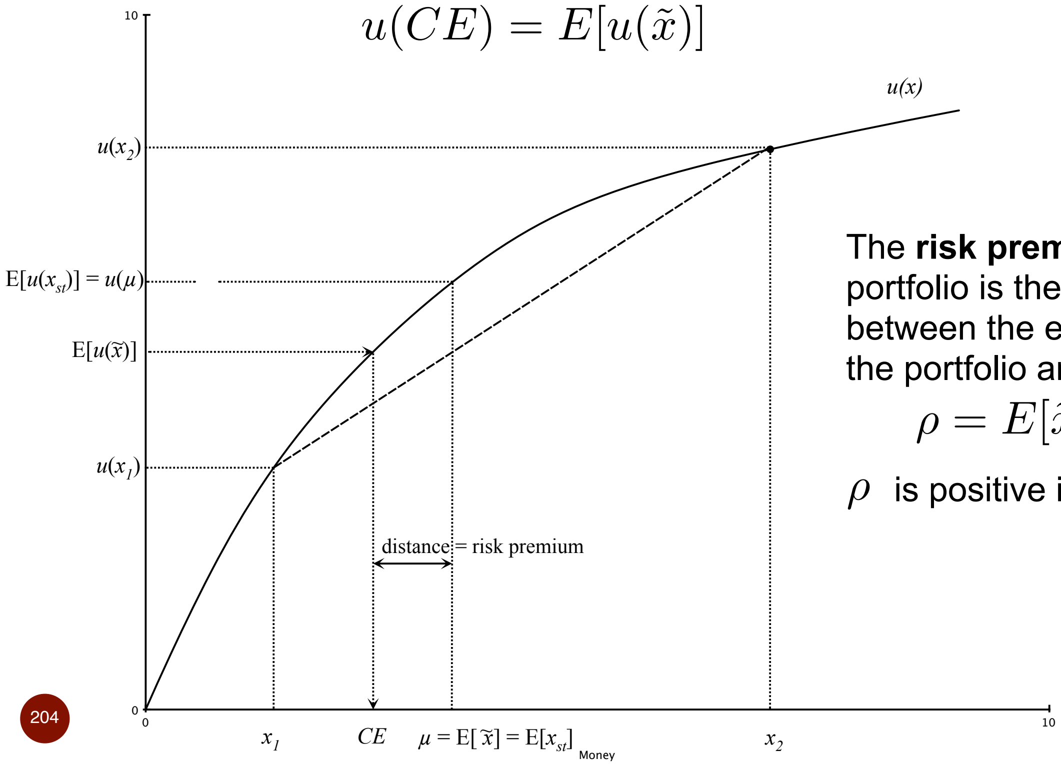
- Graphically, we can illustrate a similar idea (again assuming that there are just two outcomes). Suppose *u(x)* is concave.



Certainty equivalence, risk premium

- The **certainty equivalent (CE)** of a portfolio, $\tilde{x}$, is the monetary value of a sure-thing that would generate equal expected utility: it is the value of CE such that

$$u(CE) = E[u(\tilde{x})]$$



The **risk premium**, ρ, for the portfolio is the difference between the expected value of the portfolio and its CE:

$$\rho = E[\tilde{x}] - CE$$

ρ is positive if risk averse.

Remarks about risk aversion and concavity

- The graphical argument we used to establish the connection between concavity (decreasing marginal utility of money) and risk aversion relied on the fact that a concave function lies above all of its chords.
- A similar argument implies that increasing marginal utility of money (or convexity) will lead to reverse implications: the risky asset will always generate more utility than the sure-thing with equal expected payoff. Risk premium is negative.
- If the marginal utility of money is constant, the utility function is a line and it lies on top of all of its chords. In this case, risk has not effect on the ranking of assets. Risk premium is zero.
- The graphical argument generalizes to an arbitrary number of outcomes using an important mathematical result called Jensen's inequality. The intuition is the same, however.

Numerical example to apply the ideas

- Suppose that we knew a consumer's utility-for-money function is

$$u(x) = \sqrt{x}$$

- (I choose this for illustration because it is the simplest concave function that I know.)
- Suppose that our individual has initial wealth of W=£20,000, and she is offered an asset for p=£1000 that pays £10,000 with probability 1/3, and is worthless otherwise.
- Should she buy the asset? We need to know the her expected utility if she buys the asset and compare it to the expected utility of not buying the asset.
    - Expected utility of not buying the asset. This is just the expected utility of the sure-thing of £20,000. Hence,

$$EU_{\text{no asset}} = \sqrt{20000} = 141.4$$

Numerical example, continued

- Should she buy the asset?
  - Expected utility of not buying the asset:
$$EU_{\text{no asset}} = \sqrt{20000} = 141.4$$
  - Expected utility of the asset. Now we need to be careful to construct our so that they include all income in that state. Thus, if *i*=1 is the state in which the asset pays out,
$$x_1 = 20,000 - 1000 + 10,000 = 29,000$$
$$x_2 = 20,000 - 1000 = 19,000$$
The expected utility of the portfolio with the asset is
$$EU_{\text{asset}} = \frac{1}{3}\sqrt{29000} + \frac{2}{3}\sqrt{19000} = 148.658$$
Because 148.658 > 141.4, the asset should be purchased.

Numerical example, continued

- What is the risk premium for the portfolio with the asset?
  - Expected utility of buying the asset is
$$EU_{\text{asset}} = \frac{1}{3}\sqrt{29000} + \frac{2}{3}\sqrt{19000} = 148.658$$
  - The certainty equivalent for this portfolio satisfies
$$u(CE) = 148.658 \implies \sqrt{CE} = 148.658$$
$$\implies CE = (148.658)^2 = 22,099$$
  - The expected value of the portfolio is
$$E[\tilde{x}] = \frac{1}{3}(29000) + \frac{2}{3}(19000) = 22,333$$
You could also have calculated the expected net value of the asset, which is £2333 and then added £20,000 to it.
  - Thus,
$$\rho = 22,333 - 22,099 = \pounds 234.$$

Numerical example, continued

- If we raise the individual's initial wealth to W=£100,000, what happens to the risk premium?
  - Expected utility of the portfolio rises to 319.812
  - The certainty equivalent for the new portfolio is £102,280.
  - The risk premium becomes

$$\rho = 102,333 - 102,280 = \pounds 53.$$

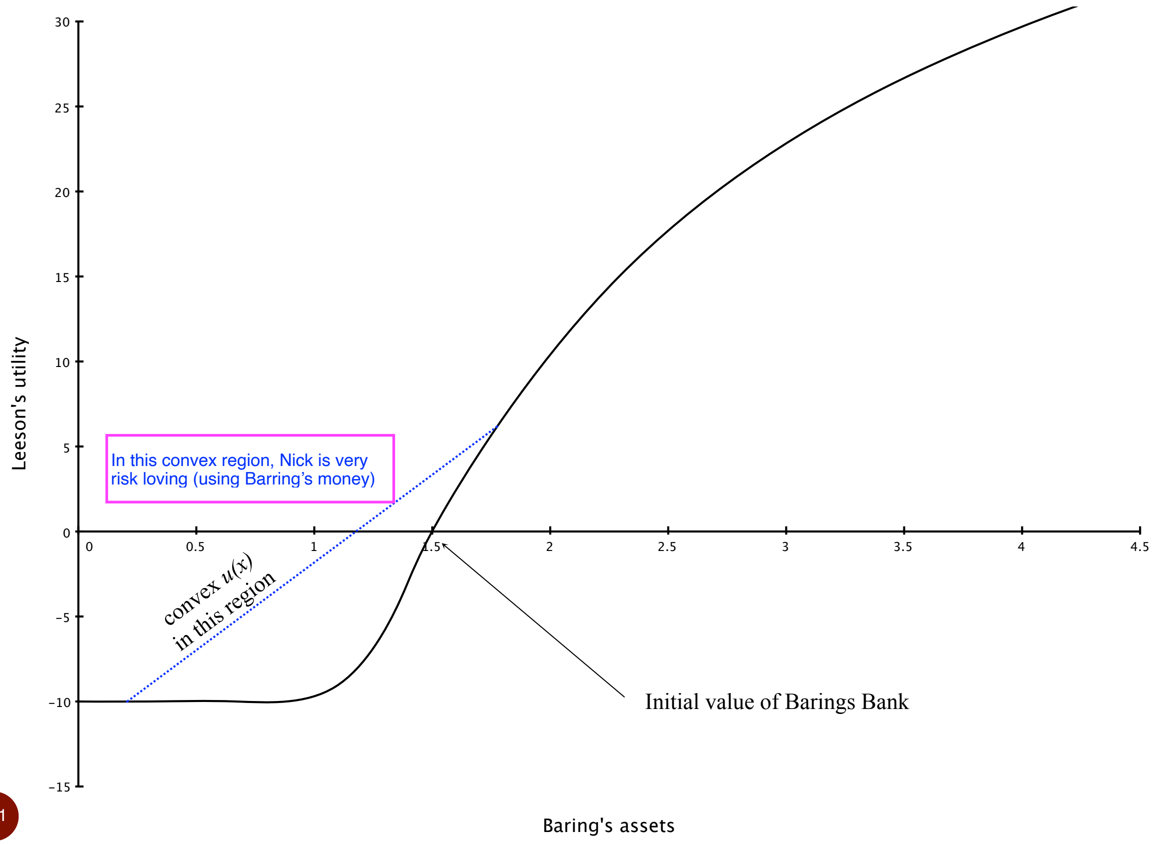
  - It is an empirical regularity that wealthier people require a lower premium for risk.

    This example illustrates the phenomena: £53 < £234.

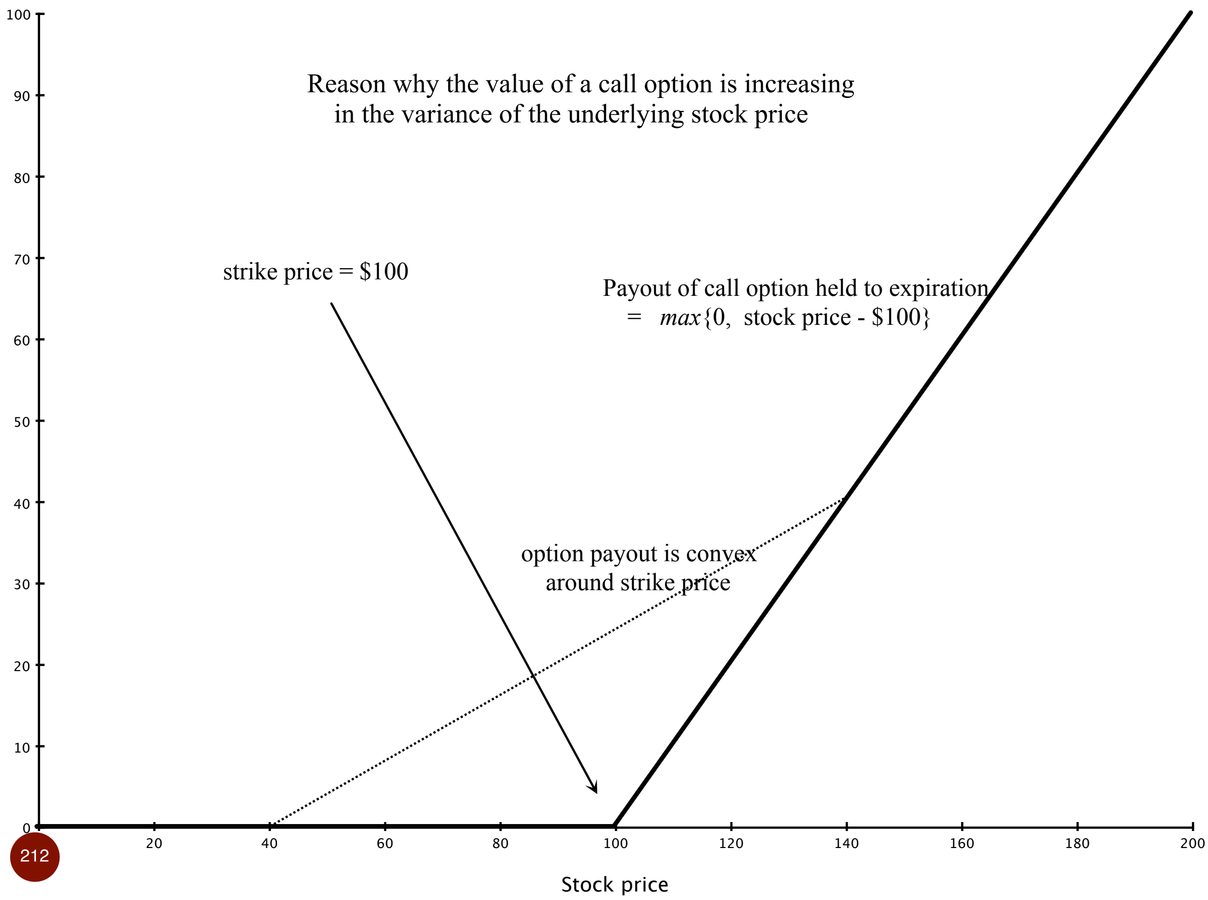
Induced risk-loving preferences

- While it is rare for someone to be naturally risk loving, often times the incentives people face cause them to become risk loving.
  - Anytime there is a lower bound to the outcome of a random variable, people will behave risk-loving around the bound.
    - E.g., Nick Leeson and Barings bank (1995). After losing \$1.4B, Leeson may rightly assume that losing \$2B wouldn't make things much worse. (Do we think his prison sentence will increase going from \$1.4B to \$2B in losses?) Thus, his worst case utility is bounded below and he has reason to gamble with Barings money to try to get out of the red.
    - E.g., call options. Suppose that you own a call option on a stock. Then you would like the company to change to a riskier strategy, holding expected value constant.

Nick Leeson and Barings



Value of a call option



Recent financial crisis and bank failures

- Same story arguably caused bank executives to take out imprudent risks using other people's money.
- Hedge funds don't suffer to the same extent from this problem given that hedge-fund managers typically large fraction of their own wealth invested.

Behavioral economics and psychology

- We mentioned the IIA assumption before and its importance for the expected utility result. Let me say more about that now. Suppose that you prefer portfolio ***a*** to portfolio ***b***. Now consider a new pair of portfolios. Portfolio ***a'*** pays exactly the same as a 50 percent of the time, and it pays of according to portfolio ***c*** otherwise. Similarly, portfolio ***b'*** pays exactly the same as ***b*** 50 percent of the time, and according to portfolio ***c*** otherwise. The IIA assumption requires that you prefer ***a'*** to ***b'*** as well. In other words, portfolio ***c*** is independently mixed into the portfolios in the same way, so it is irrelevant to the individual's ranking.
- Seems reasonable, but this assumption is typically violated in laboratory experiments when people are assessing small probabilities. (Kahneman and Tversky (1979) showed this.)

What remains after this criticism

- While EU theory provides a clear prescription for choice and it provides a clear focus on one determinant of choice (MU of $), it is probably too narrow to fit the data in many settings.
- Alternative theories of choice under uncertainty like prospect theory also have difficulties in that there are arguably too many degrees of freedom.
- Regardless of the criticism in experiments, the previous arguments about the connection between marginal utility of money and attitudes towards risk are compelling.

Mean-variance analysis

- In many finance models, it is assumed that people only care about two statistics -- the mean return of a portfolio and its variance (or standard deviation). The implication is that utility can be represented in mean-variance form:

$$U(\tilde{x}) = v(\bar{x}, \sigma^2)$$

Mean-variance analysis

- How does mean-variance relate to expected utility theory?
  - If $\tilde{x}$ is either normally or log-normally distributed, then preferences have a mean-variance form.
  - If $u(x)$ is quadratic, then preferences have a mean-variance form. Reasonable?
  - If $\tilde{x}$ is normally distributed and $u(x)$ is exponential , then
  $$U(\tilde{x}) = v(\bar{x}, \sigma^2) = \bar{x} - \lambda\sigma^2$$
  I.e., mean-variance indifference curves are linear (i.e., return, variance are perfect substitutes), and $\rho = \lambda\sigma^2$.

Problems of incomplete information

- There are two problems that arise in markets in which some individuals have private information:
  - moral hazard (hidden endogenous actions): if a buyer contracts for a service from a seller but cannot be certain of the effort the seller has provided, the seller has an incentive to shirk on this input to reduce cost. An incentive contract may be able to solve this problem and ensure efficient effort, but incentives will impose residual risk on the seller which is costly in itself. An insurance-incentives tradeoff exists.
  - adverse selection (hidden exogenous information): if a buyer wants to buy an object of unknown quality (the seller knows it, the buyer doesn't), the sellers with the lowest quality goods may be the most eager to sell. (cont.)

Problems of incomplete information

- **Adverse selection, continued**: Indeed, markets may unravel to the point where only the lowest quality goods are traded. The key driver of these sorts of problems has to do with conditional expectations.

Aside: conditional expectations

- We have previously defined the expectation of a random variable with the notation $E[\tilde{x}]$
- Suppose that we can split the population into segments and calculate a more precise expectation for each segment.
  - E.g., suppose the random variable is height of adults, and suppose the expectation is 1.65 meters. If we split the population into two groups, women and men, we will find that the expectation for men is an higher amount than the expectation for women.

    E[height|male] > E[height] > E[height|female]
- Notationally, we add the condition in the expectation bracket after a vertical line.

Adverse selection

- Adverse selection arises when some aspect of a market or other economic transaction differentially attracts some segments over others.
- The classic example is the market for used cars (Akerlof).
  - Suppose that there are three kinds of used cars: great, mediocre , and bad (known as "lemons"). Each exists in an equal proportion in the population of used cars
  - Potential sellers know which kind of car they have to sell; buyers do not (hidden exogenous information)
  - The opportunity cost of a seller for his car is the value he would obtain if he continued driving it instead. Thus, owners of "great" cars have higher cost than owners of "mediocre" cars or "lemons", etc.

Adverse selection, example

- Buyers and sellers have the following valuations:

car type:	seller value:	buyer value:
great	7	10
mediocre	5	6
lemon/bad	0	2

- Within each category, there are gains from trade if the car quality is observable. All car types would trade.
- Suppose that buyers optimistically think that all types are cars are being sold in the used car market. The expected value of a random used car is E[car value] = (1/3)10 + (1/3)6 + (1/3)2 = 6. At a price of p=6, however, "great" car owners prefer not to sell.
- Suppose buyers figure this out. The expected value of used cars for sale is now E[car value|p≤6]=(1/2)(6+2)=4. But at p=4, mediocre cars are not sold. Only "lemons" are offered for sale. The market adversely selects low quality.

Adverse selection

- There are many other examples of adverse selection that are based on the same notion of an economic transaction or market causes the conditional expectation to be different from the original.
  - E.g., winner's curse in auctions:

    Suppose you will bid on the rights to drill oil. You hire a geologist who gives you an estimate of the oil field value:

    $$E[\text{value of oil field}|\text{geology report}]$$

    Your geologist is good and unbiased, but her estimates can be too high or too low.
    - Suppose you win the sealed-bid auction for the oil field. How does your previous expectation compare to

$$E[\text{value}|\text{geology report \& you bid more than others}]$$

Adverse selection

- The winner's curse is that the expectation conditional on winning the auction is lower than your expectation before bidding. If you bid close to your initial expectation and win the auction, you should expect losses.
- Smart bidders (e.g., oil companies, for example) don't make the mistake of over bidding. They anticipate the change in conditional expectations and bid less.
- Smart sellers (who design auctions) may want to choose an auction format that reduces the winner's curse so the smart bidders will bid more.
- In general, public bidding (open-outcry auctions, Sotheby's, etc.) have a smaller winner's curse than sealed-bid auctions and generate more revenue for sellers.

Day 4:
Production, cost minimization, profit maximization

Production functions

- Virtually every good has many inputs, perhaps thousands. For each input, one can ask "How long would it take a firm to change the level of input (up or down)?"
  - **energy, raw materials**?

    perhaps a few days, maybe even a few hours
  - **labor:** line workers, sales force, etc., maybe just a few weeks
  - **labor:** management (operations, production, legal, accountants, HR): months?
  - **production machinery** and equipment: months to years?
  - **physical plant**, buildings: years?
- Clearly, the adjustment time of the inputs varies

Topic 1: Production functions

- For our purposes, we will focus on just two inputs: Labor (L) and Capital (K). This is entirely unrealistic, but like with consumer choice, more than two inputs generalizes nicely.
- The production function indicates how much output, q, is produced given labor (L) and capital (K):

$$q = f(L,K)$$

Production is increasing in each input.

Topic 1: Production functions

- **Time frames:**
  - Labor (L) we assume to be instantly flexible. It can be changed the morning just before production takes place. "Labor" does not imply it is about workers, but rather it is a proxy for quickly adjustable inputs.
  - Capital (K) we assume to require one full period to adjust. E.g., let's say the period is a quarter; then a decision to change capital made at the start of the quarter will be put into effect only at the start of next quarter.
- Production decisions made at the start of the current period that are immediately in effect are short-run decisions.
- Production decisions made at the start of the current period which take effect in the next period are long-run decisions.

Topic 1: Production functions

- In the short run when capital is fixed, we will use K to be clear.
- The marginal product of labor, MP_L, is the additional output produced by the last unit if labor, holding K fixed.

$$MP_L = \frac{\Delta q}{\Delta L} = \frac{\partial f(L, K)}{\partial L}$$

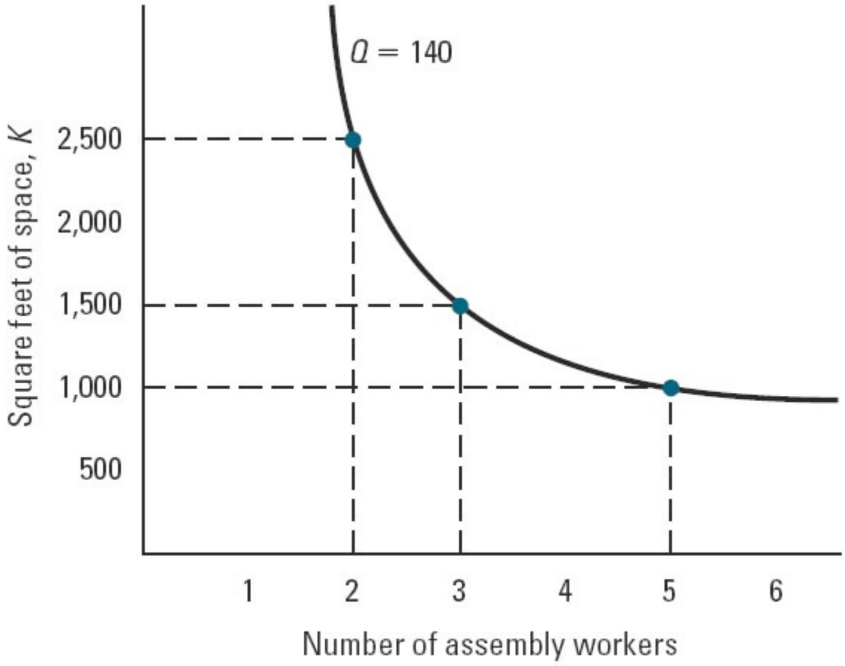
- Similarly, we define the marginal product of capital, MP_K:

$$MP_K = \frac{\Delta q}{\Delta K} = \frac{\partial f(L, K)}{\partial K}$$

Production functions

- Similarities with consumer choice:
  - if we change *(x,y)* to *(L,K)* and change utility functions to production functions, then MU_x and MU_y are analogous to MP_L and MP_K.
- Similarly, the concept of indifference curve has an analogue as an isoquant in production theory. An **isoquant** is a curve representing all combinations of inputs that generate the same output. There is a different isoquant for every level of output.
- Isoquants have the same properties as indifference curves:
  - downward sloping
  - thin and non-intersection
  - slopes give us an exchange rate
  - examples: perfect complements, perfect substitutes, etc.

Isoquant Example



(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

Substitution/exchange rates

- The slope of the isoquant is a rate of substitution for inputs. It is called the marginal rate of technical substitution, but it is otherwise the same as our familiar MRS concept.

$$MRTS_{L,K} = -\left.\frac{\Delta K}{\Delta L}\right|_{\substack{\text{along}\\\text{isoquant}}}$$

- In consumer theory, we used a utility function to obtain an alternative derivation of MRS. Here, we do the same with production functions:

$$MRTS_{L,K} = \frac{MP_L}{MP_K}$$

Returns to scale

- Returns to scale. A new concept for production theory is the idea that expanding scale sometimes increases returns.
  - E.g., suppose all inputs are doubled and output more than doubles, then we say we have increasing returns to scale.
- The mathematical definition is harder to digest, but it says the same thing. For any scaling factor, s, larger than 1, the production function exhibits increasing returns to scale if

$$f(sL,sK) > s\, f(L,K).$$

- If the inequality is reverse, we say that production exhibits decreasing returns to scale.
- If production scales up evenly, $f(sL,sK) = s\, f(L,K)$, we say it has constant returns to scale.
- Some technology (e.g., pipelines, boilers, vats, etc.) has increasing returns to scale from basic geometry of volume.

Opportunity Cost & Cost Functions

- The **opportunity cost** (also known as economic cost) of using a resource for some economic activity (e.g., producing widgets) is the highest value that the resource would have earned in its best alternative use (e.g., producing something else). It is nothing more than the answer to the hypothetical question: “how much would you actually save if you did not use the resource in the activity that you are considering?”
  - Example 1: Suppose you have an inventory of DRAM memory chips that you purchased for \$10/unit a few weeks ago, but the market price has risen to \$20/unit. Suppose that if you do not use them in the laptops you produce, your best alternative use is to sell them on the spot market for \$20/unit. Therefore, the opportunity cost is \$20/unit, not the price you paid of \$10/unit.

Opportunity cost

- Example 2: Suppose that you use a warehouse for storage that cost you $1M to build. The interest rate is 10 percent and the warehouse does not depreciate. If you decide not to use the warehouse this period, the best alternative use is to leave it empty (e.g., you cannot find anyone to rent it in the current period). In the long-run, however, the best alternative use is to sell the warehouse, but it will take 1full period to find a buyer and the price will be only $600K.
  - What is the short-run opportunity cost of the warehouse for the present period? $0
  - In the short run, how much of the $1M warehouse cost is sunk? $1M.
  - What is the long-run opportunity cost? $60,000/period. Why?
  - In the long run, how much of the $1M is sunk? $400K.

Opportunity cost

- **Sunk costs** are the unavoidable or non-recoverable (non-salvageable) expenses on inputs. You always ignore them in economic decision making! The opposite of sunk costs are expenditures which are recoverable or **avoidable**; these (by definition) are opportunity costs. It may be that only a fraction of an expenditure is sunk, like in the warehouse example. Moreover, how much is sunk is always relative to the time frame considered.
  - Sunk versus recoverable costs depend on time horizon
    - textbook model: capital is sunk in the short-run, fully recoverable in long run
    - warehouse example: short run, cost is entirely sunk; long-run only a fraction (40 percent) is sunk
  - Understand and avoid the “sunk-cost fallacy” (i.e., do not consider sunk costs in your decision making)
  - Beyond the scope of this course: if only a fraction of capital costs are recoverable and there is demand uncertainty, then physical capital becomes a “real option”

Total cost function

- We will denote the cheapest cost of producing *q* units by the total cost function: *C(q)*
  - It is denominated in money. E.g., if *q*=100 tons and *C(q)* is $1.5 million, then it costs $1.5 million to produce 100 tons. It is NOT a unit cost. It is a total cost of production.
  - It also represents the smallest cost of production. E.g., another process may produce 100 tons at a cost of $2 million. *C(q)* gives us the cheapest cost. It represents the most efficient mode of production. It is the minimum cost of producing *q* units.
  - In textbooks, *C(q)* is not always an economic (or opportunity) cost function because it may include sunk costs. In that case, when making economic decisions we must remember subtract the sunk costs and focus only on the economics/opportunity cost that remains. In this class we will often refer to the modified cost function which only contains economic (i.e., opportunity) costs with a superscript "*O*" for *opportunity* costs.

Fixed and variable cost components

- It is sometimes useful to decompose a total cost function it into three components:

$$C(q) = F_S + F_A + V(q)$$

The first two components are fixed costs; the third is a variable cost.

  - F_S is a fixed cost that is **sunk**; it must be paid regardless of production, and therefore it is not recoverable. It is not an economic/opportunity cost.
  - F_A is a fixed cost that is recoverable or **avoidable** if zero production is chosen. That is, the firm can avoid this fixed cost by shutting down.
  - *V(q)* is a **variable cost** of production. The value of *V(q)* depends on the amount of production.

- NB: In a sense, F_A is also a variable cost that depends upon q. If q>0, then $F_A > 0$ is paid, but if q=0, then it is as if $F_A = 0$. F_A is sometimes referred to as a quasi-fixed cost. The key to remember is that it is avoidable even in the short run, and therefore an opportunity cost.

Fixed and variable cost components

- Suppose all relevant opportunity costs are included in F_A and $V(q)$
- With our decomposition
$$C(q) = F_S + F_A + V(q)$$
it should now be clear that with respect to opportunity costs, we have (using an "*o*" superscript to denote opportunity costs)
$$C^o(q) = F_A + V(q) \text{ and } C(q) = F_S + C^o(q)$$
- Note, $C^o(0) = 0$, although C^o may jump just beyond *q=0.*
-

Aside: Most textbooks begin by assuming that F_A =0, and they use the notation F in place of F_S.
$$C(q) = F + V(q)$$
Importantly, the F is fixed but it is <u>also sunk</u>! Later, when textbooks allow for avoidable fixed costs, they either introduce notation like F_A, or they embed F_A in V(q).

Fixed and variable cost components

- For now, I will use an "*O*" superscript to indicate that a total cost function is representing opportunity cost.
- Every opportunity cost function has the property that
$$C^o(0) = 0.$$
If this were not the case, then there would be an unavoidable cost of production, and we know that those costs are sunk.
- With a total cost function *C(q)*, note that
$$C^o(q) = C(q) - C(0).$$

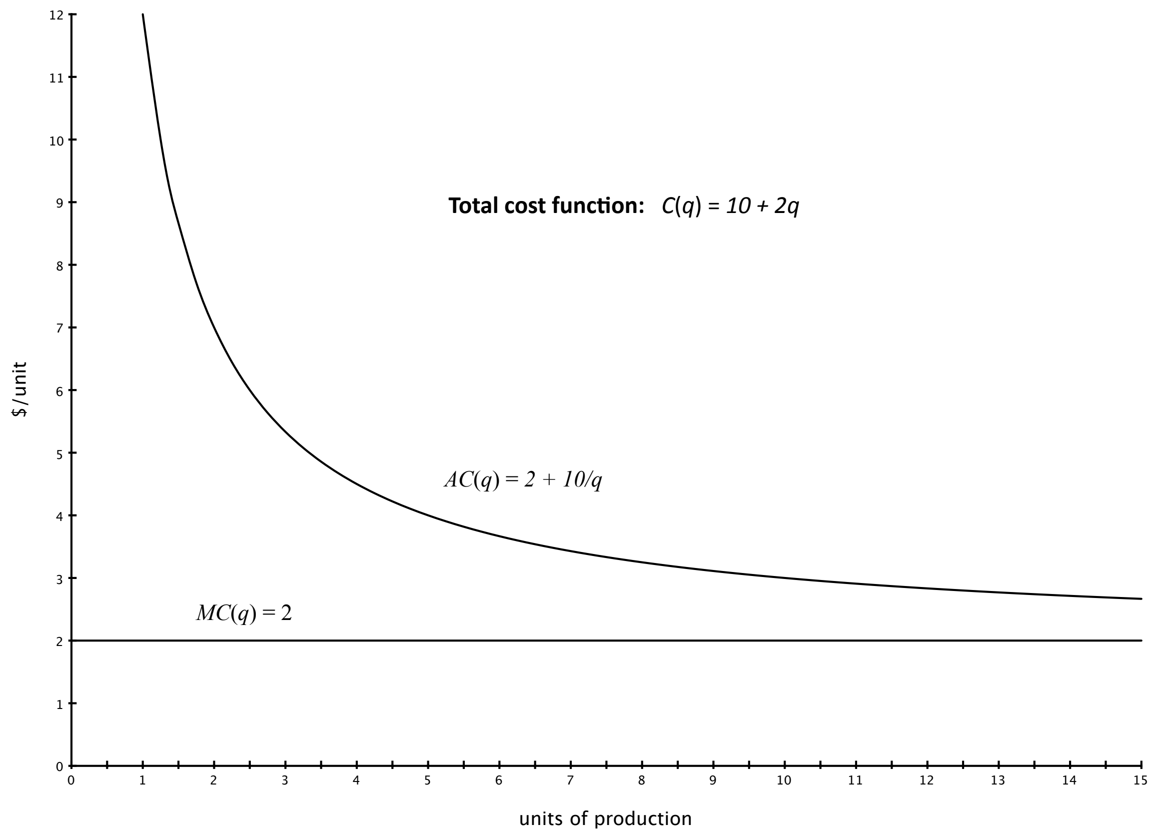
Unit cost functions

- For any total cost function, *C(q)*, we can construct two important unit-cost functions: average cost and marginal cost.
  - Average cost is
    $$AC(q) = \frac{C(q)}{q}$$
    It is measured in money/unit.
  - Marginal cost: The marginal cost function is the derivative of the total cost function. It represents the increase in cost for producing the last unit.
    $$MC(q) = C'(q)$$

Relationship between AC and MC

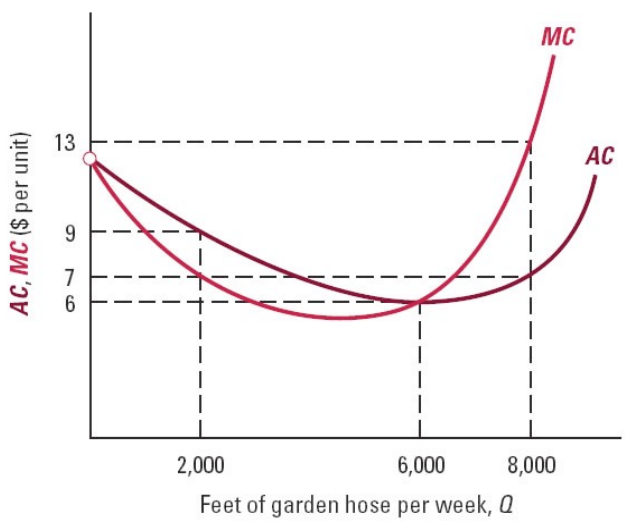
- There is an important relationship between *AC* and *MC:*
  - If $MC(q) > AC(q)$, then average cost is rising in *q*;
    If $MC(q) < AC(q)$, then average cost is falling in *q*.
- The reason for this has nothing to do with costs and everything to do with averages.
  - Suppose we measured the average height of everyone in the classroom. As late “marginal” students come into the lecture, the average height will rise if the person is taller than the average and it will fall if the person is shorter than the average.
  - One implication of this is that if *AC(q)* has a minimum, then necessarily it occurs where *MC* and *AC* cross:
    $$MC(q) = AC(q).$$

A few examples of unit cost curves:



Relationship Between AC and MC

- AC slopes downward where it lies above the MC curve
- AC slopes upward where it lies below the MC curve
- Where AC and MC cross, AC is neither rising nor falling



(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

Relationship Between AC and MC

- The relationship between *MC* and *AC* holds for any pair of unit-cost functions derived from the same total cost function.
- Thus, if we are looking at total opportunity cost, $C^o(q)$, we will find the relationship holds between the marginal opportunity cost and the average opportunity cost functions.

Short-run and Long-run Cost Functions

- The **short-run cost function**, $C_{SR}(q)$, gives the total cost of producing *q* units taking into account that some inputs (e.g., capital) may be fixed in the short-run.
- The **long-run cost function**, $C_{LR}(q)$, gives the total cost of producing *q* units assuming sufficient lead time to adjust inputs that are variable in the long run (but possibly fixed in the short run).
- The **short-run opportunity cost function**, $C^o_{SR}(q)$, gives the total opportunity cost of producing *q* units taking into account that some inputs (e.g., capital) may be fixed in the short-run. E.g., short-run sunk costs are excluded.

Short-run and Long-run Cost Functions

- The **long-run opportunity cost function**, $C_{LR}^o(q)$, gives the total opportunity cost of producing *q* units assuming sufficient lead time to adjust inputs that are variable in the long run (but possibly fixed in the short run).
- In virtually all microeconomics textbooks (including B&B), (1) capital is assumed fixed in the short run and variable in the long run, and (2) capital is bought and sold at the same rental price, *r,* in the long run. As a consequence, whenever we assume these unrealistic assumptions,
  - $C_{SR}(q)$ exceeds $C_{SR}^o(q)$ by the (sunk) cost of capital
  - $C_{LR}(q) = C_{LR}^o(q)$
- The world is more interesting and complex.

Relationship Between AC and MC

- We now have four different total cost functions defined on two dimensions: Accounting cost versus opportunity cost; short-run versus long-run. In terms of notation, we write
$$C_{SR}(q),\ C_{SR}^o(q),\ C_{LR}(q),\ C_{LR}^o(q)$$
- For each of these total cost functions, we can derive a corresponding pair of unit cost functions:
$$AC_{SR}(q),\ MC_{SR}(q),\ AC_{SR}^o(q),\ MC_{SR}^o(q),$$
$$AC_{LR}(q),\ MC_{LR}(q),\ AC_{LR}^o(q),\ \text{and}\ MC_{LR}^o(q).$$
- The relationship between each corresponding pair *MC* and *AC* functions holds as before.

Derivation of (textbook) cost functions

- We first derive $C_{SR}(q)$ and its properties using the textbook assumption that there is only one short-run input, *L*, that can be varied.
- We want to determine the least cost of producing *q* units as a function of economic variables, *w* and *r*, given a production function, *f*, and a fixed level of capital, $\overline{K}$.
- We proceed in two steps:
  - what is the cheapest combination of inputs to produce q?
  - what is the cost of those inputs?

Derivation of short-run cost

- What is the cheapest combination of inputs to produce *q*?
  - This is almost trivial. We should employ the least amount of labor such that
$$q = f(L, \bar{K}).$$
    We can solve this expression for the critical value of labor which we denote $L^*_{SR}(q, \bar{K})$.
    - E.g., suppose $f(L, K) = \sqrt{LK}$. Then we need to solve $q = \sqrt{L\bar{K}}$ for *L*. The solution is
$$L^*_{SR}(q, \bar{K}) = \frac{q^2}{\bar{K}}$$
  - Because there is only one short-run input, there is no interesting substitution available between inputs.

Derivation of short-run cost

- What is the cost of these inputs?
  - Now we just price out the inputs from the previous step.
$$C_{SR}(q) = wL^*_{SR}(q, \bar{K}) + r\bar{K}.$$
  - That is all there is to do in the short-run with one input.
  - E.g. for the production function in our previous example
$$C_{SR}(q) = w\frac{q^2}{\bar{K}} + r\bar{K}$$
- With more than one input, the analysis will be similar to the long-run cost minimization analysis we will see; i.e., we must use input prices and *MRTS* to choose the best input mix with more than one variable input.

Derivation of short-run cost

- Notice at this stage the total cost function includes the sunk capital costs.
$$C_{SR}(q) = wL^*_{SR}(q, \bar{K}) + r\bar{K}.$$
  - Using our previous notation,
$$F_S = r\bar{K} \quad \text{and} \quad V_{sr}(q) + F_A = wL^*_{SR}(q, \bar{K}),$$
  where we have added the subscripts "sr" to the *V*(*q*) function to make clear it is derived for the short-run.
  - Hence, we have
$$C_{SR}(q) = F_S + F_A + V_{sr}(q)$$
- What about short-run unit cost functions?

Derivation of short-run unit cost functions

- Our short-run cost function is

$$C_{SR}(q) = F_S + F_A + V_{sr}(q)$$

and our short-run opportunity cost function is

$$C^o_{SR}(q) = wL^*_{SR}(q, \bar{K}) = F_A + V_{sr}(q)$$

- Average cost functions are constructed by dividing by q:

$$AC_{SR}(q) = \frac{C_{SR}(q)}{q} \quad \text{and} \quad AC^o_{SR}(q) = \frac{C^o_{SR}(q)}{q}$$

- Note that

$$AC^o_{SR}(q) = \frac{F_A + V_{sr}(q)}{q}$$

When FA = 0,

$$AC^o_{SR}(q) = \frac{V_{sr}(q)}{q} = AVC(q)$$

Beware: Textbooks often implicitly assume F_A =0 and use *AVC* when $AC^o_{SR}(q)$ is the appropriate concept.

Derivation of short-run unit cost functions

- Our short-run cost function is

$$C_{SR}(q) = F_S + F_A + V_{sr}(q)$$

- The short-run marginal cost function does not depend upon the existence of fixed costs; it depends only on the variable cost function. Hence, we have

$$MC_{SR}(q) = MC^o_{SR}(q) = MVC(q) = V'_{sr}(q).$$

- Can we say more?
  - How much extra input, L, do you need to make one unit of output?

$$\Delta q = MP_L \cdot \Delta L$$

Thus,

$$\frac{\partial L^*_{SR}(q, \bar{K})}{\partial q} = \frac{\Delta L}{\Delta q} = \frac{1}{MP_L}.$$

  - How much does this cost?

$$MC_{SR}(q) = w\frac{1}{MP_L}$$

Derivation of short-run unit cost functions

- So we have short-run marginal cost is

$$MC_{SR}(q) = \frac{w}{MP_L}$$

What does this mean?

  - Let's take a simple example. Suppose you drive a car that gets 15 km/liter and the price of petrol is £1/liter. In the short run, petrol is your only operating expense. What is your marginal cost of driving 1 kilometer?
  - First, ask how much petrol is needed to drive 1km. The answer is 1/(15km/liter)=0.067 liters/km.
  - Next, ask how much that amount of petrol will cost:

    (1£/liter)(0.067 liters/km)=£0.067/km. That's the answer.

    Do you recognize the components in the expression for short-run MC?

Derivation of long-run cost

- Everything is more interesting because we have more than one input to choose. Marginal rates of substitution matter.
- Given *q*, we ask (1) what is the least-cost combination of inputs to achieve *q*, and then (2) we ask how much do those inputs cost.
- Least-cost combination?
  - The problem is very much like that of consumer choice. Instead of starting with a cost constraint and maximizing output (i.e., finding the highest iso-quant that touches the input budget set), the problem is reversed. We start with an iso-quant giving *q*, and then we find the cheapest budget line that touches it.
  - These "budget" lines are called iso-cost lines. The formula for an iso-cost line with cost C is $C = wL + rK$.

Derivation of long-run cost

- The formula for an iso-cost line with cost C is

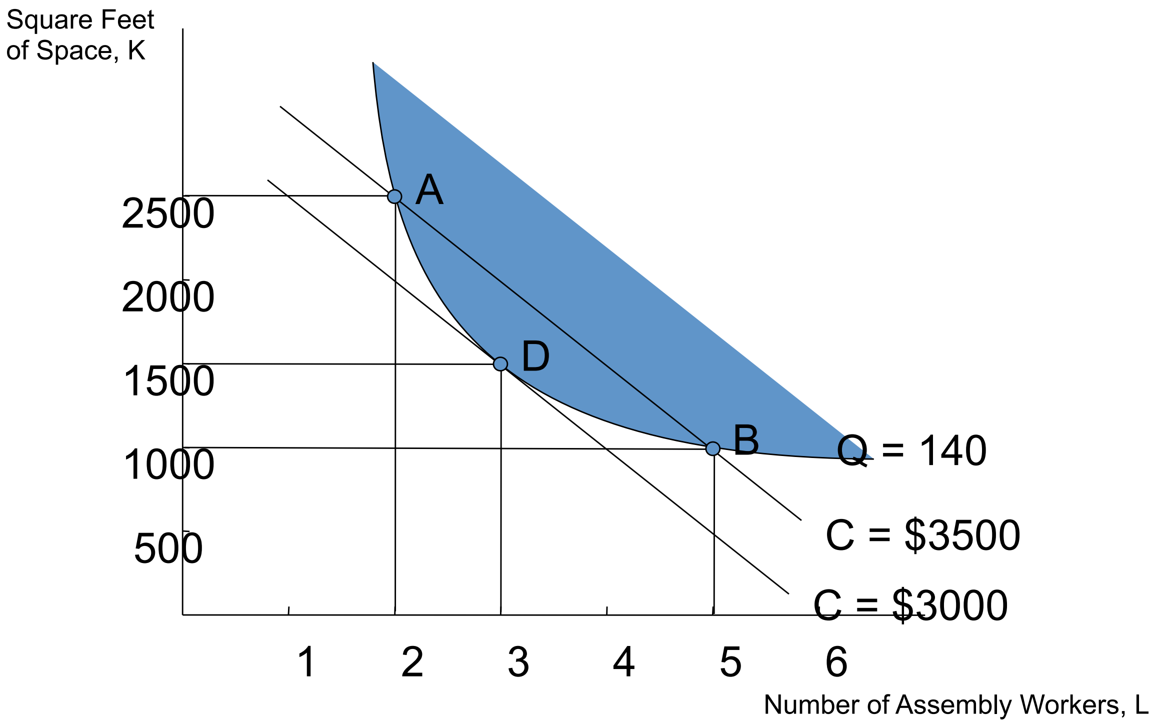
  $$C = wL + rK.$$

  Solving for K, we have $K=C/r - (w/r)\,L.$
  - The slope of each iso-cost line is $-w/r$, the input-price ratio
- Conditions for cost minimization:
  - No waste condition. Produce exactly q, not more.

    $$q = f(L, K).$$
  - No overlap/tangency condition. As in consumer theory, wherever possible, choose inputs so that

    $$MRTS_{L,K} = \frac{w}{r} \quad \text{or alternatively} \quad \frac{MP_L}{MP_K} = \frac{w}{r}.$$
  - Alternative interpretations again: equating rates of return, arbitrage across markets, etc.

Least-Cost Method, No-Overlap Rule Example



(figure from from Bernheim, Whinston, Microeconomics, 1st ed, 2008)

Derivation of long-run cost

- The two conditions for cost minimization give us the cost-minimizing inputs,
$$L^*_{LR}(q) \text{ and } K^*_{LR}(q).$$
- The long-run cost function is therefore
$$C_{LR}(q) = wL^*_{LR}(q) + rK^*_{LR}(q).$$
- Because capital can be bought and sold at *r*, all costs are avoidable in the long-run, and therefore
$$C_{LR}(q) = C^o_{LR}(q).$$
- More realistically, some component of capital is sunk in the long run. If so, remember to subtract it to obtain opportunity cost.

Derivation of long-run unit cost functions

- The long-run average cost function is derived as standard:
$$AC_{LR}(q) = \frac{C_{LR}(q)}{q}$$
- The long-run marginal cost function is found via differentiation of the input functions. Is there a simple economic intuition like we had in the case of short-run MC?
  - If the firm increased production using only labor, then the same short-run argument would apply and
$$\boxed{MC_{LR}(q) = \frac{w}{MP_L}}$$
  - Similarly, if the firm increased output using only capital,
$$MC_{LR}(q) = \frac{r}{MP_K}$$
  - But the no-overlap/tangency condition tells us that the right-hand sides of these expressions are identical!
  - Thus,
$$MC_{LR}(q) = \frac{w}{MP_L} = \frac{r}{MP_K}$$

Derivation of long-run unit cost functions

- Let's be very clear: the equation for marginal cost is the same for both short-run and long-run costs. The values of these functions will generally be different. Why?
  - Remember that the marginal products are evaluated at a specific input combination. In general, to produce a specific *q*, the short-run input combination with some fixed *K*, will not be the same as the long-run input combination. If the given short-run capital happens to equal the long-run cost-minimizing capital level, then the value of the marginal costs will be the same:
  - That is, if at some *q*

$$L^*_{SR}(q, \bar{K}) = L^*_{LR}(q) \text{ and } \bar{K} = K^*_{LR}(q)$$

then

$$MC_{SR}(q) = MC_{LR}(q)$$

Comparing long and short-run unit costs

- The previous derivations provide a few useful comparisons that can ultimately be summarized in a single graph.
  - For a given capital stock, $\bar{K}$, define $\bar{q}$ as the output for which the capital stock is at the long-run optimal level:

$$\bar{K} = K^*_{LR}(\bar{q})$$

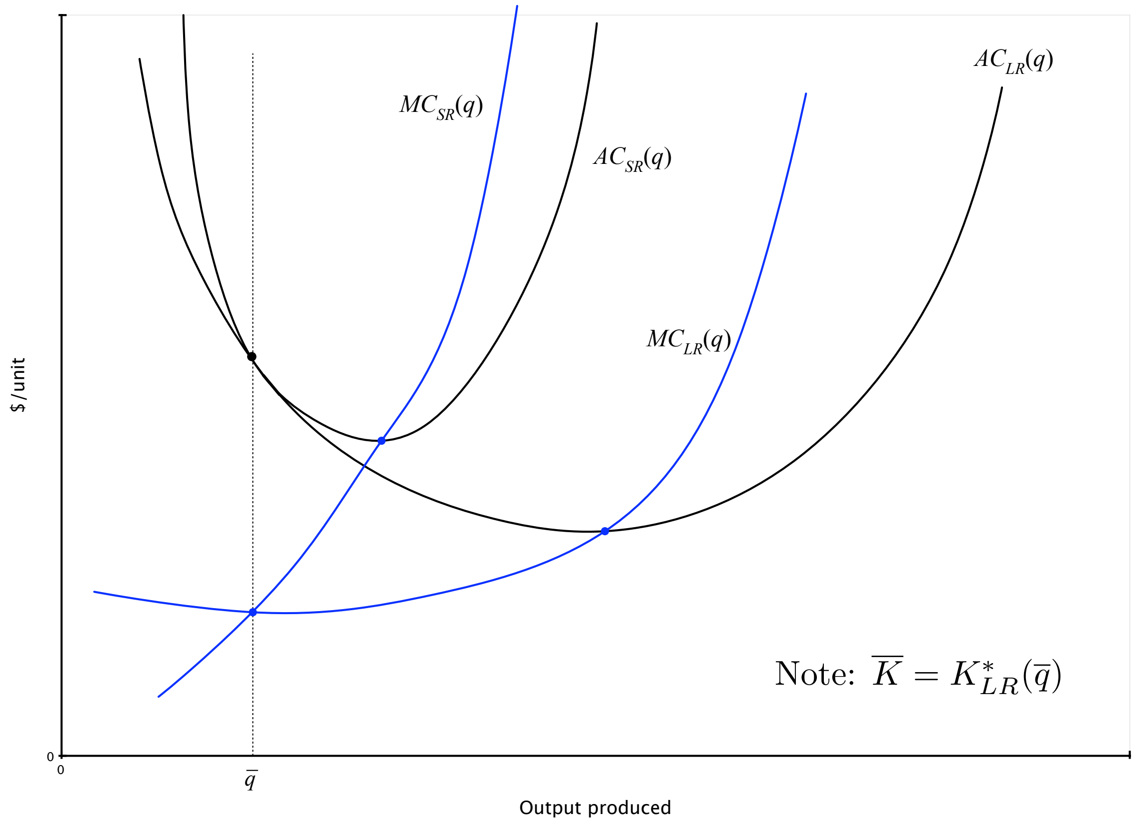
  - Then we know that at $\bar{q}$,

$$C_{SR}(\bar{q}) = C_{LR}(\bar{q}) \implies AC_{SR}(\bar{q}) = AC_{LR}(\bar{q})$$

and we know that

$$MC_{SR}(\bar{q}) = MC_{LR}(\bar{q})$$

Comparing long and short-run unit costs



Detour: A simple example of a real option

- Suppose that you have an opportunity to invest K=£1600 which can be exercised either this period or next.
- The project generates *p* in returns each period forever. *p* is presently unknown.
  - with a 50-50 probability it will either be *p*=£100 or *p*=£300
  - the discount rate is 10%
- Once p is determined, it will remain constant forever. Thus, the present value of the returns is either £1100 or £3300.
- Standard NPV analysis says invest immediately because

  0.50(1100)+0.50(3300)-1600=600 > 0.

  This is the correct answer if your choice is between investing today and never investing. But, you have an option.

Real options, continued

- Suppose you wait one period to learn p and only then make your investment decisions. Upon learning that p=100, you choose not to invest; if p=300, you invest. Your PV is now

  $$0.50(3300-1600)/(1+.10) = 773 > 600.$$

- In essence, the value of having the flexibility for waiting is £773-600=173.
- To correctly perform the NPV analysis, you would have to charge yourself £773 as an opportunity cost for exercising the option in the first period.

  $$NPV=0.50(1100)+0.50(3300)-1600-773 = -173 < 0.$$

- Indeed, the upshot of this analysis is that the option component of the opportunity cost can be huge and decisive.
- If you are interested in these issues, take a look at Dixit and Pindyck's book, Investment under Uncertainty, ch 2 (on chalk)

Other useful cost concepts

- **Economies of scale**. If average cost is decreasing, there are economies of scale in that region of output. If average cost is increasing, then there are diseconomies of scale in the region.
  - More useful idea than returns to scale as it includes the possibility that different levels of production should be undertaken with different input ratios.
  - Example. Mutual funds. Latzko article.
- Related idea: **minimum efficient scale (MES)**. The MES is generally defined to be the minimum point on the long-run average cost curve. If there is no minimum, it is usually defined as the level at which virtually all significant economies of scale have been realized (e.g., 99% of the economies, etc.)

Topic 4: Other useful cost concepts

- **Economies of scope**. Like economies of scale, but applied to multiple product lines. Synergies.
  - Simple example with two products:

$$C(q_1, q_2) < C(q_1, 0) + C(0, q_2)$$

- **Learning/experience curves** (learning by doing). Arises when current total cost function depends upon cumulative output produced in recent past.

$$C(q_t, Q_{t-1}), \quad \text{where} \quad Q_{t-1} = \sum_{i=t_0}^{t-1} q_i$$

  - E.g., Aerospace production, Argote-Epple article.

Definitions of economic profit, etc.

- We will continue with our assumption that there are two periods of analysis: short-run and long-run.
- Note: we will have two decisions to make in every period
  - This period, you decide how much to produce to maximize short-run profits. Current production decision.
  - This period you also make a planning decision for the long run (next period). Specifically, we will make a decision today about the capital stock you will have next period. Future production decision.
- So what exactly do we maximize? Economic profit. Economic profit is revenues minus opportunity cost. We denote the revenue function by R(q) and denote economic profit as

$$\pi(q) = R(q) - C^o(q)$$

Economic profit

- Economic profit

$$\pi(q) = R(q) - C^o(q)$$

  is a net benefit concept, like net present value.
  - An economic profit that is zero is not a bad thing.If the profit is positive, you make strictly positive producer surplus (another term for economic profit); if it is negative, you are better off employing your inputs in their best alternative uses.
- The textbook unfortunately, uses $\pi(q)$ to mean accounting profits. In the long-run, given the assumption of no sunk costs, there is no difference between accounting profit and economic profit (producer surplus). In the short run, accounting profits and economic profits diverge. We will not use the concept of accounting profit in this class.

Economic profit

- With respect to the different time frames, we will be precise and use the notation

$$\pi_{SR}(q) = R(q) - C^o_{SR}(q)$$
$$\pi_{LR}(q) = R(q) - C^o_{LR}(q)$$

- At the start of each production period, *q* is chosen to maximize short-run economic profit and *K* is chosen to maximize long-run economic profit.

More about revenue functions

- What is the revenue function telling us?
- $R(q)$ gives the amount of revenue a firm earns from sales of q units. Embedded in this function are the interesting aspects of competition:
  - Under perfect competition, the firm is a price taker. If the market price is p, then the revenue function is simply

    $$R(q) = p\,q.$$

    Thus, $MR(q) = p$ and $AR(q) = p$. Simple.
  - Now suppose that the firm is the only producer in the market and that the market demand curve is $p=P(q)$. Note, I am using capital letters to denote functions. We have

    $$R(q) = P(q)\,q,$$

    $$MR(q) = P(q) + P'(q)\,q.$$

    [Note: This should look familiar from our topic connecting the relationship between market revenues and equilibrium output to the elasticity of demand.]

Rules for profit maximization

- There are two rules for profit maximization:
  - Marginal output rule: if you are going to produce any output, it should be at a point where

    $$MR(q^*) = MC(q^*)$$

  - Shutdown/operation rule. You should produce the best output derived from the marginal output rule only if

    $$\pi(q^*) \geq 0.$$

- These rules are applied to find the current production level and the future capital level.
- How and why do these rules work?

Rules for profit maximization

- Marginal output rule:
  - If *MR* > *MC*, then increasing output yields additional revenues that more than cover additional costs. Profit rises from increasing *q*.
  - If *MR* < *MC*, then reducing output causes a loss in revenues that is less than the gain in reduced cost. Profit rises from decreasing *q*.
  - Different decision horizons may have different answers:

$$MR(q^*_{SR}) = MC_{SR}(q^*_{SR})$$
$$MR(q^*_{LR}) = MC_{LR}(q^*_{LR})$$

Rules for profit maximization

- Shutdown. It is entirely possible that the avoidable fixed costs so large that producing anything at all generates losses.
  - Time frames. It is possible that it is optimal to shutdown in the current period but operate in the long run. (E.g., deciding to enter a new market.) It is also possible that operating today but shutting down in the long run is optimal.
  - Restatement of rule: (using price and average cost)

$$\pi_{SR}(q^*_{SR}) \geq 0 \iff \frac{R(q^*_{SR})}{q} \geq AC^o_{SR}(q^*_{SR}) \iff p \geq AC^o_{SR}(q^*_{SR})$$
$$\pi_{LR}(q^*_{LR}) \geq 0 \iff \frac{R(q^*_{LR})}{q} \geq AC^o_{LR}(q^*_{LR}) \iff p \geq AC^o_{LR}(q^*_{LR})$$

NB: we use the appropriate average opportunity cost here.

Numerical example

- Example: Suppose *R(q) = q(10 - q)*. Thus, *MR(q) = 10 - 2q.* [This is the derivative of *R(q)*.] The total cost functions are given by

$$C_{SR}(q) = 10 + 4q, \text{ but } C^o_{SR}(q) = 4q$$
$$C_{LR}(q) = C^o_{LR}(q) = 10 + 4q.$$

  These are very simple cost functions. The fixed component 10 is sunk in the short run but recoverable in the long run.

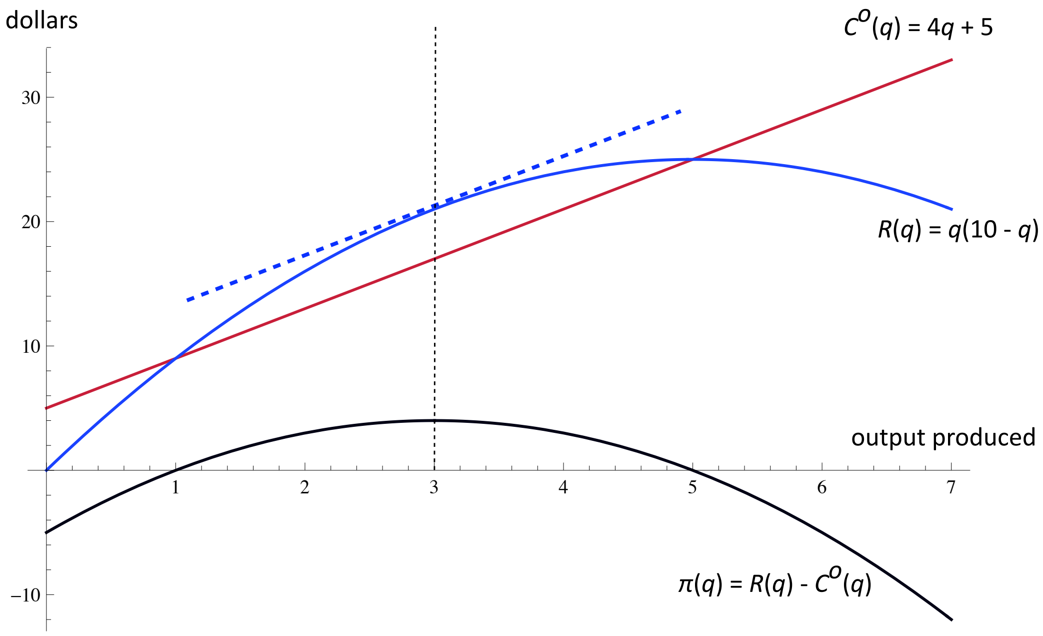
- Short run decision:
  - marginal output rule:
  $$MR(q_{SR}) = MC_{SR}(q_{SR}) \implies 10 - 2q_{SR} = 4 \implies q^*_{SR} = 3.$$
  - shutdown/operation rule:
  $$\pi_{SR}(q^*_{SR}) = 3(10-3) - 3 \cdot 4 = 21 - 12 = 9 > 0.$$
  Therefore, produce $q^*_{SR} = 3$ this period.

Numerical example

- Long run decision:
  - marginal output rule:
  $$MR(q_{LR}) = MC_{LR}(q_{LR}) \implies 10 - 2q_{LR} = 4 \implies q^*_{LR} = 3.$$
  This is the same in the short-run because we the marginal cost functions are identical in the example.
  - shutdown/operation rule:
  $$\pi_{LR}(q^*_{LR}) = 3(10-3) - 10 - 3 \cdot 4 = 21 - 10 - 12 = -1 < 0.$$
  Therefore, it is better to shutdown in the long run. Hence, the correct long-run decision is not $q^*_{LR} = 3$, but instead $q^*_{LR} = 0$.

Numerical example

- Let's change our example a little and graph it. Suppose that the opportunity cost of production is $C^O(q) = 5 + 4q$.



Another numerical example

- Example: Suppose $R(q) = 200q$. Thus, $MR(q) = 200$. The total cost function (same in short and long runs) is given by

$$C^o_{SR}(q) = C^o_{LR}(q) = 4000 + q^2.$$

Here, marginal cost (the derivative of the cost function) is $MC(q)=2q$.

- Profit-maximizing output:
  - marginal output rule:

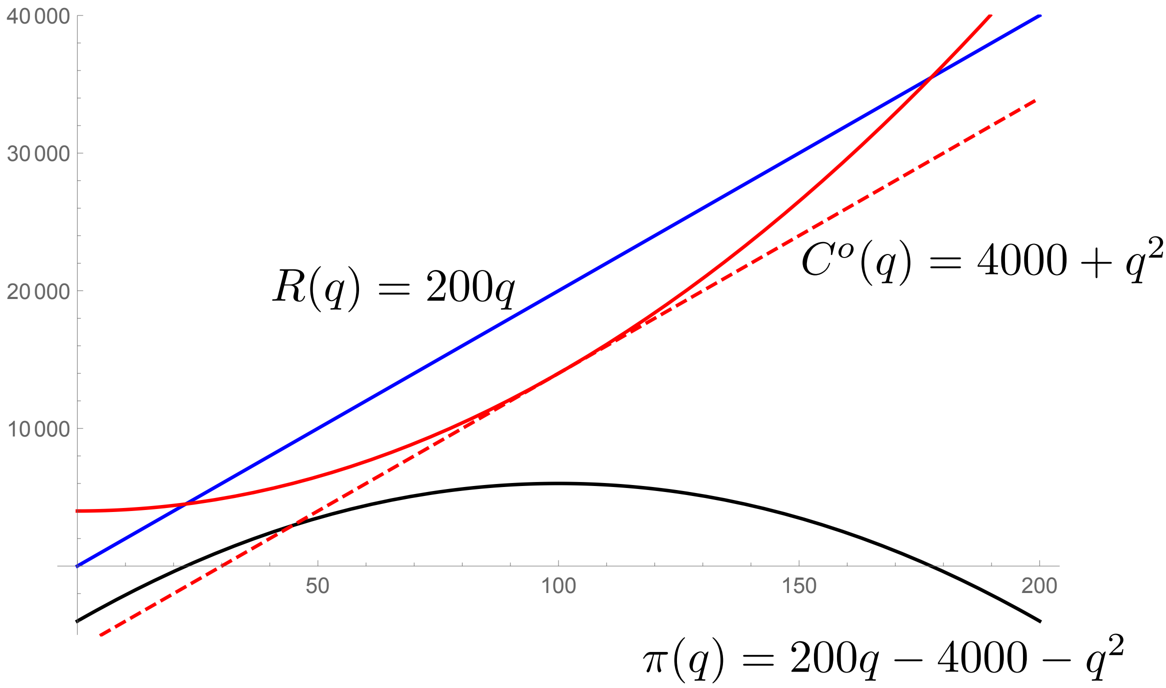
$$MR(q) = MC(q) \implies 200 = 2q \implies q^* = 100.$$

  - shutdown/operation rule:

$$\pi(q^*) = 200(100) - 4000 - (100)^2 = 6000.$$

Therefore, produce 100 units.

Another numerical example



Applying the rules in practice

- The article “Airline takes the marginal route” (1963) emphasizes the short-run decision and argues that you should ignore the cost of capital when making short-run production decisions.
- The article “Real key to creating wealth” (1993) focuses on the long run: make certain that each unit of capital creates enough value to justify its opportunity cost; otherwise, sell the capital.
- There is no contradiction between these articles, although a quick reading might suggest otherwise.
- George Burns (comedian) once quoted a business friend who said, “I lose money on each unit I sell, but I make up for it in volume.” Does the above discussion show that there is some sense to this statement?

Day 5:
Supply functions and competitive equilibrium, revisited

Perfect Competition (review)

- When we refer to the model of perfect competition, we are referring to singular assumption that buyers and sellers are price takers.
  - Some textbooks argue that perfect competition further requires there be a very large number of buyers and sellers. Unquestionably, this makes it more likely that buyers and sellers act as price takers. But it is also possible that there is price taking behavior with just a few firms.
  - Textbooks usually add the requirement that entry and exit is unrestricted. But again, even if there are entry-exit restrictions (e.g., the government only allows a few firms to operate), providing there is enough competition so that incumbent firms are price takers, the competitive model correctly describes the equilibrium price and allocation. With entry-exit restrictions, it is as if all but a few firms have infinite costs of production and so do not enter; removing the entry restriction has the effect of lowering these costs, so the equilibrium may be improved upon by removing the restrictions.

Perfect Competition (review)

    - Finally, some textbooks reserve the definition of perfect competition to the case of homogenous firms (e.g., constant or increasing cost industries). In this case, we get the further property that firms do not make positive economic profits. But I do not know of a single price-taking market for which sellers are homogenous. Typically, firms have different advantages, core competencies, geographical locations, etc.

- In sum, it is most useful to organize our thinking by

  "Buyer and Seller price-taking = perfect competition"

  Thus, we include the possibility of a perfectly competitive market with differences between firms, government restrictions, smaller number of highly competitive firms, etc.

    - Of course, when considering government restrictions, we must modify the conclusion that the competitive equilibrium is efficient because such restrictions implicitly affect costs.

Overview of Perfect Competition

- The main upshot of perfect competition is price-taking behavior which in turn gives us a simple formula for marginal revenue:

$$MR(q) = p.$$

- We can apply our usual rules for profit maximization using this relationship for marginal revenue.

Derivation of supply functions

Roadmap of supply function analysis:

- Derivation of individual firm supply functions
  - short run
  - long run
- Derivation of market supply functions
  - short run
  - long run
- Examples:
  - short run Aluminum supply (circa 1993) using real data
  - constant-cost industry, both short run and long run

Recall: Two rules for profit maximization

- Marginal output rule: if the firm produces, it should choose outputs for each time horizon such that

$$MR(q^*_{SR}) = MC_{SR}(q^*_{SR}),$$
$$MR(q^*_{LR}) = MC_{LR}(q^*_{LR}).$$

- Operation/shutdown rule: at the most profitable positive outputs (which satisfy the marginal output rules above) ...

$$\begin{array}{lcll} AR(q^*_{SR}) & > & AC^o_{SR}(q^*_{SR}) = AVC(q^*_{SR}) & \Longrightarrow \quad \text{operate in short run,} \\ AR(q^*_{SR}) & < & AC^o_{SR}(q^*_{SR}) = AVC(q^*_{SR}) & \Longrightarrow \quad \text{shutdown in short run;} \end{array}$$

$$\begin{array}{lcll} AR(q^*_{LR}) & > & AC^o_{LR}(q^*_{LR}) = AC_{LR}(q^*_{LR}) & \Longrightarrow \quad \text{operate in long run,} \\ AR(q^*_{LR}) & < & AC^o_{LR}(q^*_{LR}) = AC_{LR}(q^*_{LR}) & \Longrightarrow \quad \text{shutdown in long run.} \end{array}$$

Derivation of individual firm's supply curve

- We will apply the two rules of profit maximization. Because we are considering markets with perfect competition, we have the simplification

$$MR(q) = AR(q) = p.$$

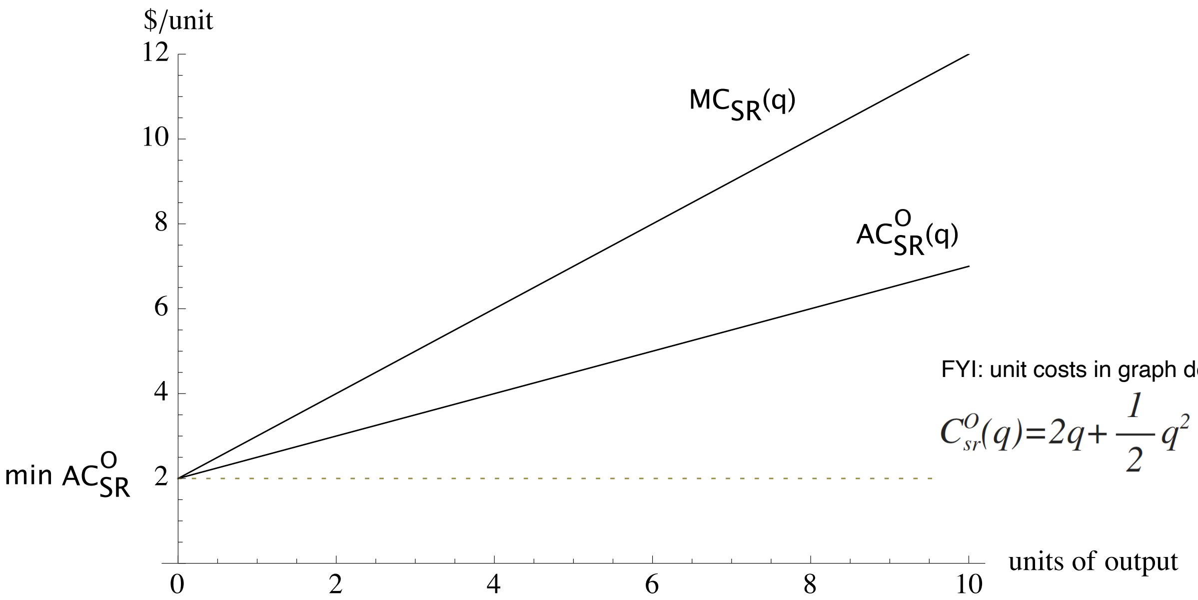
- Short-run profit maximization rules:
  - if producing positive output, then $p = MC_{SR}(q^*_{SR})$.
  - produce only if $p \geq AC^o_{SR}(q^*_{SR}) = AVC_{SR}(q^*_{SR})$.

- Long-run profit maximization rules:
  - if producing positive output, then $p = MC_{LR}(q^*_{LR})$.
  - produce only if $p \geq AC^o_{LR}(q^*_{LR}) = AC_{LR}(q^*_{LR})$.

- Given a firm's unit cost curves, it is straightforward to apply these rules and derive the firm's optimal supply response.

A firm's short-run supply curve

- Short-run unit cost curves for a firm with zero *MES* (i.e., $AC^O_{SR}(q)$ is lowest at $q=0$):



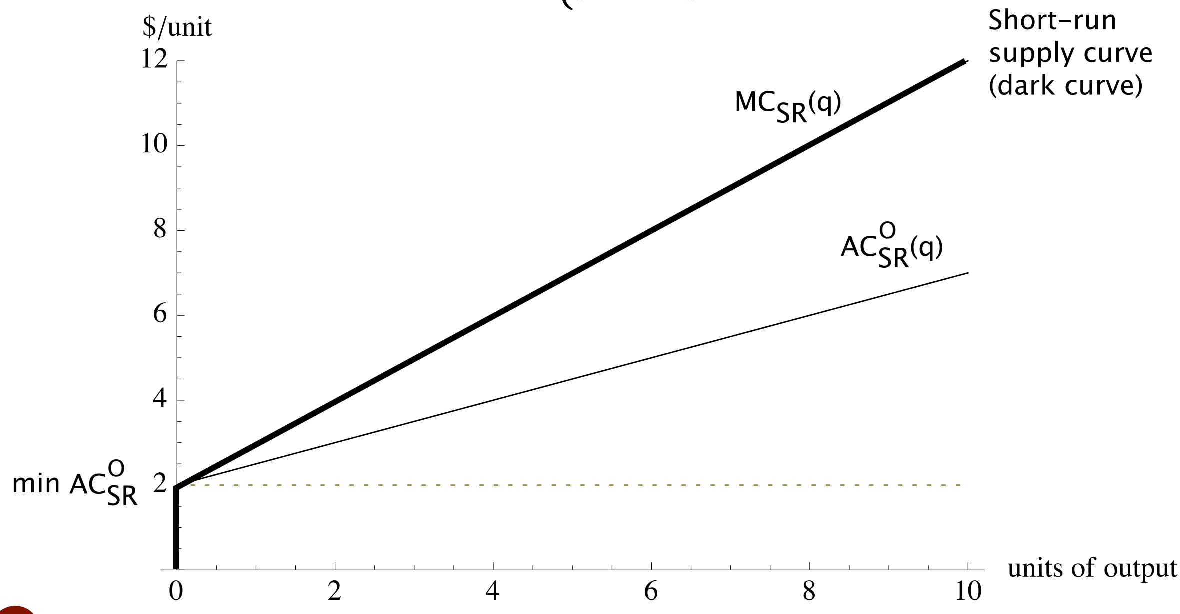
FYI: unit costs in graph derived from

$$C^O_{sr}(q) = 2q + \frac{1}{2}q^2$$

A firm's short-run supply curve, continued

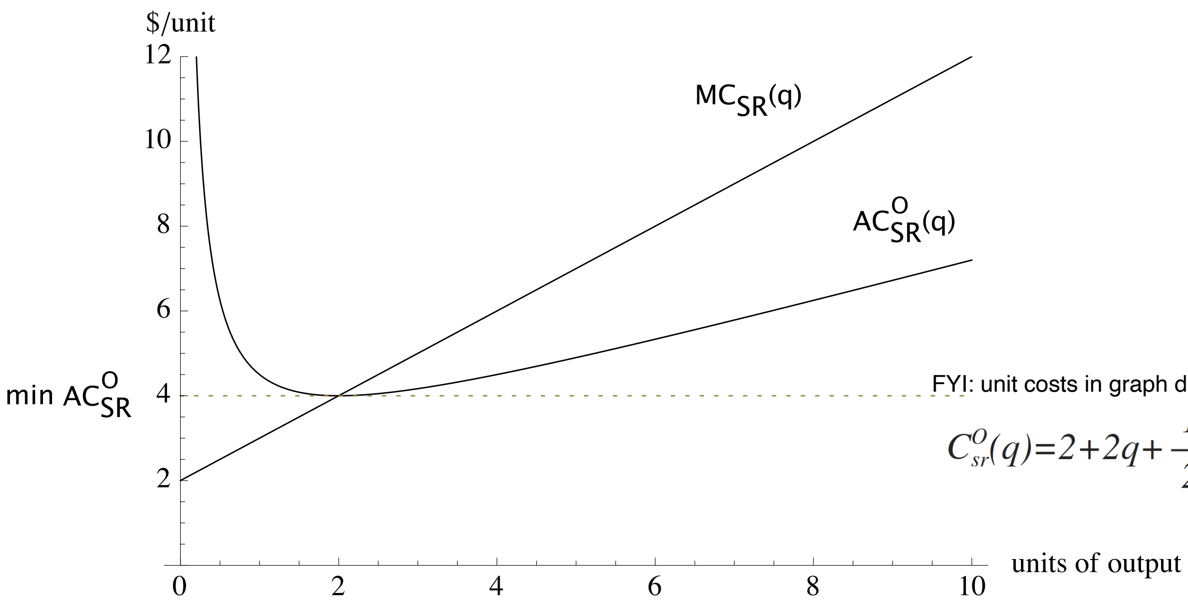
- Short-run supply curve for a firm with zero *MES* (i.e., $AC^{O}_{SR}(q)$ is lowest at $q=0$):

$$q^s = \begin{cases} 0 & \text{if } p < 2 \\ p-2 & p \geq 2 \end{cases}$$



A firm's short-run supply curve, continued

- Short-run unit cost curves for a firm with positive *MES* (i.e., $AC^{O}_{SR}(q)$ is lowest at a positive level of output):

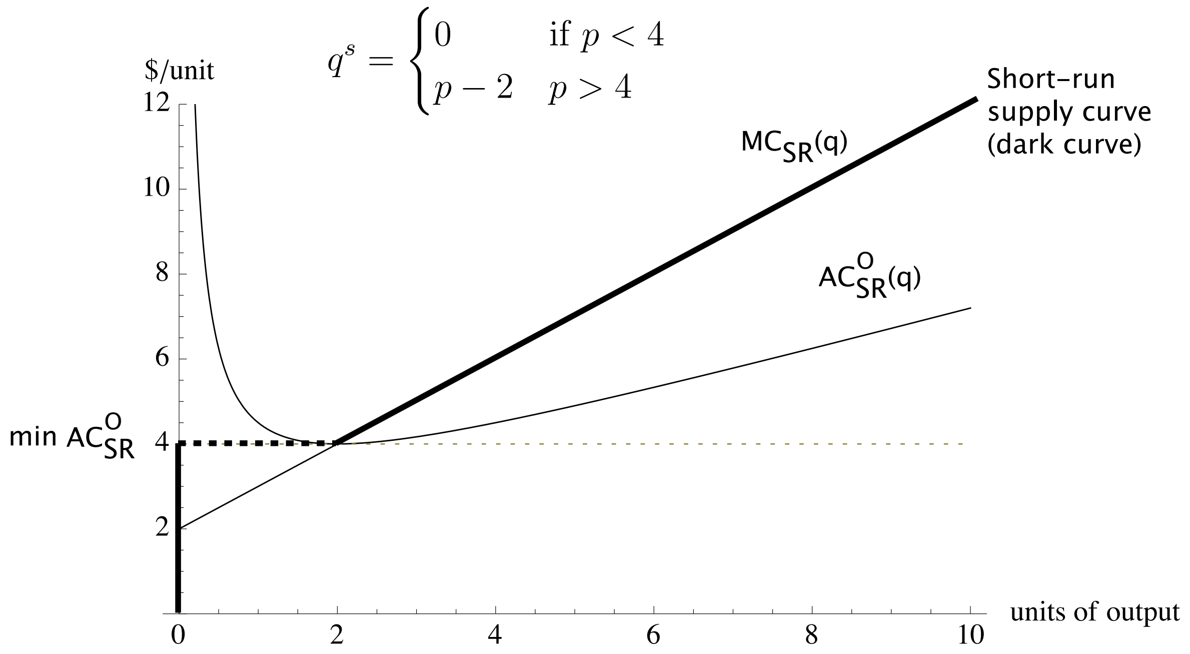


FYI: unit costs in graph derived from

$$C^{O}_{sr}(q) = 2 + 2q + \frac{1}{2}q^2$$

A firm's short-run supply curve, continued

- Short-run supply curve for a firm with positive *MES* (i.e., $AC^{O}_{SR}(q)$ is lowest at a positive level of output):



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Aside:

- Question: Will a competitive firm ever produce in a region where there are economies of scale?

- Answer: NO. A firm produces at $p = MC(q^*)$ if and only if it makes nonnegative profits, so $p > AC^o(q^*)$. It thus follows that

$$p = MC(q^*) \geq AC^o(q^*).$$

Because *MC* is not less than *AC*, it can not be the case that *AC* is falling.

- This will not necessarily be true for a monopolist.

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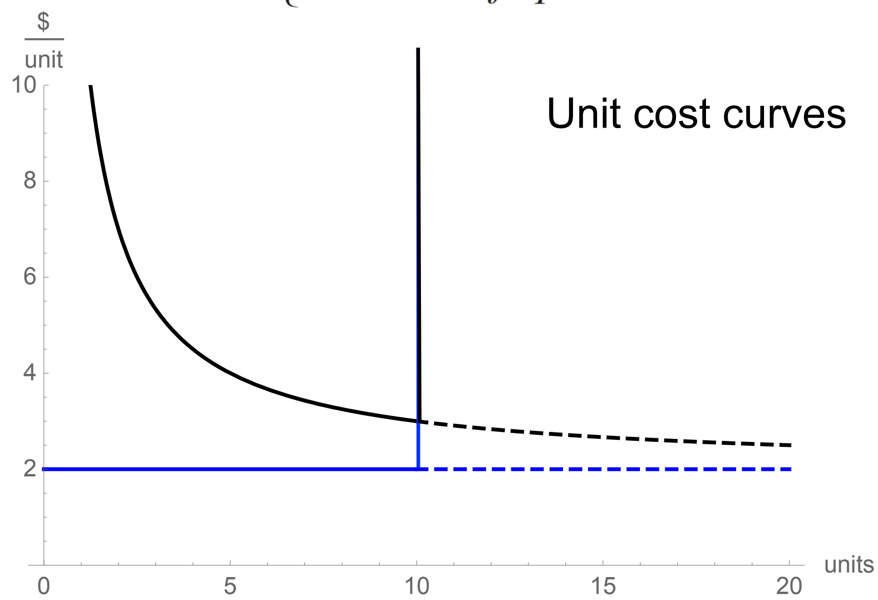
Another example of firm's SR supply

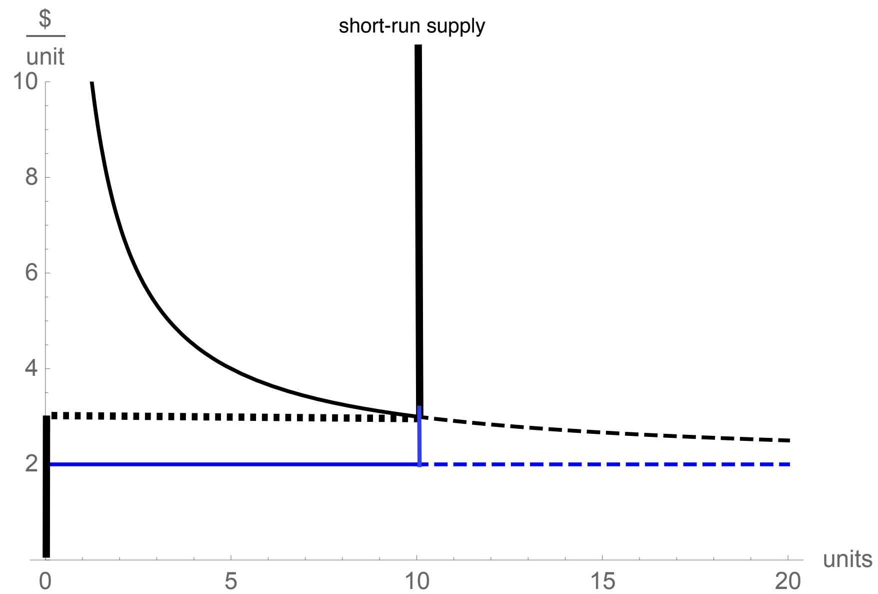
- Suppose that a firm's short run cost is given by

$$C_{sr}^{O}(q)=\begin{cases} 0 & \text{if } q=0 \\ 10+2q & \text{if } 0<q\leq 10 \\ \infty & \text{if } q>10 \end{cases}$$



Another example of firm's SR supply

- It is still the case that MC intersects AC at its minimum.
- The firm's supply curve is again defined by minimum AC.



A firm's long-run supply curve

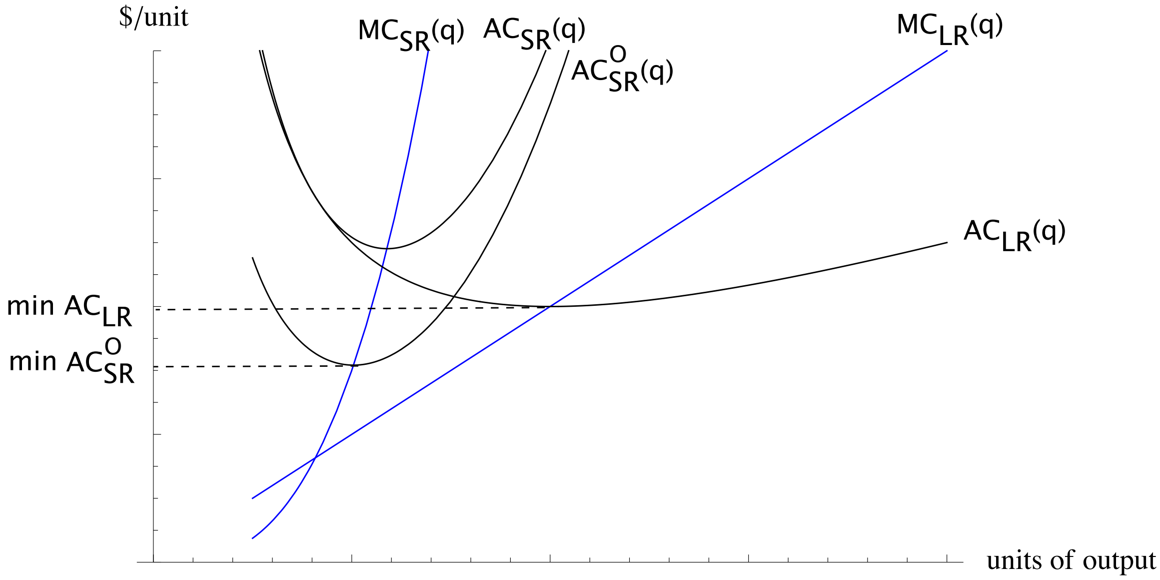
- The graphical analysis is the same as in the short-run setting, except for the use of long-run marginal and average cost functions to derive the supply curve.
- A firm's long-run supply curve is zero for all prices below its long-run minimum average cost, and equal to its long-run marginal cost curve for all prices exceeding the long-run minimum average cost.

Summary of firm's supply curves

- In both the short-run and long run, note that the supply curve is entirely determined by the firm's relevant unit cost functions.
- The supply curve summarizes the relevant unit cost of production for the firm. This deserves emphasis.
  - For the units of production between 0 and the firm's *MES*, the relevant unit cost is the minimum average cost of production; in a sense, the "marginal cost" of the *MES* output is the minimum average cost;
  - For the units exceeding the *MES* output, the relevant unit cost is the marginal cost of production.

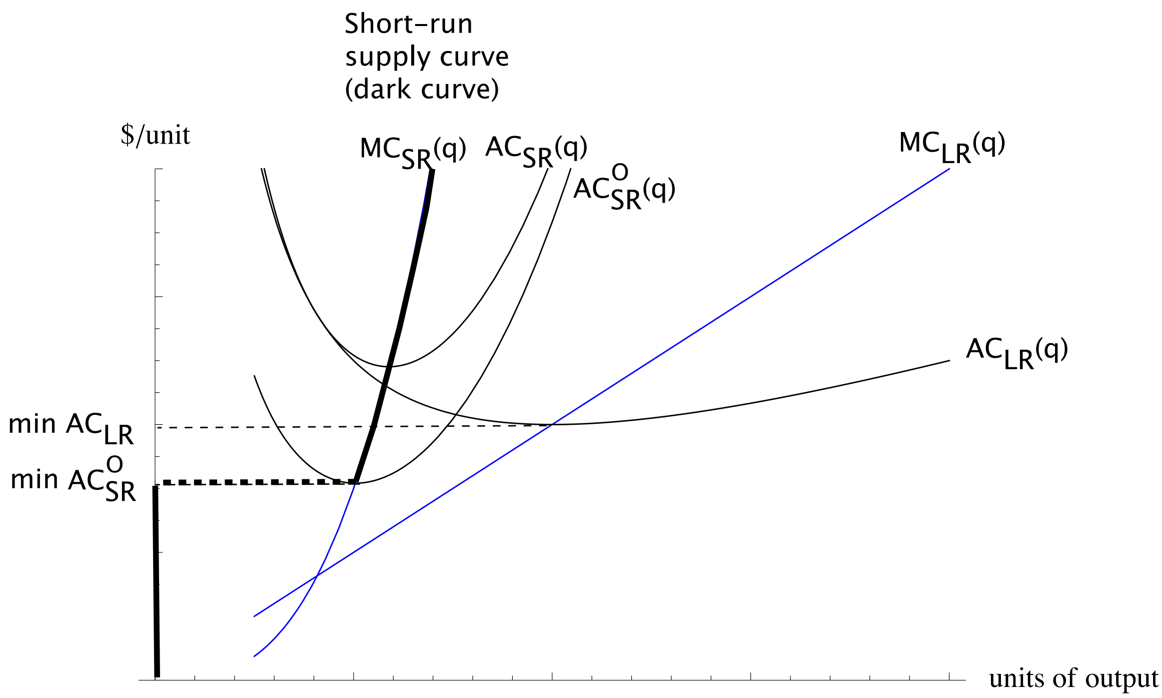
Summary of firm's supply curves

- Putting it altogether in one graph



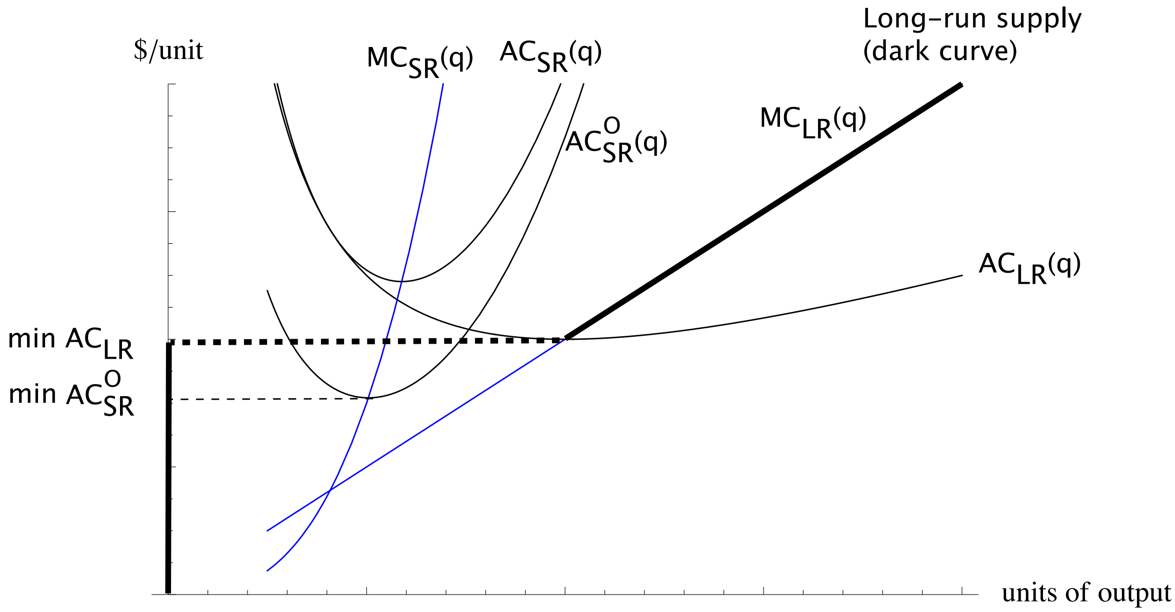
Summary of firm's supply curves

- Putting it altogether in one graph -- short-run supply



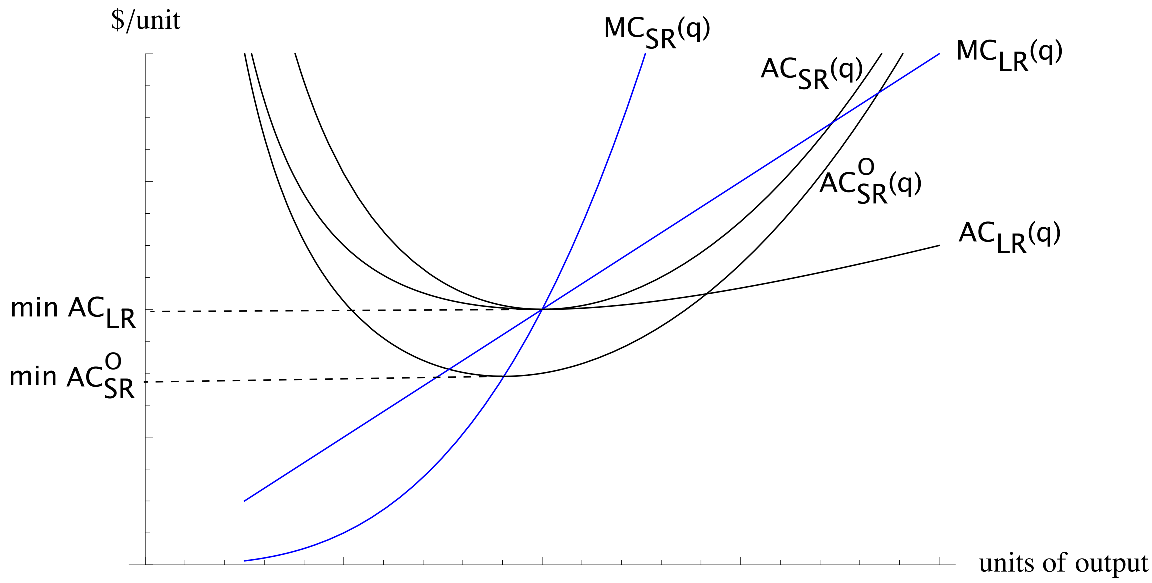
Summary of firm's supply curves

- Putting it altogether in one graph -- long-run supply curve



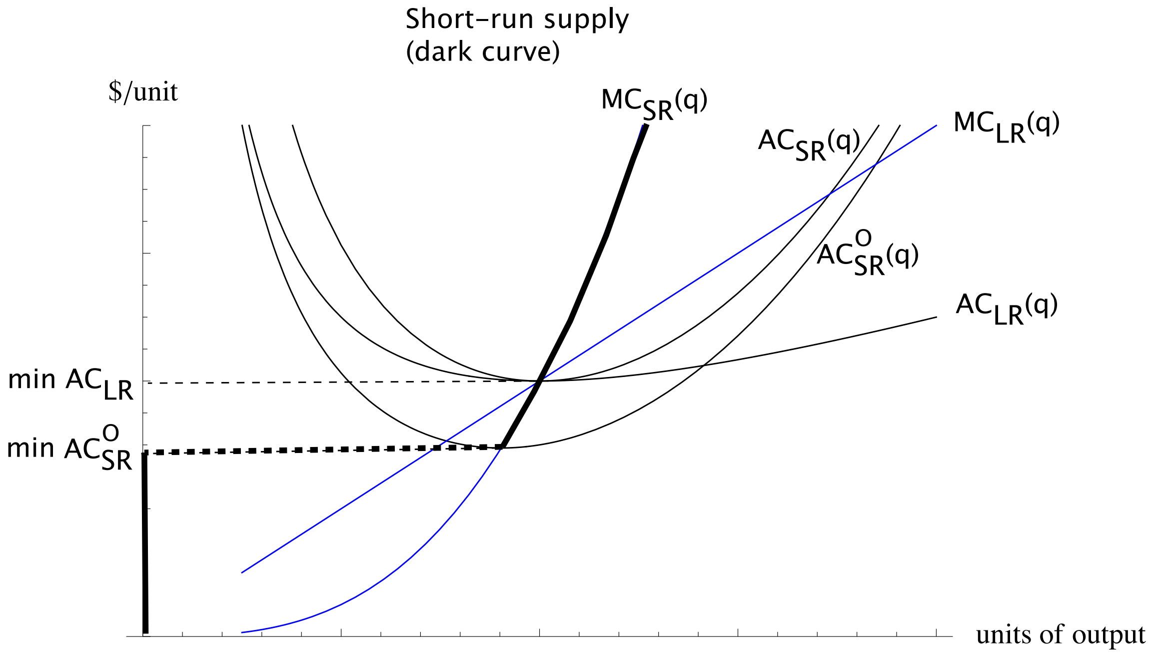
Summary of firm's supply curves

- In the previous graphs, capital stock was set for a low level of production. Suppose now it is set at the long-run *MES* level:



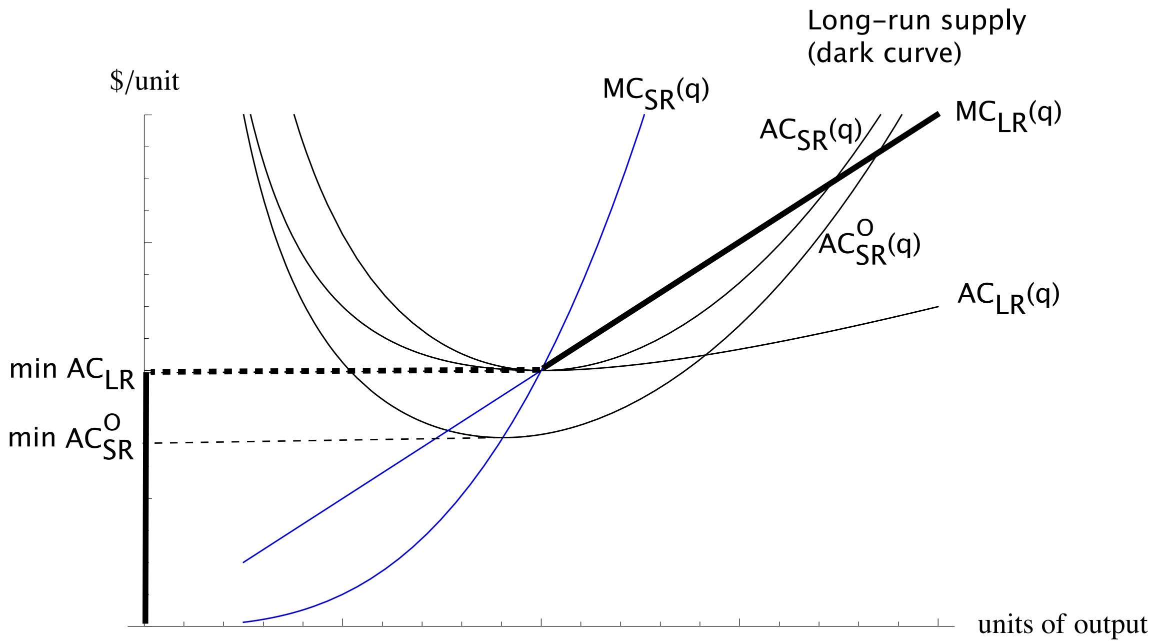
Summary of firm's supply curves

- Short-run supply curve



Summary of firm's supply curves

- Long-run supply curve

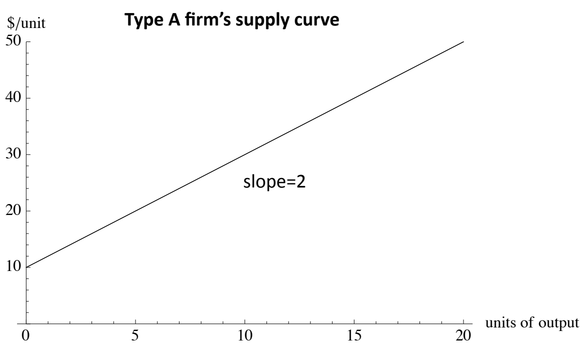


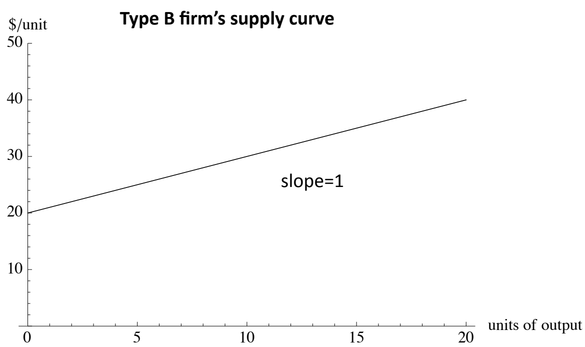
Deriving the short-run market supply curve

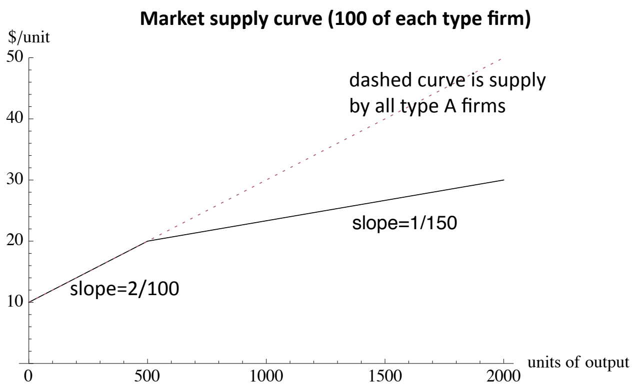
- We assume that some capital is necessary for production.
  - Thus, in the short run, only existing firms can produce.
- The market supply curve is therefore simply the horizontal sum of the supply curves of the currently active firms.
- If different firms have different technologies or different capital levels, then the firm supply curves are different and the market supply curve will be “scalloped”

Short-run market supply curve, continued

- Simple example: 100 “type A” firms and 100 “type B” firms:







E.g.: at a price of $p=30$, each type A firm produces $q_A=10$ and each type B firm produces $q_B=10$, so total market supply is $Q^S=2000$.

Short-run market supply curve, continued

- Real example: World supply of Aluminum in 1993

  (Discussed in class. Data on canvas.)

Long-run market supply curve

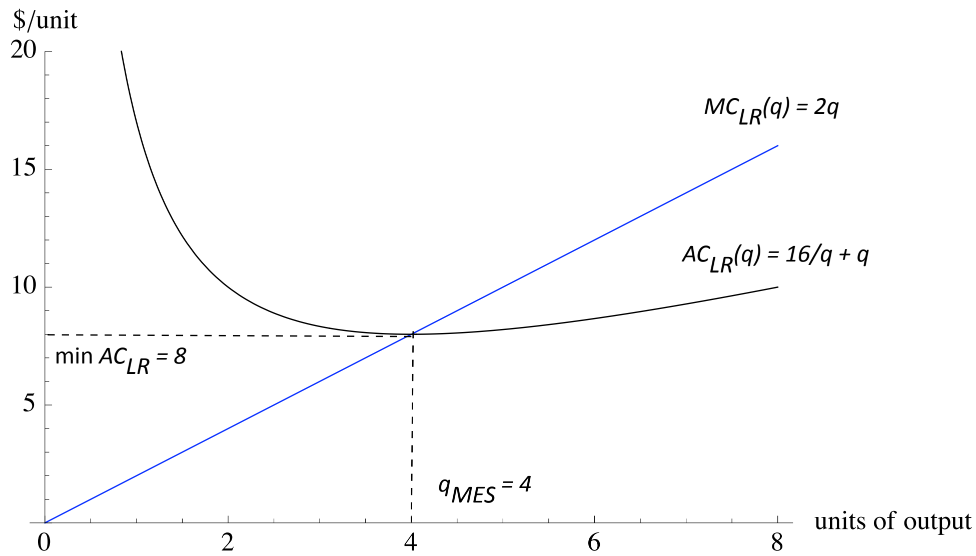
- In the long run, the market supply curve includes **every potential supplier**, not just those currently in the market.
- Thus, we add together the supply curves of existing firms and all potential entrants.
- A few benchmark cases that will help us think about the more general case. There are two dimensions to our taxonomy.
  - Homogenous firms versus heterogenous firms
  - Constant input costs versus non-constant (typically increasing) input costs

Long-run market supply curve

- Case 1: Homogenous firms, constant input costs. Together, this is called a "constant-cost industry."
- Case 2: Homogenous firms, increasing input costs. Together, this is called an "increasing-cost industry."
  - The assumption of "increasing input costs" is made at the industry level. That is, each firm takes its input prices as given, but if every firm in the industry increases production simultaneously, then input prices rise.
  - The assumption of "decreasing input costs" is the reverse (e.g., the more laptops produced, the cheaper LCD screens become). Together with the assumption of homogenous firms, we have a "decreasing cost industry."
- General case: Heterogeneous firms, non-constant input costs.

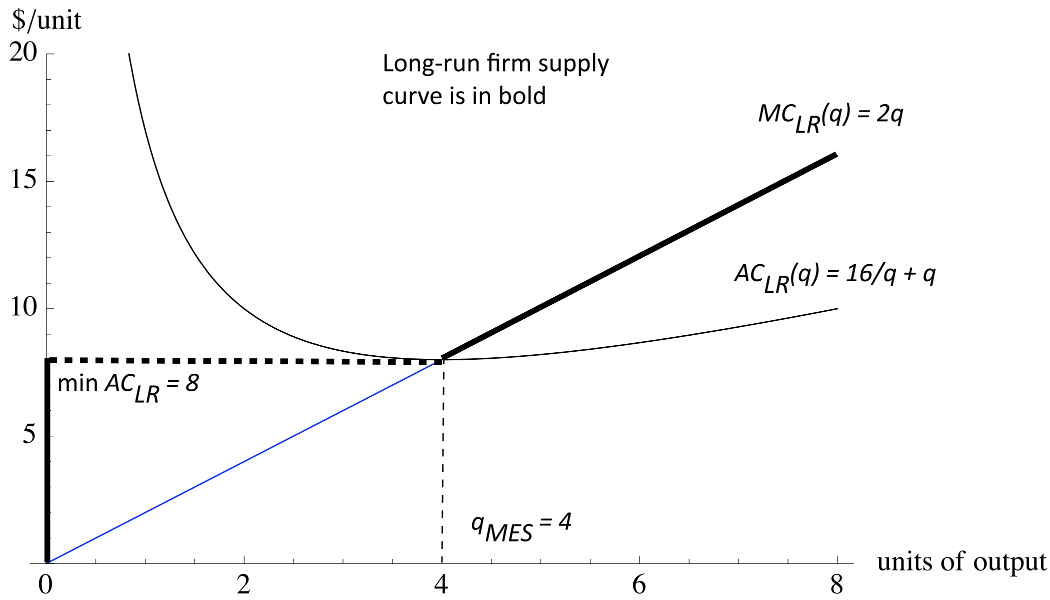
Case 1: Constant-cost industry

- Because all firms are identical (homogeneous), we need only consider a single set of unit cost curves. As an example, suppose that $C_{LR}(q) = 16 + q^2$. Each firm's unit costs are...



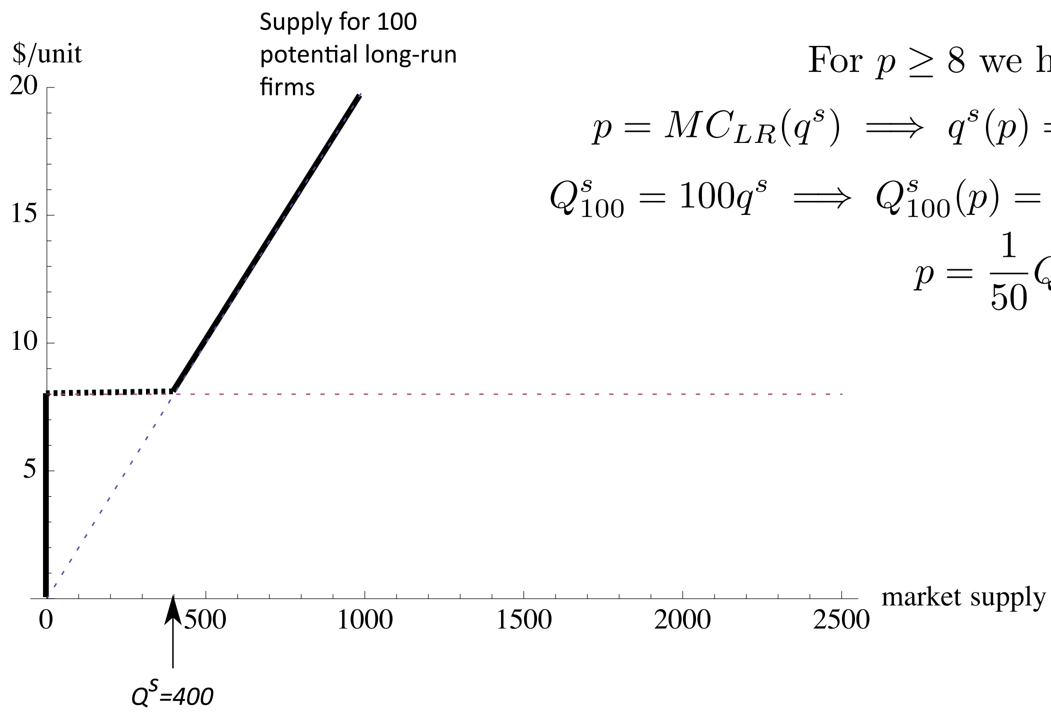
Case 1: Constant-cost industry

- Each firm's long-run supply curve is given by the bold curve. For $p < 8$ no firm supplies output; at a price of $p=8$, every firm is just willing to supply $q=4$ units.



Case 1: Constant-cost industry

- No suppose that there only 100 potential long-run suppliers.



For $p \geq 8$ we have

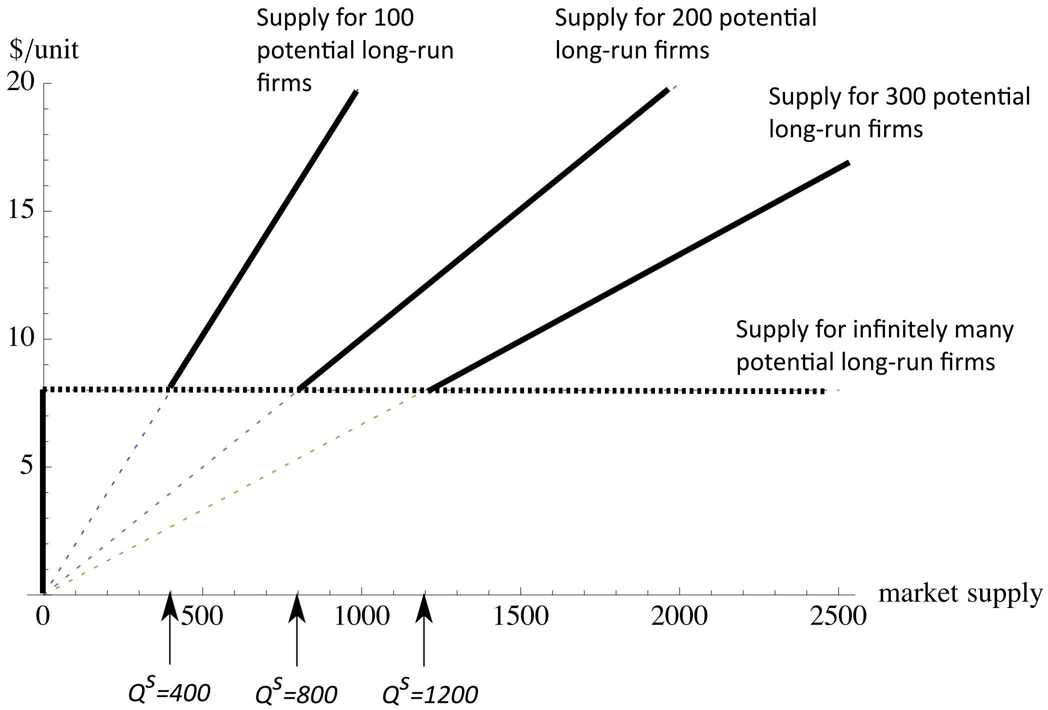
$$p = MC_{LR}(q^s) \implies q^s(p) = \frac{p}{2}$$

$$Q^s_{100} = 100q^s \implies Q^s_{100}(p) = 50p$$

$$p = \frac{1}{50}Q^s_{100}$$

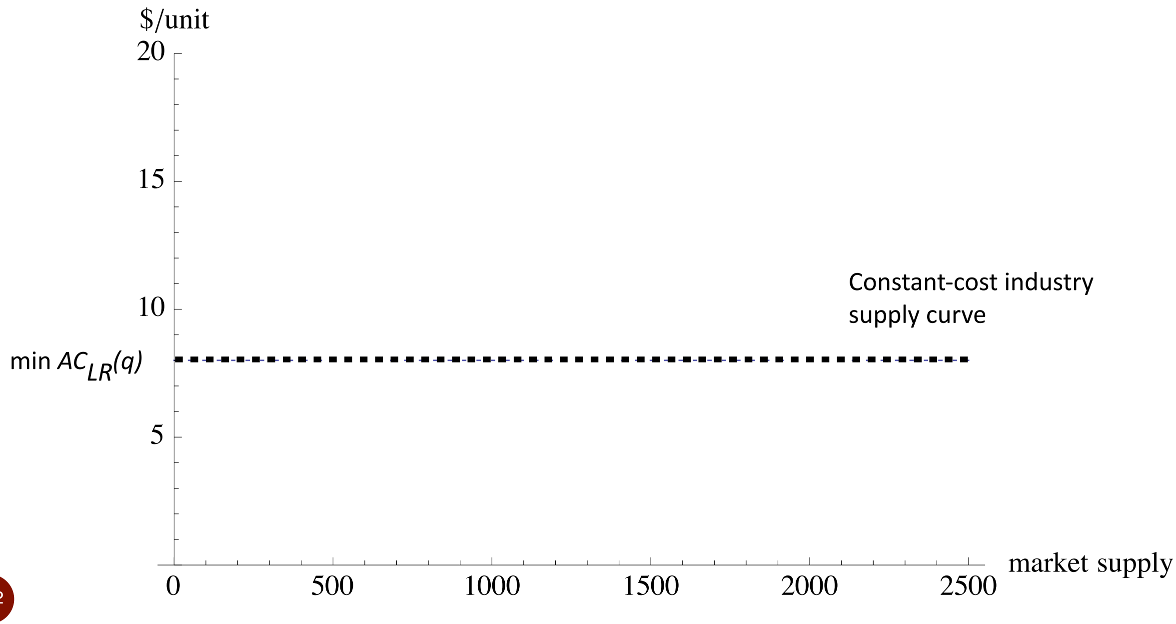
Case 1: Constant-cost industry

- Consider also 200, 300 and an infinite number of firms.



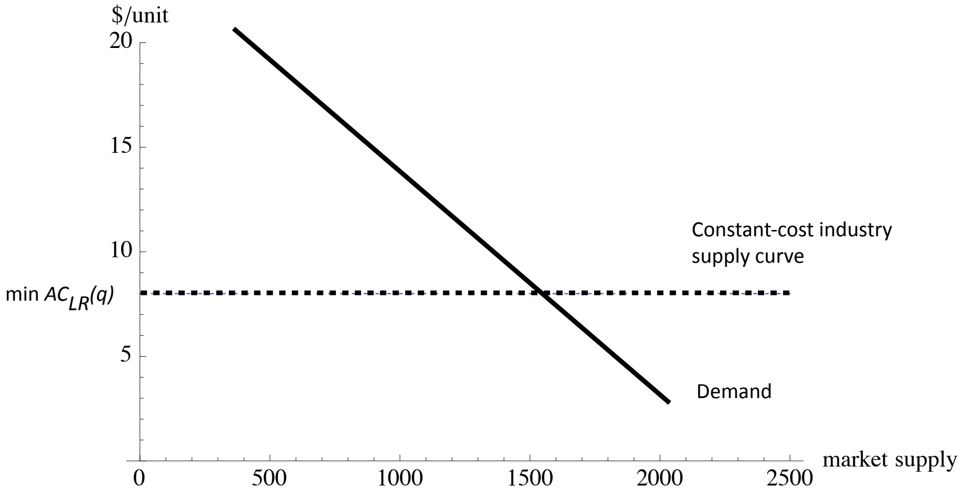
Case 1: Constant-cost industry

- With an unlimited number of potential entrants, the constant cost industry supply curve is a horizontal line at the minimum long-run average cost of the firms. The dots represent firms entering with the MES output. Often we draw a solid line.



Case 1: Constant-cost industry

- We do not need to know the market demand function to predict that the equilibrium price in a constant-cost industry is $p = min\ AC$ (assuming that the market is active).



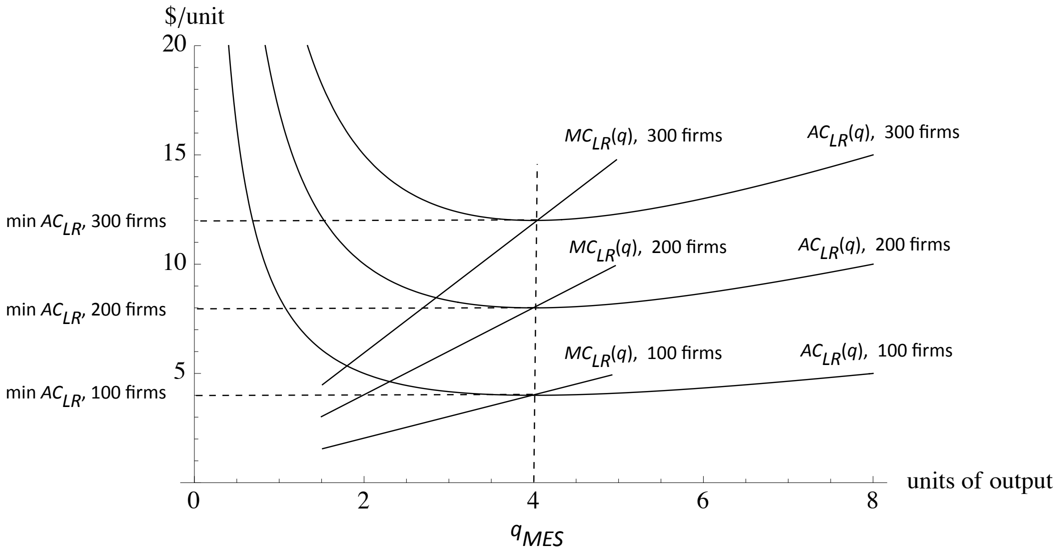
- We ignore the technical requirement that equilibrium output is a multiple of the MES. (Introduces slight rounding error.)

Case 2: Increasing-cost industry

- We continue to suppose an unlimited number of homogenous firms in the long-run.
- We now consider the case where the size of the industry output has an impact on input prices, although no individual firm has any perceived impact.
- These assumptions, in tandem, imply that every firm has the same unit cost functions, but the unit cost functions depend upon the number of firms producing in the industry.

Case 2: Increasing-cost industry

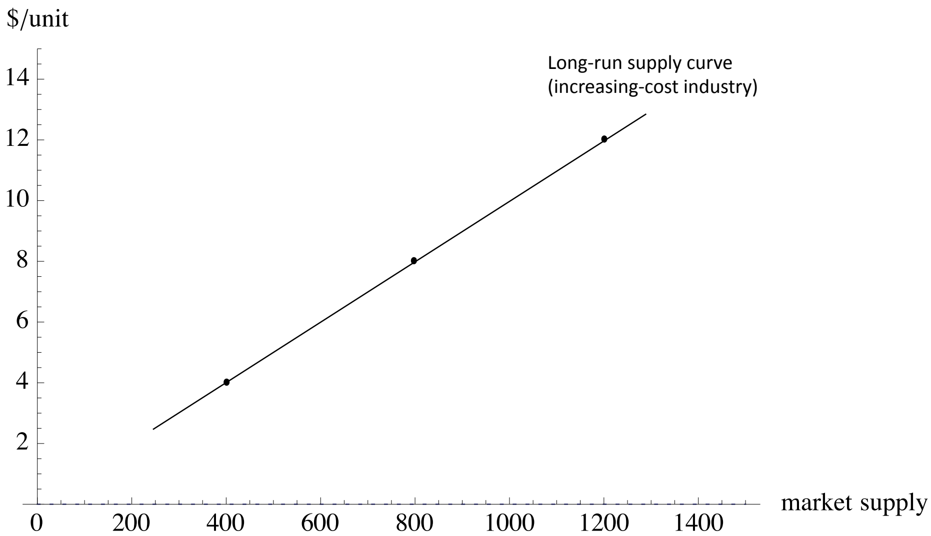
- Consider three configurations: 100 firms, 200 firms and 300 firms. For each setting, there is a set of *MC* and *AC* curves.



- While this is drawn with the same *MES* for each configuration, this need not be the case in general.

Case 2: Increasing-cost industry

- At a price of *p=4*, 100 firms are willing to supply *Q=400*; at a price of *p=8*, 200 firms are willing to supply *Q=800*; at a price of *p=12*, 300 firms are willing to supply a total of *Q=1200*.
- Thus, the three points (*p=4,Q=400*), (*p=8,Q=800*), and (*p=12,Q=1200*) must be on the long-run market supply curve.



Case 2: Increasing-cost industry

- **Remarks**
  - Firms enter or exit to the point where the price equals the minimum average cost for the given firm configuration.
  - Firms make zero economic profit, just like in the constant-cost industry case.
  - The case of **decreasing-cost industry** is similar to the increasing-cost industry except that now the long-run supply curve is downward sloping. Empirically uncommon.
    - Example: notebook prices fall as industry output expands because prices of inputs (LCD screens, flash memory, etc.) fall as industry scale rises. Still, no economic profits.
  - Zero economic profit arises because firms are identical. No firm has a competitive advantage over another, so there can be no economic profit in the long run.

Case 2: Increasing-cost industry

- **Producer surplus?** Note that there is area between the equilibrium price and the supply curve in an increasing-cost industry.
- **Puzzle:** None of the firms make economic profit/surplus so where is the producer surplus going?
- **Answer**: the producer surplus is the economic profit going to the scarce factors of production (the input suppliers that are causing the unit costs of the final output to rise).
- **Conclusion:** Producer surplus is economic profit going to someone along the vertical chain of production. But in the case of an increasing-cost industry where all the downstream firms are identical and numerous, this economic profit flows to the upstream input suppliers.

General case: firm heterogeneity and non-constant input costs

- A more general model of costs allow some firms to have a cost advantage over others. (E.g., access to cheap inputs, distribution networks, geographical location, subsidies, different technologies, experience, etc.). With firm heterogeneity, economic profits may accrue to the firms with cost advantages.
- Cost heterogeneity and non-constant input prices provide two reasons why supply curves may slope upward:
  - input prices rise with industry output
  - higher output supply requires production by higher-cost firms, which requires a higher market price

General case: firm heterogeneity and non-constant input costs

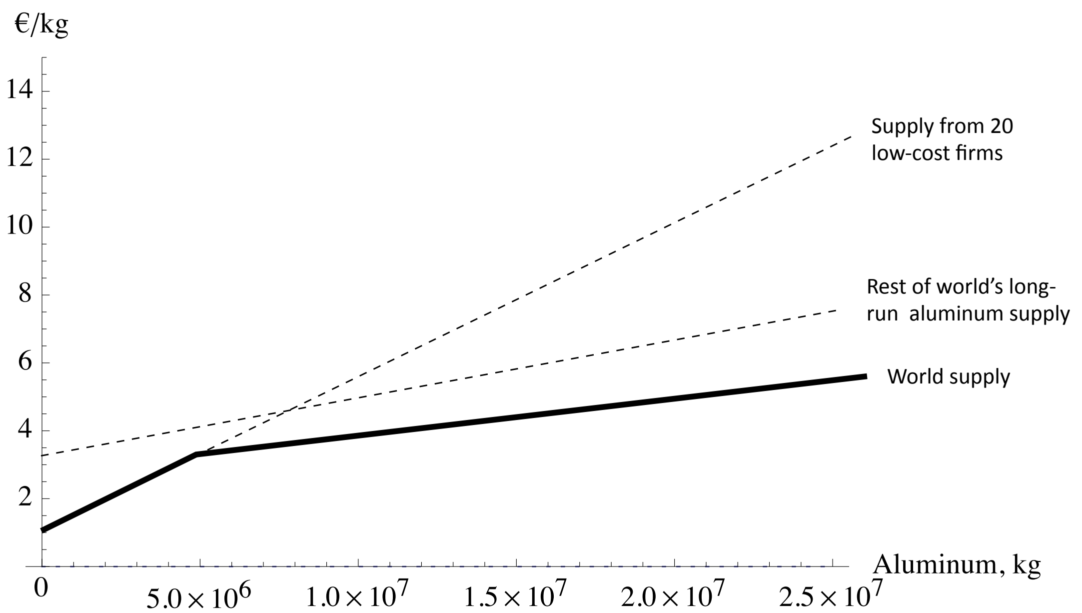
- With cost heterogeneity, there are a finite number of firms with cost advantages and rising long-run supply functions.
- The supply curves for these firms are added together to obtain the lower part of the market supply curve.
- When prices rise on this curve to the point where an unlimited number of identical firms are willing to produce, we can construct the residual supply curve for this segment of high-cost firms just as in the homogenous case (constant-cost, increasing-cost or decreasing-cost industry).
- We obtain the market supply curve by adding together the low-cost supply segment with the unlimited capacity, higher-cost segment.

General case: firm heterogeneity and non-constant input costs

- For example: Consider the world market for aluminum and suppose that it is an increasing-cost industry (bauxite prices rise with increased aluminum production). Furthermore, suppose that all smelters have the same technology, except that 20 existing smelters in Eastern Europe receive subsidized electricity. The subsidy applies only to these 20 existing plants; no new smelters will obtain the subsidy.
- Economic profit will accrue to these advantaged smelting operations. (More on this later.)

General case: firm heterogeneity and non-constant input costs

- Aluminum long-run supply:



Producer surplus

- We defined the area under the market price and above the market supply curve as **producer surplus.**
- **Where does it come from?**

Source of rents (producer surplus)

- When all firms are identical, as in the case of a constant-industry, no firm earns economic profit. Economic profits only accrue to scarce factors that generate value. To repeat …

Scarcity is the source of profit in competitive markets

- Thus, if input costs are non-constant but firms are numerous and identical, no firm can earn an economic profit.

Source of rents (producer surplus)

- In an increasing-cost industry, the economic profits accrue to the scarce inputs.
  - We need to be careful about who gets the producer surplus: “The area above the market supply curve and below the market price is producer surplus earned by the **vertical-production chain**.”
  - Aluminum smelting example: When the demand for Aluminum increases, most of the long-run economic profits go to the Bauxite/ Alumina operations. This is because (i) Bauxite is scarce (upward sloping supply curve) and (ii) building smelters and generating electricity are largely constant-cost activities (not scarce factors).

Source of rents (producer surplus)

- So why is this connection between scarcity and economic profit often referred to as the source of “Ricardian rents”?
  - In short, David Ricardo was interested in understanding why the import tariffs on grain in the 19th century (called the “Corn Laws”) did not help farmers, but instead simply raised the rent of farmland and created economic profits for the land owners.
  - The reason for the outcome is that the farmers were all equally skilled and in great supply (not scarce) but the land for growing grain was scarce. When the prices of grain increased so farmers earned greater revenues, the land owners raised the rent of the farmland to capture the economic profit entirely for themselves.
  - An abundance of land and insufficient labor would reverse the result: farmers (the scarce input) would earn profit.

Source of rents (producer surplus)

- To economists, “Ricardian rent” or simply “rent” is used to mean economic profit going to the owners of scarce inputs. The scarce input doesn’t have to be land; it could be a low-cost production process known by only a single firm, etc.
- You can read more about Ricardo and the corn laws in Harford (ch 1). Harford uses Ricardo’s idea to explain why Starbucks does not get all of the profit from selling coffee. A large fraction of profit is captured by owners of scarce retail space.
- Ricardo’s idea is absolutely fundamental to thinking about business in competitive markets. The idea is properly emphasized by Porter throughout his book, Competitive strategy.
- “Scarcity is the unique source of profit in competitive markets” is one of the five most valuable ideas from microeconomics. Other top ideas are “Supply=Demand in competitive markets”, “ignore sunk costs” and “think on the margin”. We will have to wait until Lectures 9 and 10 to finish our list.